

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\411-V-0017%.pvdb

Date of Analysis : Aug 11, 2026 5:39pm

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 3000.000 mm ID
 Vessel Design Length, Tangent to Tangent 8500.00 mm
 Specified Datum Line Distance 0.00 mm
 Internal Design Temperature 95 °C
 Internal Design Pressure 0.101 MPa
 External Design Temperature 95 °C
 External Design Pressure 0.074 MPa
 Maximum Allowable Working Pressure 0 MPa
 External Max. Allowable Working Pressure 0 MPa
 Shop Test Pressure 0.427 MPa
 Field Test Pressure 0.427 MPa
 Required Minimum Design Metal Temperature 11.4 °C
 Warmest Computed Minimum Design Metal Temperature 11.0 °C

Wind Design Code IS 875 (Part 3):2015 Amd. 1&2 2020
 Earthquake Design Code IS 1893 RSM (Part 4): 2015 + 2018 Amendment 2

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-240 316L	...	...	S31603	
Head	SA-240 316L	...	...	S31603	
Flange	SA-240 316L	...	...	S31603	
Nozzle	SA-312 TP316L	...	...	S31603	+IT
Nozzle	SA-312 TP316	...	...	S31600	
Nozzle	SA-182 F316L	...	<= 5	S31603	+IT
Nozzle	SA-240 316L	...	...	S31603	+IT
Re-Pad	SA-240 316L	...	...	S31603	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-182 F316L	...	<= 5	S31603	
Nozzle Flg	SA-182 F316	...	> 5	S31600	
Rings	SA-240 316L	...	...	S31603	
Rings	SA-240 304	...	...	S30400	+IT
Flg Bolting	SA-193 B7	...	<= 2 1/2	G41400	
Hrz Bolting	SA-193 B8	1	...	S30400	
Saddles	SA-240 316L	...	...	S31603	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Left head	0.139	0.07	1.000	0.0000	No	No
Shell-1	0.139	0.07	1.000	0.0000	N/A	No
Shell-2	0.139	0.07	1.000	0.0000	N/A	No
Shell-3	0.139	0.07	1.000	0.0000	N/A	No
Right head	0.139	0.07	1.000	0.0000	No	No
N8 flg	0.101	0.07	2.000	0.0000	N/A	No
N8 flg	0.101	0.07	1.000	0.0000	No	No

Liquid Level: 3000.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 1.268



Stiffener Ring Specifications:

Elevation mm	Selected Type	Ring Description
2050.00	Bar 150.0 x 14.0	Ring 1
415.00	Bar 200.0 x 28.0	SR1
4050.00	Bar 150.0 x 14.0	Ring 2
6550.00	Bar 150.0 x 14.0	Ring 3
8135.00	Bar 200.0 x 28.0	SR, 2085.0 Fr40

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	50.0	50.0	12.0	10.0	1.8	...	1.00	1.00
Cylinder	3050.0	3000.0	10.0	10.0	1.8	...	1.00	1.00
Cylinder	6050.0	3000.0	10.0	10.0	1.8	...	1.00	1.00
Cylinder	8450.0	2400.0	10.0	10.0	1.8	...	1.00	1.00
Ellipse	8500.0	50.0	12.0	10.0	1.8	...	1.00	1.00
Body Flg	7285.0	0.0	68.0	66.8	58.6	...	1.00	1.00
Ellipse	6050.0	0.0	8.0	6.0	1.5	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	10	6.85365	0.074458	0.15853
20	Ring	10	4.61714	0.074458	0.26811
Ring	Ring	10	6.75664	0.074458	0.19805
Ring	30	10	7.36901	0.074458	0.16024
30	Ring	10	7.36901	0.074458	0.16024
Ring	40	10	8.04739	0.074458	0.127
40	Ring	10	8.04739	0.074458	0.127
Ring	Ring	10	6.66971	0.074458	0.20469
Ring	50	10	4.46155	0.074458	0.27296
50	60	10	6.85365	0.074458	0.15853
60	70	66.802	58.5724	0.074458	No Calc
70	80	6	2.33294	0.074458	0.35557

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	Ring	665	4174.63	No Calc	No Calc
Ring	Ring	1635	4175.75	1731284	34394484
Ring	30	2000	4176.31	2731322	10357825
30	Ring	2000	4176.31	No Calc	No Calc
Ring	40	2500	4175.37	3375710	10357825
40	Ring	2500	4175.37	No Calc	No Calc
Ring	Ring	1585	4175.66	3072123	10357825
Ring	50	615	4176.42	1652883	34394572
50	60	No Calc	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1 6"	WRC-107/537	0.499	n/a	Passed
N3 2"	WRC-297	0.354	Shell	Passed



N6 3"	WRC-297	0.685	Nozzle	Passed
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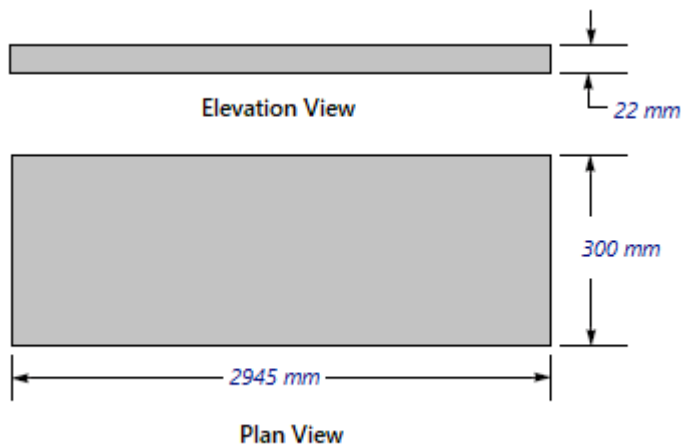
Foundation/Support Summary

Saddle Parameters:

Saddle Width	300.000	mm
Saddle Bearing Angle	150.000	deg.
Centerline Dimension	1715.000	mm
Wear Pad Width	350.000	mm
Wear Pad Thickness	10.000	mm
Wear Pad Bearing Angle	162.000	deg.
Distance from Saddle to Tangent	365.000	mm

Baseplate Length	2945.000	mm
Baseplate Thickness	22.000	mm
Baseplate Width	300.000	mm
Number of Ribs (including outside ribs)	5	
Rib Thickness	10.000	mm
Web Thickness	10.000	mm
Height of Center Web	179.000	mm
Number of Bolts in Baseplate	8	

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	677291.31	N
Maximum Transverse Saddle Shear Load	67264.06	N
Maximum Longitudinal Saddle Shear Load	29885.46	N

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	447026.12	N
Maximum Transverse Saddle Shear Load	10383.19	N
Maximum Longitudinal Saddle Shear Load	8038.17	N

Weights:

Fabricated - Bare w/o removable internals	14314.2	kgm
Shop Test - Fabricated + water (full)	81614.5	kgm
Shipping - Fab. + rem. intls.+ shipping app.	19135.1	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	19135.1	kgm
Empty - Fab. + intls. + details + wghts.	19135.1	kgm
Operating - Empty + operating liquid (no ca)	104246.5	kgm
Field Test - Empty weight + test medium (full)	86879.9	kgm

Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1 6"	1	OK	...	OK	1.56	OK	Passed
N3 2"	1	...	...	OK	1.50	OK	No Calc[*]
N4 4"	1	OK	...	OK	1.50	OK	Passed
N5 1.5"	1	...	...	OK	1.50	OK	No Calc[*]
N6 3"	1	...	...	OK	1.50	OK	No Calc[*]
N7 4"	1	OK	...	OK	1.50	OK	Passed
M1 24"	...	OK	...	...		OK	Passed
N8 26"	...	OK	...	OK	1.50	OK	Passed
Nozzle boot	...	OK	...	OK	2.56	OK	Passed
N2 3"	2	OK	...	OK	5.26	OK	Passed
N9 3"	1	...	...	OK	1.50	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 0 Nozzle : Nozzle boot
 Minimum MAWP Shells/Flanges : 0 Element : Shell-3
 Minimum MAPnc Shells/Flanges : 1 Element : Shell-3

Computed Vessel M.A.W.P. : 0 MPa

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
10	N1 6"	0.000	180.000	308.102
20	N3 2"	250.000	270.000	101.600
20	N4 4"	550.000	270.000	194.361
20	N5 1.5"	250.000	326.100	78.548
20	N6 3"	1400.000	270.000	155.854
20	N7 4"	1100.000	270.000	203.200
30	M1 24"	4550.000	270.000	1219.200
40	N8 26"	7285.000	270.000	1320.800
40	Nozzle boot	6050.000	90.000	2032.000
60	N2 3"	0.000	0.000	241.300
70	N9 3"	0.000	0.000	155.854

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

Checking Multiple Nozzles on Flat Head per ASME Sec. VIII Div. 1 UG-39

Comparing Nozzles on Element: N8 flg

Note: No Nozzle pairs found on this element.

UG-39 Nozzle Diameter and Distance to Edge Checks:



Nozzle Description	Nozzle dia. mm	Head Dia. /2 mm	Distance from Edge mm	Nozzle dia./4 mm
N2 3"	76.2	345.33	381.13	19.05

No Multiple Nozzle spacing violations detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N5 1.5"	1.500 in	Actual	LWN	2.620	14.224	...	...	274.43	150
N3 2"	2.000 in	Actual	LWN	3.060	13.462	...	...	310.50	300
N6 3"	3.000 in	40	WNF	3.500	5.486	188.90	6.00	310.66	300
N2 3"	3.000 in	Actual	LWN	4.240	15.748	...	...	271.64	150
N9 3"	3.000 in	40	WNF	3.500	5.486	...	...	307.08	300
N4 4"	4.000 in	80	WNF	4.500	8.560	214.30	10.00	3160.00	150
N7 4"	4.000 in	Actual	LWN	5.500	19.050	...	...	3160.00	150
N1 6"	6.000 in	40	WNF	6.625	7.112	288.27	8.00	406.48	300
M1 24"	24.000 in	Actual	WNF	24.787	10.000	729.60	10.00	293.40	150
N8 26"	26.000 in	Actual	WNF	26.945	12.000	762.00	10.00	349.56	150
Nozzle boot	40.000 in	Actual	None	40.945	12.000	1220.00	10.00	803.02	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N5 1.5"	SA-182 F316L	4.000	10.000	...	...	...
N3 2"	SA-312 TP316	4.000	10.000	...	...	...
N6 3"	SA-312 TP316	6.000	10.000	6.000	6.000	...
N2 3"	SA-312 TP316	66.802	10.000	...	...	...
N9 3"	SA-312 TP316L	6.000	10.000	...	...	...
N4 4"	SA-312 TP316L	10.000	10.000	8.000	10.000	...
N7 4"	SA-312 TP316L	6.000	10.000	...	...	...
N1 6"	SA-312 TP316L	6.350	10.000	6.000	8.000	...
M1 24"	SA-240 316L	6.000	10.000	6.000	10.000	...
N8 26"	SA-240 316L	6.000	10.000	8.000	10.000	...
Nozzle boot	SA-240 316L	10.000	10.000	8.000	10.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N5 1.5"	250.000	326.1	211.14	0.00	Shell-1
N3 2"	250.000	270.0	300.00	0.00	Shell-1
N6 3"	1400.000	270.0	300.00	0.00	Shell-1
N2 3"	...	0.0	204.79	0.00	N8 flg
N9 3"	...	0.0	300.00	0.00	N8 flg
N4 4"	550.000	270.0	300.00	2850.00	Shell-1
N7 4"	1100.000	270.0	300.00	2850.00	Shell-1
N1 6"	...	180.0	350.00	0.00	Left head
M1 24"	4550.000	270.0	250.00	0.00	Shell-2
N8 26"	7285.000	270.0	300.00	0.00	Shell-3
Nozzle boot	6050.000	90.0	700.00	0.00	Shell-3



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 12.0mm THK X 3000.0mm ID X 50.0mm	SA-240 316L
2	CYLINDER: 10.0mm THK X 3000.0mm ID X 3000.0mm	SA-240 316L
1	CYLINDER: 10.0mm THK X 3000.0mm ID X 2400.0mm	SA-240 316L
1	BLIND FLANGE: 68.0mm THK X 870.0mm OD	SA-240 316L
1	ELLIPTICAL HEAD: 2.0 X 1, 8.0mm THK X 1016.0mm ID X 50.0mm	SA-240 316L
2	SADDLE: 300.0mm X 150 DEG	SA-240 316L
1	PLATFORM: 1000mm X 3012mm WIDE	
1	LADDER: 4000.0mm	...
1	BAR RING STIFFENER: 3024.0mm ID X 3324.0mm OD X 14.0mm	SA-240 304
2	BAR RING STIFFENER: 3020.0mm ID X 3420.0mm OD X 28.0mm	SA-240 316L
2	BAR RING STIFFENER: 3020.0mm ID X 3320.0mm OD X 14.0mm	SA-240 304
1	CLASS 150 GR 2.3, 24.0" BLIND FLANGE(S)	SA-182 F316L
1	GASKET: 704.8mm OD X 673.1mm ID	...
24	BODY FLANGE BOLTS: 31.8mm DIA	SA-193 B7
24	NUTS FOR BODY FLANGE BOLTS: 31.8mm DIA	...
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
N8 flg	[11]	-196						
Left head	[16]	-196						
Shell-1	[16]	-196						
Shell-2	[16]	-196						
Shell-3	[16]	-196						
Right head	[16]	-196						
N8 flg	[16]	-196						
SR1	[23] !	11	11		1.000	10.000	1.00	No
SR, 2085.0 Fr	[23] !	11	11		1.000	10.000	1.00	No
N1 6"	[1] !	11	-104		0.013	6.223	1.00	No
N3 2"	[15]	-196						
N4 4"	[1] !	11	-104		0.006	7.490	1.00	No
N5 1.5"	[1] !	11	-104		0.238	6.000	1.00	No
N6 3"	[1] !	11	-104		0.006	4.801	1.00	No
N7 4"	[1] !	11	-104		0.132	10.000	1.00	No
M1 24"	[1] !	11	-104		0.027	10.000	1.00	No
N8 26"	[1] !	11	-104		0.024	12.000	1.00	No
Nozzle boot	[1] !	11						No
N2 3"	[15]	-196						
N9 3"	[1] !	11	-104		0.007	4.801	1.00	No
Bolting	[21]	-48						

Warmest MDMT: 11 11

Required Minimum Design Metal Temperature 11.4 °C
 Warmest Computed Minimum Design Metal Temperature 11.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.



Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	0.1013	MPa
Design Internal Temperature	95.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	11.4	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.081
0.050	0.123
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.104
0.800	0.088
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000



Consider Vortex Shedding

N

Shop Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Field Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Vertical
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	0	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	1.500
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.081
0.050	0.123
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.104



0.800	0.088
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration Yes

Design Pressure + Static Head Y

Consider MAP New and Cold in Noz. Design N

Consider External Loads for Nozzle Des. Y

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) No

Do not use Nozzle MDMT Interpretation VIII-1 01-37 No

Use Table G instead of exact equation for "A" Yes

Shell Head Joints are Tapered Yes

Compute "K" in corroded condition Yes

Use Code Case 2286 No

Use the MAWP to compute the MDMT Yes

For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes

For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D No

Calculate B31.3 type stress for Nozzles with Loads Yes

Reduce the MDMT due to lower membrane stress Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	Left head
Distance "FROM" to "TO"	50 mm
Inside Diameter	3000 mm
Element Thickness	10 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.1013 MPa
Design Temperature Internal Pressure	95 °C
Design External Pressure	0.07446 MPa
Design Temperature External Pressure	95 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 316L
Allowable Stress, Ambient	115.15 MPa
Allowable Stress, Operating	115.15 MPa
Allowable Stress, Hydrotest	155.14 MPa
Material Density	0.008027 kg/cm³
P Number Thickness	0 mm
Yield Stress, Operating	146.37 MPa
External Pressure Chart Name	HA-4
UNS Number	S31603
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No



Element From Node	10
Detail Type	Liquid
Detail ID	Liquid 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	3000 mm
Liquid Density	0.001266 kg/cm ³

Element From Node	10
Detail Type	Nozzle
Detail ID	N1 6"
Dist. from "FROM" Node / Offset dist	1240 mm
Nozzle Diameter	6 in.
Nozzle Schedule	40
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	249.37 N
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-312 TP316L +IT

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Shell-1
Distance "FROM" to "TO"	3000 mm
Inside Diameter	3000 mm
Element Thickness	10 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.1013 MPa
Design Temperature Internal Pressure	95 °C
Design External Pressure	0.07446 MPa
Design Temperature External Pressure	95 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 316L
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Saddle
Detail ID	LeftSaddle
Dist. from "FROM" Node / Offset dist	365 mm
Width of Saddle	300 mm
Height of Saddle at Bottom	1715 mm
Saddle Contact Angle	150.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	350 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	162.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	20
Detail Type	Platform
Detail ID	PL
Dist. from "FROM" Node / Offset dist	0 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	11640000 mm ²
Platform Weight	44181 N
Platform Railing Weight	0 N/mm



Platform Grating Weight	0.017	kg/cm ²
Platform Width	3012	mm
Platform Height	1000	mm
Platform Clearance or End Offset	230	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.37	N/mm
Platform Length (top head platform)	8500	mm

Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	3000	mm
Liquid Density	0.001266	kg/cm ³

Element From Node	20	
Detail Type	Ring	
Detail ID	Ring 1	
Dist. from "FROM" Node / Offset dist	2000	mm
Inside Diameter of Ring	3024	mm
Thickness of Ring	14	mm
Outside Diameter of Ring	3324	mm
Material Name	SA-240 304	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element From Node	20	
Detail Type	Ring	
Detail ID	SR1	
Dist. from "FROM" Node / Offset dist	365	mm
Inside Diameter of Ring	3020	mm
Thickness of Ring	28	mm
Outside Diameter of Ring	3420	mm
Material Name	SA-240 316L +IT	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N3 2"	
Dist. from "FROM" Node / Offset dist	200	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	108.49	N
Grade of Attached Flange	GR 2.2	
Nozzle Matl	SA-312 TP316	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N4 4"	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	4	in.
Nozzle Schedule	80	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	808.01	N
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-312 TP316L +IT	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N5 1.5"	
Dist. from "FROM" Node / Offset dist	200	mm



Nozzle Diameter 1.5 in.
 Nozzle Schedule None
 Nozzle Class 150
 Layout Angle 326.10001
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 62.111 N
 Grade of Attached Flange GR 2.3
 Nozzle Matl SA-182 F316L +IT

Element From Node 20
 Detail Type Nozzle
 Detail ID N6 3"
 Dist. from "FROM" Node / Offset dist 1350 mm
 Nozzle Diameter 3 in.
 Nozzle Schedule 40
 Nozzle Class 300
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 127.68 N
 Grade of Attached Flange GR 2.3
 Nozzle Matl SA-312 TP316

Element From Node 20
 Detail Type Nozzle
 Detail ID N7 4"
 Dist. from "FROM" Node / Offset dist 1050 mm
 Nozzle Diameter 4 in.
 Nozzle Schedule None
 Nozzle Class 150
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 1873.8 N
 Grade of Attached Flange GR 2.3
 Nozzle Matl SA-312 TP316L +IT

Element From Node 20
 Detail Type Weight
 Detail ID Lifting lug 1
 Dist. from "FROM" Node / Offset dist 950 mm
 Miscellaneous Weight 686.42 N
 Offset from Element Centerline 1500 mm

Element From Node 20
 Detail Type Weight
 Detail ID Lifting lug 2
 Dist. from "FROM" Node / Offset dist 950 mm
 Miscellaneous Weight 686.42 N
 Offset from Element Centerline 1500 mm

Element# 3/7
 Element From Node 30
 Element To Node 40
 Element Type Cylinder
 Description Shell-2
 Distance "FROM" to "TO" 3000 mm
 Inside Diameter 3000 mm
 Element Thickness 10 mm
 Internal Corrosion Allowance 0 mm
 Nominal Thickness 10 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 0.1013 MPa
 Design Temperature Internal Pressure 95 °C
 Design External Pressure 0.07446 MPa
 Design Temperature External Pressure 95 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-240 316L
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0



Weld is pre-Heated

No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	3000 mm
Liquid Density	0.001266 kg/cm ³

Element From Node	30
Detail Type	Ring
Detail ID	Ring 2
Dist. from "FROM" Node / Offset dist	1000 mm
Inside Diameter of Ring	3020 mm
Thickness of Ring	14 mm
Outside Diameter of Ring	3320 mm
Material Name	SA-240 304
Height of Section Ring	0 mm
Using Custom Stiffener Section	No

Element From Node	30
Detail Type	Nozzle
Detail ID	M1 24"
Dist. from "FROM" Node / Offset dist	1500 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	270.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	3519.3 N
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-240 316L +IT

Element From Node	30
Detail Type	Weight
Detail ID	Cleat weight
Dist. from "FROM" Node / Offset dist	1200 mm
Miscellaneous Weight	4903 N
Offset from Element Centerline	0 mm

Element From Node	30
Detail Type	Weight
Detail ID	Manway davit
Dist. from "FROM" Node / Offset dist	1500 mm
Miscellaneous Weight	490.3 N
Offset from Element Centerline	0 mm

Element#	4/7
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell-3
Distance "FROM" to "TO"	2400 mm
Inside Diameter	3000 mm
Element Thickness	10 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.1013 MPa
Design Temperature Internal Pressure	95 °C
Design External Pressure	0.07446 MPa
Design Temperature External Pressure	95 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 316L
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No



Element From Node	40	
Detail Type	Saddle	
Detail ID	New Saddle	
Dist. from "FROM" Node / Offset dist	2085	mm
Width of Saddle	300	mm
Height of Saddle at Bottom	1715	mm
Saddle Contact Angle	150.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	350	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	162.0	
Friction coefficient	0.1	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	3000	mm
Liquid Density	0.001266	kg/cm ³
Element From Node	40	
Detail Type	Ring	
Detail ID	Ring 3	
Dist. from "FROM" Node / Offset dist	500	mm
Inside Diameter of Ring	3020	mm
Thickness of Ring	14	mm
Outside Diameter of Ring	3320	mm
Material Name	SA-240 304	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	
Element From Node	40	
Detail Type	Ring	
Detail ID	SR, 2085.0 Fr40	
Dist. from "FROM" Node / Offset dist	2085	mm
Inside Diameter of Ring	3020	mm
Thickness of Ring	28	mm
Outside Diameter of Ring	3420	mm
Material Name	SA-240 316L +IT	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N8 26"	
Dist. from "FROM" Node / Offset dist	1235	mm
Nozzle Diameter	26	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	2129.7	N
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-240 316L +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	Nozzle boot	
Dist. from "FROM" Node / Offset dist	0.0005822	mm
Nozzle Diameter	40	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	2417.4	N



Grade of Attached Flange None
Nozzle Matl SA-240 316L +IT

Element From Node 40
Detail Type Weight
Detail ID Lifting lug 3
Dist. from "FROM" Node / Offset dist 1450 mm
Miscellaneous Weight 686.42 N
Offset from Element Centerline 1500 mm

Element From Node 40
Detail Type Weight
Detail ID Lifting lug 4
Dist. from "FROM" Node / Offset dist 1450 mm
Miscellaneous Weight 686.42 N
Offset from Element Centerline 1500 mm

Element# 5/7
Element From Node 50
Element To Node 60
Element Type Elliptical
Description Right head
Distance "FROM" to "TO" 50 mm
Inside Diameter 3000 mm
Element Thickness 10 mm
Internal Corrosion Allowance 0 mm
Nominal Thickness 12 mm
External Corrosion Allowance 0 mm
Design Internal Pressure 0.1013 MPa
Design Temperature Internal Pressure 95 °C
Design External Pressure 0.07446 MPa
Design Temperature External Pressure 95 °C
Effective Diameter Multiplier 1.2
Material Name SA-240 316L
Efficiency, Longitudinal Seam 1.0
Efficiency, Circumferential Seam 1.0
Elliptical Head Factor 2.0
Weld is pre-Heated No

Element From Node 50
Detail Type Liquid
Detail ID Liquid 50
Dist. from "FROM" Node / Offset dist 0 mm
Height/Length of Liquid 3000 mm
Liquid Density 0.001266 kg/cm³

Element# 6/7
Element From Node 60
Element To Node 70
Element Type Flange
Description N8 flg
Distance "FROM" to "TO" 66.802 mm
Flange Outside Diameter 869.95 mm
Element Thickness 66.802 mm
Internal Corrosion Allowance 0 mm
Nominal Thickness 68 mm
External Corrosion Allowance 0 mm
Design Internal Pressure 0.1013 MPa
Design Temperature Internal Pressure 95 °C
Design External Pressure 0.07446 MPa
Design Temperature External Pressure 95 °C
Effective Diameter Multiplier 1.2
Material Name SA-240 316L
Perform Flange Stress Calculation (Y/N) Y
Weld is pre-Heated No



Element From Node	60
Detail Type	Nozzle
Detail ID	N2 3"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	3 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	154.07 N
Grade of Attached Flange	GR 2.2
Nozzle Matl	SA-312 TP316

Element#	7/7
Element From Node	70
Element To Node	80
Element Type	Elliptical
Description	N8 flg
Distance "FROM" to "TO"	50 mm
Inside Diameter	1016 mm
Element Thickness	6 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	8 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.1013 MPa
Design Temperature Internal Pressure	95 °C
Design External Pressure	0.07446 MPa
Design Temperature External Pressure	95 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 316L
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Nozzle
Detail ID	N9 3"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	3 in.
Nozzle Schedule	40
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	117.38 N
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-312 TP316L +IT



XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Left head		50	...	50	...
Shell-1		3050	...	3000	...
Shell-2		6050	...	3000	...
Shell-3		8450	...	2400	...
Right head		8500	...	50	...
N8 flg		7285	1800	...	66.802
N8 flg		6050	-2200	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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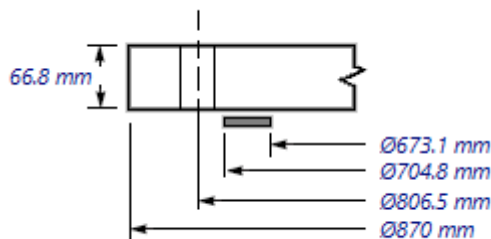


Flange Input Data Values

Description: FLANGE :

N8 flg

Flange Type		Blind	
Design Pressure	P	0.10	MPa
Design Temperature		95	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	869.950	mm
Flange Thickness	t	66.8020	mm
Flange Material		SA-240 316L	
Flange Material UNS number		S31603	
Flange Allowable Stress At Temperature	Sfo	115.15	MPa
Flange Allowable Stress At Ambient	Sfa	115.15	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.38	MPa
Bolt Allowable Stress At Ambient	Sa	172.38	MPa
Perimeter along the Center of the Bolts	L	2533.538	mm
Diameter of Bolt Circle	C	806.450	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	749.300	mm
Flange Face Inside Diameter	Fid	641.350	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	704.850	mm
Gasket Inside Diameter	Gi	673.100	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	15.875 mm
Basic Gasket Width,	$bo = N/2$	7.938 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.100 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	690.651 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 690.6509^2 \cdot 0.101$$

$$= 37956.559 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$\begin{aligned} &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P \\ &= 2 \cdot 7.0995 \cdot \pi \cdot 690.6509 \cdot 3.0 \cdot 0.1 \\ &= 9364.146 \text{ N} \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max(W_{o,1} + W_{o,2} + H'p, 0) \\ &= \max(37957 + 9364 + 0, 0) \\ &= 47320.707 \text{ N} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\ &= 68.95 \cdot 7.0995 \cdot \pi \cdot 690.651 + 0.0 \cdot 0.0 \cdot 0.0 \\ &= 1062024.500 \text{ N} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a) \\ &= \max((47321 + 0 + 4 \cdot 0 / 690.65) / 172, 1062025 / 172) \\ &= 6161.652 \text{ mm}^2 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	6161.652	14384.487	
Radial Distance between Bolts and Edge:	31.750	31.750	
Circ. Spacing between the Bolts:	71.450	105.263	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned} &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\ &= 14384.487 \cdot 172.38 / (68.95 \cdot \pi (704.85 + 673.1)) \\ &= 8.307 \text{ mm} \end{aligned}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned} &= S_a \cdot A_b \\ &= 172.38 \cdot 14384.4873 \\ &= 2479315.50 \text{ N} \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned} &= W_o \\ &= 47320.71 \text{ N} \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned} &= (C - G) / 2 \\ &= (806.45 - 690.6509) / 2 \\ &= 57.8995 \text{ mm} \end{aligned}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$\begin{aligned} &= 1.9 \cdot W_{m1} \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2 \\ &= 1.9 \cdot 47321 \cdot 57.8995 \cdot 1.0 / (66.802^2 \cdot 690.6509) + 0.3 \cdot 1.0 \cdot 0.1 \cdot 690.6509^2 / 66.802^2 \\ &= 4.94 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$\begin{aligned} &= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G) \\ &= 1.9 \cdot 2479316 \cdot 57.8995 \cdot 1.0 / (66.802^2 \cdot 690.6509) \\ &= 88.50 \text{ MPa} \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned} &= W_{m1} / A_b \\ &= 47321 / 14384.4873 \\ &= 3.29 \text{ MPa} \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (W_{m2} / A_b) \\ &= (1062025 / 14384.4873) \\ &= 73.84 \text{ MPa} \end{aligned}$$



Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	4.94	115.15	88.50 115.15
Bolting	3.29	172.38	73.84 172.38

Reqd. Blind Flange Thickness at Center 58.566 mm
 Reqd. Blind Flange Thickness at Gasket 58.566 mm

Estimated M.A.W.P. (Operating) 2 MPa
 Estimated Finished Weight of Flange at given Thk. 318.7 kgm
 Estimated Unfinished Weight of Forging at given Thk 318.7 kgm

SA-240 316L, Min Metal Temp without impact per UHA-51: -196 °C

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	4.94	115.15	88.50 115.15
Bolting	3.29	172.38	73.84 172.38

Minimum Req'd Thickness, ambient, no Corrosion: 58.572 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Left head		0.1386	12	...	3000	115.15	1
Shell-1		0.1386	10	...	3000	115.15	2
Shell-2		0.1386	10	...	3000	115.15	3
Shell-3		0.1386	10	...	3000	115.15	4
Right head		0.1386	12	...	3000	115.15	5
N8 flg		0.1013	68	...	869.95	115.15	6
N8 flg		0.1013	8	...	1016	115.15	7

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Left head		0.10132	0.72984	0.76713	10	1.80593	1
Shell-1		0.10132	0.7273	0.76458	10	1.80701	2
Shell-2		0.10132	0.7273	0.76458	10	1.80701	3
Shell-3		0.10132	0.7273	0.76458	10	1.80701	4
Right head		0.10132	0.72984	0.76713	10	1.80593	5
N8 flg		0.10132	2.3625	2.36256	66.802	58.5658	6
N8 flg		0.10132	1.35839	1.35839	6	1.5	7

Minimum 0 1

MAWP: 0 MPa, limited by: Nozzle Reinforcement.

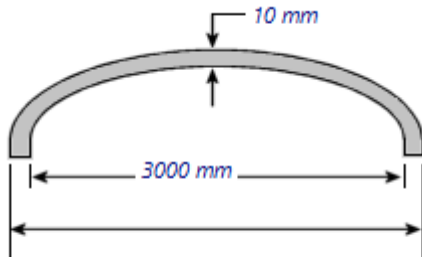
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Left head

Elliptical Head Design Information:					
Design Pressure	P	0.139 MPa	Int. Design Temperature		95 °C
Material		SA-240 316L	Ext. Design Temperature		95 °C
Length		50.000 mm	External Pressure Chart		HA-4
			UNS Number		S31603
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	115.1 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	115.1 MPa
Inside Diameter	Di	3000.000 mm	Actual Stress		20.8 MPa
Spec. Min. Thk t,e, or sa		10.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		10387596.0 mm ²	SG of Contents		1.268
Element Volume		3888.4 ltr	Weight of Contents		48319.1 N
Empty Weight		10703.3 N	Operating Weight		59022.3 N
			Comp. All. Ope. Stress		66.1 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 37.500 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.750 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (0.139 \cdot 3000.0 \cdot 1.0) / (2 \cdot 115.15 \cdot 1.0 - 0.2 \cdot 0.139)$$

$$= 1.8059 + 0.0000 = 1.8059 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.037 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 115.15 \cdot 1.0 \cdot 10.0) / (1.0 \cdot 3000.0 + 0.2 \cdot 10.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 115.15 \cdot 1.0 \cdot 10.0) / (1.0 \cdot 3000.0 + 0.2 \cdot 10.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.139 \cdot (1.0 \cdot 3000.0 + 0.2 \cdot 10.0)) / (2 \cdot 1.0 \cdot 10.0)$$

$$= 20.806 \text{ MPa}$$



Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (0.139 \cdot 1500.0) / (115.15 \cdot 1.0 - 0.6 \cdot 0.139) + 0.0 \\ &= 1.807 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.037 MPa

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (115.15 \cdot 1.0 \cdot 12.0) / (1500.0 + 0.6 \cdot 12.0) \\ &= 1 - 0 = 1 \text{ MPa} \end{aligned}$$

% Elongation per Table UG-79-1 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.744 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 316L, Min Metal Temp without impact per UHA-51: -196 °C

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 37.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.750 mm

EXTERNAL PRESSURE RESULTS for: Left head

Elliptical Head From 10 to 20 Ext. Chart: HA-4 at 95 °C

Elastic Modulus from Chart: HA-4 at 95 °C: 187379 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
10.000	3020.00	302.00	0.0004599	43.09

$$MAEP = B / (K_0 \cdot Do / t) = 43.0876 / (0.9 \cdot 302.0) = 0.1585 \text{ MPa}$$

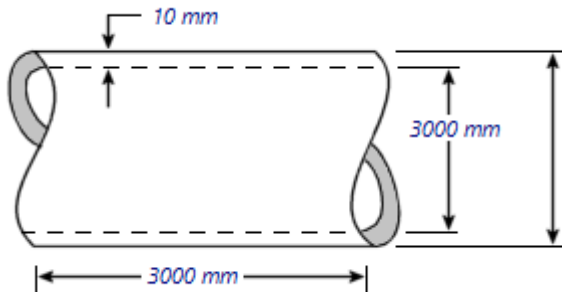
Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
6.854	3020.00	440.64	0.0003152	29.53

$$MAEP = B / (K_0 \cdot Do / t) = 29.5307 / (0.9 \cdot 440.6413) = 0.0745 \text{ MPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	0.139 MPa	Int. Design Temperature		95 °C
Material		SA-240 316L	Ext. Design Temperature		95 °C
Length		3000.000 mm	External Pressure Chart		HA-4
			UNS Number		S31603
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	115.1 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	115.1 MPa
Inside Diameter	Di	3000.000 mm	Actual Stress		20.9 MPa
Spec. Min. Thk t _e , or s _a		10.0000 mm	Spec. Nominal Thk.		10.000 mm
Surface Area		28462830.0 mm ²	SG of Contents		1.268
Element Volume		21209.6 ltr	Weight of Contents		263558.5 N
Empty Weight		22330.4 N	Operating Weight		285889.0 N
			Comp. All. Ope. Stress		66.1 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.139 \cdot 1500.0) / (115.15 \cdot 1.0 - 0.6 \cdot 0.139)$$

$$= 1.8070 + 0.0000 = 1.8070 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.037 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (115.15 \cdot 1.0 \cdot 10.0) / (1500.0 + 0.6 \cdot 10.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (115.15 \cdot 1.0 \cdot 10.0) / (1500.0 + 0.6 \cdot 10.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.139 \cdot (1500.0 + 0.6 \cdot 10.0)) / (1.0 \cdot 10.0)$$

$$= 20.875 \text{ MPa}$$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 0.332 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 316L, Min Metal Temp without impact per UHA-51: -196 °C



EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 20 to SR1 Ext. Chart: HA-4 at 95 °C

Elastic Modulus from Chart: HA-4 at 95 °C: 187379 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	665.00	302.00	0.2202	0.0012749	60.73

$$MAEP = (4*B)/(3*(Do/t)) = (4*60.7272)/(3*302.0) = 0.2681 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.617	3020.00	665.00	654.08	0.2202	0.0003899	36.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*36.5282)/(3*654.0846) = 0.0745 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	4174.63	302.00	1.3823	0.0001802	16.88

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.8806)/(3*302.0) = 0.0745 \text{ MPa}$$

Cylindrical Shell From SR1 to Ring 1 Ext. Chart: HA-4 at 95 °C

Elastic Modulus from Chart: HA-4 at 95 °C: 187379 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	1635.00	302.00	0.5414	0.0004788	44.86

$$MAEP = (4*B)/(3*(Do/t)) = (4*44.8589)/(3*302.0) = 0.1981 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.757	3020.00	1635.00	446.97	0.5414	0.0002664	24.96

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.9614)/(3*446.9677) = 0.0745 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	4175.75	302.00	1.3827	0.0001801	16.88

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.8759)/(3*302.0) = 0.0745 \text{ MPa}$$

Cylindrical Shell From Ring 1 to 30 Ext. Chart: HA-4 at 95 °C

Elastic Modulus from Chart: HA-4 at 95 °C: 187379 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	2000.00	302.00	0.6623	0.0003874	36.29



$$MAEP = (4*B)/(3*(Do/t)) = (4*36.2948)/(3*302.0) = 0.1602 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.369	3020.00	2000.00	409.82	0.6623	0.0002443	22.89

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.8873)/(3*409.8243) = 0.0745 \text{ MPa}$$

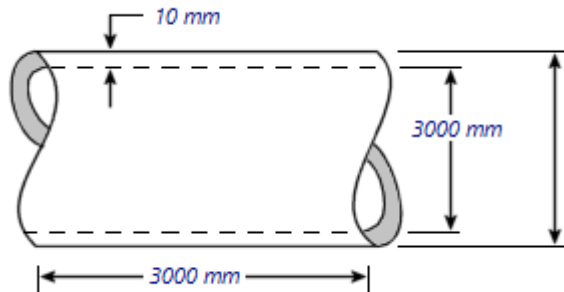
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	4176.31	302.00	1.3829	0.0001801	16.87

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.8735)/(3*302.0) = 0.0745 \text{ MPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	0.139 MPa	Int. Design Temperature		95 °C
Material		SA-240 316L	Ext. Design Temperature		95 °C
Length		3000.000 mm	External Pressure Chart		HA-4
			UNS Number		S31603
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	115.1 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	115.1 MPa
Inside Diameter	Di	3000.000 mm	Actual Stress		20.9 MPa
Spec. Min. Thk t _e , or s _a		10.0000 mm	Spec. Nominal Thk.		10.000 mm
Surface Area		28462830.0 mm ²	SG of Contents		1.268
Element Volume		21209.6 ltr	Weight of Contents		263558.5 N
Empty Weight		22330.4 N	Operating Weight		285889.0 N
			Comp. All. Ope. Stress		66.1 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.139 \cdot 1500.0) / (115.15 \cdot 1.0 - 0.6 \cdot 0.139)$$

$$= 1.8070 + 0.0000 = 1.8070 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.037 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (115.15 \cdot 1.0 \cdot 10.0) / (1500.0 + 0.6 \cdot 10.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (115.15 \cdot 1.0 \cdot 10.0) / (1500.0 + 0.6 \cdot 10.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.139 \cdot (1500.0 + 0.6 \cdot 10.0)) / (1.0 \cdot 10.0)$$

$$= 20.875 \text{ MPa}$$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 0.332 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 316L, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: Shell-2

Cylindrical Shell From 30 to Ring 2 Ext. Chart: HA-4 at 95 °C

Elastic Modulus from Chart: HA-4 at 95 °C: 187379 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	2000.00	302.00	0.6623	0.0003874	36.29

$$MAEP = (4*B)/(3*(Do/t)) = (4*36.2948)/(3*302.0) = 0.1602 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.369	3020.00	2000.00	409.82	0.6623	0.0002443	22.89

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.8873)/(3*409.8243) = 0.0745 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	4176.31	302.00	1.3829	0.0001801	16.87

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.8735)/(3*302.0) = 0.0745 \text{ MPa}$$

Cylindrical Shell From Ring 2 to 40 Ext. Chart: HA-4 at 95 °C

Elastic Modulus from Chart: HA-4 at 95 °C: 187379 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	2500.00	302.00	0.8278	0.0003070	28.77

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.7664)/(3*302.0) = 0.127 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.047	3020.00	2500.00	375.28	0.8278	0.0002237	20.96

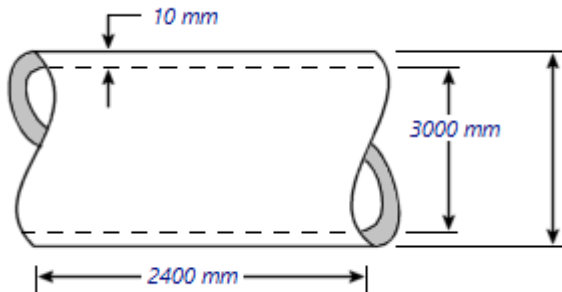
$$MAEP = (4*B)/(3*(Do/t)) = (4*20.9574)/(3*375.2767) = 0.0745 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	4175.37	302.00	1.3826	0.0001801	16.88

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.8774)/(3*302.0) = 0.0745 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	0.139 MPa	Int. Design Temperature		95 °C
Material		SA-240 316L	Ext. Design Temperature		95 °C
Length		2400.000 mm	External Pressure Chart		HA-4
			UNS Number		S31603
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	115.1 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	115.1 MPa
Inside Diameter	Di	3000.000 mm	Actual Stress		20.9 MPa
Spec. Min. Thk t,e, or sa		10.0000 mm	Spec. Nominal Thk.		10.000 mm
Surface Area		22770266.0 mm²	SG of Contents		1.268
Element Volume		16967.6 ltr	Weight of Contents		210846.8 N
Empty Weight		17864.3 N	Operating Weight		228711.1 N
			Comp. All. Ope. Stress		66.1 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.139 \cdot 1500.0) / (115.15 \cdot 1.0 - 0.6 \cdot 0.139)$$

$$= 1.8070 + 0.0000 = 1.8070 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.037 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (115.15 \cdot 1.0 \cdot 10.0) / (1500.0 + 0.6 \cdot 10.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (115.15 \cdot 1.0 \cdot 10.0) / (1500.0 + 0.6 \cdot 10.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.139 \cdot (1500.0 + 0.6 \cdot 10.0)) / (1.0 \cdot 10.0)$$

$$= 20.875 \text{ MPa}$$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 0.332 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 316L, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: Shell-3

Cylindrical Shell From 40 to Ring 3 Ext. Chart: HA-4 at 95 °C

Elastic Modulus from Chart: HA-4 at 95 °C: 187379 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	2500.00	302.00	0.8278	0.0003070	28.77

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.7664)/(3*302.0) = 0.127 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.047	3020.00	2500.00	375.28	0.8278	0.0002237	20.96

$$MAEP = (4*B)/(3*(Do/t)) = (4*20.9574)/(3*375.2767) = 0.0745 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	4175.37	302.00	1.3826	0.0001801	16.88

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.8774)/(3*302.0) = 0.0745 \text{ MPa}$$

Cylindrical Shell From Ring 3 to SR, 2085.0 Fr40 Ext. Chart: HA-4 at 95 °C

Elastic Modulus from Chart: HA-4 at 95 °C: 187379 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	1585.00	302.00	0.5248	0.0004949	46.36

$$MAEP = (4*B)/(3*(Do/t)) = (4*46.3628)/(3*302.0) = 0.2047 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.670	3020.00	1585.00	452.79	0.5248	0.0002699	25.29

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.2866)/(3*452.7931) = 0.0745 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	4175.66	302.00	1.3827	0.0001801	16.88

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.8762)/(3*302.0) = 0.0745 \text{ MPa}$$

Cylindrical Shell From SR, 2085.0 Fr40 to 50 Ext. Chart: HA-4 at 95 °C

Elastic Modulus from Chart: HA-4 at 95 °C: 187379 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	615.00	302.00	0.2036	0.0013896	61.82



$$MAEP = (4*B)/(3*(Do/t)) = (4*61.825)/(3*302.0) = 0.273 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.462	3020.00	615.00	676.89	0.2036	0.0004035	37.80

$$MAEP = (4*B)/(3*(Do/t)) = (4*37.8019)/(3*676.8946) = 0.0745 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	4176.42	302.00	1.3829	0.0001801	16.87

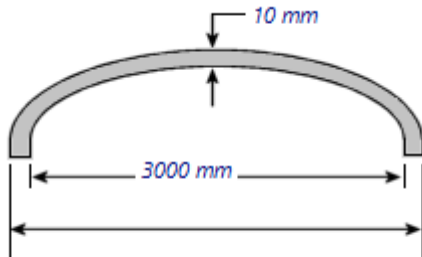
$$MAEP = (4*B)/(3*(Do/t)) = (4*16.8731)/(3*302.0) = 0.0745 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Right head

Elliptical Head Design Information:					
Design Pressure	P	0.139 MPa	Int. Design Temperature		95 °C
Material		SA-240 316L	Ext. Design Temperature		95 °C
Length		50.000 mm	External Pressure Chart		HA-4
			UNS Number		S31603
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	115.1 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	115.1 MPa
Inside Diameter	Di	3000.000 mm	Actual Stress		20.8 MPa
Spec. Min. Thk t,e, or sa		10.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		10387596.0 mm ²	SG of Contents		1.268
Element Volume		3888.4 ltr	Weight of Contents		48319.1 N
Empty Weight		10703.3 N	Operating Weight		59022.3 N
			Comp. All. Ope. Stress		66.1 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 37.500 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.750 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (0.139 \cdot 3000.0 \cdot 1.0) / (2 \cdot 115.15 \cdot 1.0 - 0.2 \cdot 0.139)$$

$$= 1.8059 + 0.0000 = 1.8059 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.037 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 115.15 \cdot 1.0 \cdot 10.0) / (1.0 \cdot 3000.0 + 0.2 \cdot 10.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 115.15 \cdot 1.0 \cdot 10.0) / (1.0 \cdot 3000.0 + 0.2 \cdot 10.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.139 \cdot (1.0 \cdot 3000.0 + 0.2 \cdot 10.0)) / (2 \cdot 1.0 \cdot 10.0)$$

$$= 20.806 \text{ MPa}$$



Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (0.139 \cdot 1500.0) / (115.15 \cdot 1.0 - 0.6 \cdot 0.139) + 0.0 \\ &= 1.807 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.037 MPa

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (115.15 \cdot 1.0 \cdot 12.0) / (1500.0 + 0.6 \cdot 12.0) \\ &= 1 - 0 = 1 \text{ MPa} \end{aligned}$$

% Elongation per Table UG-79-1 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.744 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 316L, Min Metal Temp without impact per UHA-51: -196 °C

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 37.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.750 mm

EXTERNAL PRESSURE RESULTS for: Right head

Elliptical Head From 50 to 60 Ext. Chart: HA-4 at 95 °C

Elastic Modulus from Chart: HA-4 at 95 °C: 187379 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
10.000	3020.00	302.00	0.0004599	43.09

$$MAEP = B / (K_0 \cdot Do / t) = 43.0876 / (0.9 \cdot 302.0) = 0.1585 \text{ MPa}$$

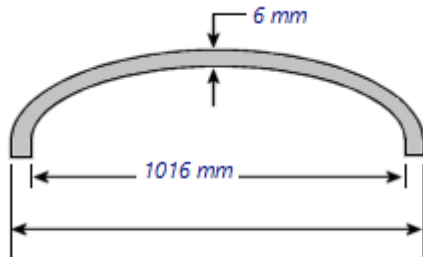
Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
6.854	3020.00	440.64	0.0003152	29.53

$$MAEP = B / (K_0 \cdot Do / t) = 29.5307 / (0.9 \cdot 440.6413) = 0.0745 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: N8 flg

Elliptical Head Design Information:					
Design Pressure	P	0.101 MPa	Int. Design Temperature	95 °C	
Material		SA-240 316L	Ext. Design Temperature	95 °C	
Length		50.000 mm	External Pressure Chart	HA-4	
			UNS Number	S31603	
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	115.1 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	115.1 MPa
Inside Diameter	Di	1016.000 mm	Actual Stress	8.6 MPa	
Spec. Min. Thk	t _e , or sa	6.0000 mm	Spec. Nominal Thk.	8.000 mm	
Surface Area		1316576.6 mm ²	SG of Contents	0.000	
Element Volume		177.9 ltr	Weight of Contents	0.0 N	
Empty Weight		891.6 N	Operating Weight	891.6 N	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 12.700 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.350 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (0.101 \cdot 1016.0 \cdot 1.0) / (2 \cdot 115.15 \cdot 1.0 - 0.2 \cdot 0.101)$$

$$= 0.4471 + 0.0000 = 0.4471 \text{ mm}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 115.15 \cdot 1.0 \cdot 6.0) / (1.0 \cdot 1016.0 + 0.2 \cdot 6.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 115.15 \cdot 1.0 \cdot 6.0) / (1.0 \cdot 1016.0 + 0.2 \cdot 6.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.101 \cdot (1.0 \cdot 1016.0 + 0.2 \cdot 6.0)) / (2 \cdot 1.0 \cdot 6.0)$$

$$= 8.589 \text{ MPa}$$



Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (0.101 \cdot 508.0) / (115.15 \cdot 1.0 - 0.6 \cdot 0.101) + 0.0 \\ &= 0.447 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (115.15 \cdot 1.0 \cdot 8.0) / (508.0 + 0.6 \cdot 8.0) \\ &= 2 \text{ MPa} \end{aligned}$$

% Elongation per Table UG-79-1 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 3.395 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 316L, Min Metal Temp without impact per UHA-51: -196 °C

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 12.700 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.350 mm

EXTERNAL PRESSURE RESULTS for: N8 flg

Elliptical Head From 70 to 80 Ext. Chart: HA-4 at 95 °C

Elastic Modulus from Chart: HA-4 at 95 °C: 187379 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
6.000	1028.00	171.33	0.0008106	54.83

$$MAEP = B / (K_0 \cdot Do / t) = 54.8291 / (0.9 \cdot 171.3333) = 0.3556 \text{ MPa}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.333	1028.00	440.65	0.0003152	29.53

$$MAEP = B / (K_0 \cdot Do / t) = 29.5304 / (0.9 \cdot 440.6461) = 0.0745 \text{ MPa}$$

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	0.427 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.132 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	0.965 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	0.361 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.145 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.145 MPa
App 27-4	= M.A.W.P.	0.328 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 0.328 • 1.0
 = 0.427 MPa

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Left head	68.5	155.1	0.441	0.46
Shell-1	68.7	155.1	0.443	0.46
Shell-2	68.7	155.1	0.443	0.46
Shell-3	68.7	155.1	0.443	0.46
Right head	68.5	155.1	0.441	0.46
N8 flg	39.5	155.1	0.254	0.47

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1 6"	Nozzle	115.10	115.10	1.000
N1 6"	Pad	115.10	115.10	1.000
N3 2"	Nozzle	117.20	117.20	1.000
N4 4"	Nozzle	115.10	115.10	1.000
N4 4"	Pad	115.10	115.10	1.000
N5 1.5"	Nozzle	115.10	115.10	1.000
N6 3"	Nozzle	117.20	117.20	1.000
N6 3"	Pad	115.10	115.10	1.000
N7 4"	Nozzle	115.10	115.10	1.000
M1 24"	Nozzle	115.10	115.10	1.000
M1 24"	Pad	115.10	115.10	1.000
N8 26"	Nozzle	115.10	115.10	1.000
N8 26"	Pad	115.10	115.10	1.000
Nozzle boot	Nozzle	115.10	115.10	1.000
Nozzle boot	Pad	137.90	137.90	1.000
N2 3"	Nozzle	117.20	117.20	1.000
N9 3"	Nozzle	115.10	115.10	1.000

Minimum 1.000

Stress ratios for Stiffening Ring Materials (MPa):

Description	Ambient	Operating	Ratio
Ring 1	137.90	114.80	1.201
SR1	115.15	115.15	1.000
Ring 2	137.90	114.80	1.201
Ring 3	137.90	114.80	1.201
SR, 2085.0 Fr40	115.15	115.15	1.000

Minimum 1.000



Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Left head	115.15	115.15	1.000
Shell-1	115.15	115.15	1.000
Shell-2	115.15	115.15	1.000
Shell-3	115.15	115.15	1.000
Right head	115.15	115.15	1.000
N8 flg	115.15	115.15	1.000
N8 flg	115.15	115.15	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N1 6"	5.21	155.10	0.034
N3 2"	1.13	186.20	0.006
N4 4"	2.86	155.10	0.018
N5 1.5"	0.88	155.10	0.006
N6 3"	3.51	186.20	0.019
N7 4"	1.49	155.10	0.010
M1 24"	14.18	155.10	0.091
N8 26"	12.83	155.10	0.083
Nozzle boot	19.59	155.10	0.126
N2 3"	1.31	186.20	0.007
N9 3"	3.59	155.10	0.023

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	0.427 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.132 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	0.965 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	0.361 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.145 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.145 MPa
App 27-4	= M.A.W.P.	0.328 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 0.328 • 1.0
 = 0.427 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Left head	68.5	155.1	0.441	0.46
Shell-1	68.7	155.1	0.443	0.46
Shell-2	68.7	155.1	0.443	0.46
Shell-3	68.7	155.1	0.443	0.46
Right head	68.5	155.1	0.441	0.46
N8 flg	39.5	155.1	0.254	0.47

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1 6"	Nozzle	115.10	115.10	1.000
N1 6"	Pad	115.10	115.10	1.000
N3 2"	Nozzle	117.20	117.20	1.000
N4 4"	Nozzle	115.10	115.10	1.000
N4 4"	Pad	115.10	115.10	1.000
N5 1.5"	Nozzle	115.10	115.10	1.000
N6 3"	Nozzle	117.20	117.20	1.000
N6 3"	Pad	115.10	115.10	1.000
N7 4"	Nozzle	115.10	115.10	1.000
M1 24"	Nozzle	115.10	115.10	1.000
M1 24"	Pad	115.10	115.10	1.000
N8 26"	Nozzle	115.10	115.10	1.000
N8 26"	Pad	115.10	115.10	1.000
Nozzle boot	Nozzle	115.10	115.10	1.000
Nozzle boot	Pad	137.90	137.90	1.000
N2 3"	Nozzle	117.20	117.20	1.000
N9 3"	Nozzle	115.10	115.10	1.000

Minimum 1.000

Stress ratios for Stiffening Ring Materials (MPa):

Description	Ambient	Operating	Ratio
Ring 1	137.90	114.80	1.201
SR1	115.15	115.15	1.000
Ring 2	137.90	114.80	1.201
Ring 3	137.90	114.80	1.201
SR, 2085.0 Fr40	115.15	115.15	1.000

Minimum 1.000



Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Left head	115.15	115.15	1.000
Shell-1	115.15	115.15	1.000
Shell-2	115.15	115.15	1.000
Shell-3	115.15	115.15	1.000
Right head	115.15	115.15	1.000
N8 flg	115.15	115.15	1.000
N8 flg	115.15	115.15	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N1 6"	5.21	155.10	0.034
N3 2"	1.13	186.20	0.006
N4 4"	2.86	155.10	0.018
N5 1.5"	0.88	155.10	0.006
N6 3"	3.51	186.20	0.019
N7 4"	1.49	155.10	0.010
M1 24"	14.18	155.10	0.091
N8 26"	12.83	155.10	0.083
Nozzle boot	19.59	155.10	0.126
N2 3"	1.29	186.20	0.007
N9 3"	3.59	155.10	0.023

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

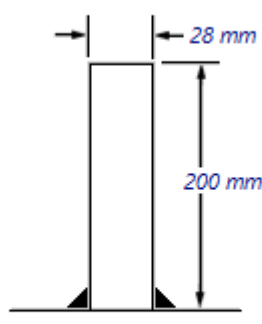
ASME Code, Section VIII Division 1, 2025

MDMT Calculations for: SR1, Shell to Ring at Ring (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Stiffening Ring Calculations for: SR1, SA-240 316L , Bar Ring: 200.0 x 28.0 mm

Material treatment options:



Effective Length of Shell: 191 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	1911.596	5.0000	9557.980
Ring :	5599.998	109.9999	615999.438
Total:	7511.594		625557.438

Centroid of Ring plus Shell: 83 mm

	Inertia	Distance	A*Dist ²
Shell:	15929.951	78.2789	11713474.000
Ring :	18666622.000	-26.7210	3998474.000
Total:	18682552.000		15711948.000

Available Moment of Inertia, Ring + Shell: 0.344E+08 mm⁴

Required Stress in Ring plus Shell Breq 11.34 MPa

Required Strain in Ring plus Shell Areq 0.0001210

Required Moment of Inertia, Ring plus Shell:

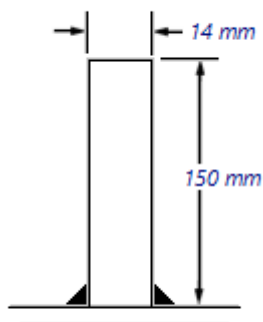
$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (3020.0^2 \cdot 1150.0(10.0 + 5599.9976/1150.0)0.000121) / 10.9$$

$$= 1731285 \text{ mm}^4$$

Stiffening Ring Calculations for: Ring 1, SA-240 304 , Bar Ring: 150.0 x 14.0 mm

Material treatment options:



Effective Length of Shell: 191 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	1911.596	5.0000	9557.980
Ring :	2100.001	85.0000	178500.109
Total:	4011.597		188058.094

Centroid of Ring plus Shell: 47 mm

	Inertia	Distance	A*Dist ²
Shell:	15929.951	41.8786	3352591.250
Ring :	3937500.000	-38.1214	3051810.000
Total:	3953430.000		6404401.000

Available Moment of Inertia, Ring + Shell: 0.104E+08 mm⁴

A = max(Areq Shell, Areq Ring) Per UG-29
 A = max(0.000161, 0.00016) = 0.000161

Required Stress in Ring plus Shell Breq 15.12 MPa
 Required Strain in Ring plus Shell Areq 0.0001610

Required Moment of Inertia, Ring plus Shell:

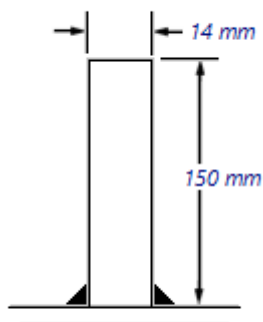
$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (3020.0^2 \cdot 1817.4999(10.0 + 2100.0007/1817.4999)0.000161) / 10.9$$

$$= 2731323 \text{ mm}^4$$

Stiffening Ring Calculations for: Ring 2, SA-240 304 , Bar Ring: 150.0 x 14.0 mm

Material treatment options:



Effective Length of Shell: 191 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	1911.596	5.0000	9557.980
Ring :	2100.001	85.0000	178500.109
Total:	4011.597		188058.094

Centroid of Ring plus Shell: 47 mm

	Inertia	Distance	A*Dist ²
Shell:	15929.951	41.8786	3352591.250
Ring :	3937500.000	-38.1214	3051810.000
Total:	3953430.000		6404401.000

Available Moment of Inertia, Ring + Shell: 0.104E+08 mm⁴

Required Stress in Ring plus Shell Breq 15.43 MPa
 Required Strain in Ring plus Shell Areq 0.0001640

Required Moment of Inertia, Ring plus Shell:

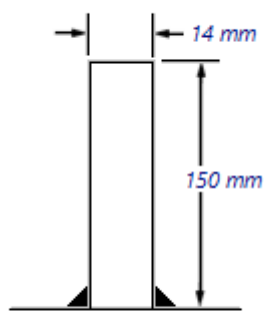
$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (3020.0^2 \cdot 2250.0(10.0 + 2100.0007/2250.0)0.000164) / 10.9$$

$$= 3375711 \text{ mm}^4$$

Stiffening Ring Calculations for: Ring 3, SA-240 304 , Bar Ring: 150.0 x 14.0 mm

Material treatment options:



Effective Length of Shell: 191 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	1911.596	5.0000	9557.980
Ring :	2100.001	85.0000	178500.109
Total:	4011.597		188058.094

Centroid of Ring plus Shell: 47 mm

	Inertia	Distance	A*Dist ²
Shell:	15929.951	41.8786	3352591.250
Ring :	3937500.000	-38.1214	3051810.000
Total:	3953430.000		6404401.000

Available Moment of Inertia, Ring + Shell: 0.104E+08 mm⁴

A = max(Areq Shell, Areq Ring) Per UG-29
 A = max(0.000163, 0.000162) = 0.000163

Required Stress in Ring plus Shell Breq 15.29 MPa
 Required Strain in Ring plus Shell Areq 0.0001630

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (3020.0^2 \cdot 2042.4999(10.0 + 2100.0007/2042.4999)0.000163) / 10.9$$

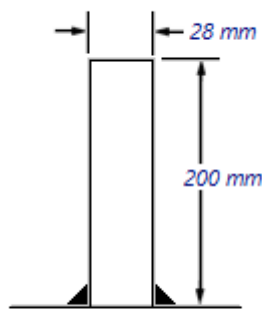
$$= 3072124 \text{ mm}^4$$

MDMT Calculations for: SR, 2085.0 Fr40, Shell to Ring at Ring (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Stiffening Ring Calculations for: SR, 2085.0 Fr40, SA-240 316L , Bar Ring: 200.0 x 28.0 mm

Material treatment options:



Effective Length of Shell: 191 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	1911.596	5.0000	9557.980
Ring :	5600.003	110.0000	616000.562
Total:	7511.599		625558.562

Centroid of Ring plus Shell: 83 mm

	Inertia	Distance	A*Dist ²
Shell:	15929.951	78.2790	11713504.000
Ring :	18666676.000	-26.7210	3998477.750
Total:	18682606.000		15711982.000

Available Moment of Inertia, Ring + Shell: 0.344E+08 mm⁴

Required Stress in Ring plus Shell Breq 11.18 MPa
 Required Strain in Ring plus Shell Areq 0.0001190

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (3020.0^2 \cdot 1100.0(10.0 + 5600.0029/1100.0)0.000119) / 10.9$$

$$= 1652884 \text{ mm}^4$$

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	No Calc	3020	10	0.0004599	43.0876
20	Ring	665	3020	10	0.0012749	60.7272
Ring	Ring	1635	3020	10	0.0004788	44.8589
Ring	30	2000	3020	10	0.00038739	36.2948
30	Ring	2000	3020	10	0.00038739	36.2948
Ring	40	2500	3020	10	0.00030704	28.7664
40	Ring	2500	3020	10	0.00030704	28.7664
Ring	Ring	1585	3020	10	0.00049485	46.3628
Ring	50	615	3020	10	0.0013896	61.825
50	60	No Calc	3020	10	0.0004599	43.0876
60	70	No Calc	...	66.802	No Calc	No Calc
70	80	No Calc	1028	6	0.00081064	54.8291

External Pressure Calculations:

		External	External	External	External
--	--	----------	----------	----------	----------



From	To	Actual T. mm	Required T. mm	Design Pressure MPa	M.A.W.P. MPa
10	20	10	6.85365	0.074458	0.15853
20	Ring	10	4.61714	0.074458	0.26811
Ring	Ring	10	6.75664	0.074458	0.19805
Ring	30	10	7.36901	0.074458	0.16024
30	Ring	10	7.36901	0.074458	0.16024
Ring	40	10	8.04739	0.074458	0.127
40	Ring	10	8.04739	0.074458	0.127
Ring	Ring	10	6.66971	0.074458	0.20469
Ring	50	10	4.46155	0.074458	0.27296
50	60	10	6.85365	0.074458	0.15853
60	70	66.802	58.5724	0.074458	No Calc
70	80	6	2.33294	0.074458	0.35557

Minimum

0

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	Ring	665	4174.63	No Calc	No Calc
Ring	Ring	1635	4175.75	1731284	34394484
Ring	30	2000	4176.31	2731322	10357825
30	Ring	2000	4176.31	No Calc	No Calc
Ring	40	2500	4175.37	3375710	10357825
40	Ring	2500	4175.37	No Calc	No Calc
Ring	Ring	1585	4175.66	3072123	10357825
Ring	50	615	4176.42	1652883	34394572
50	60	No Calc	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	1091.5	3888.42	1091.5	3888.42	76.4052
20	30	2277.22	21209.6	2277.22	21209.6	159.406
30	40	2277.22	21209.6	2277.22	21209.6	159.406
40	50	1821.77	16967.6	1821.77	16967.6	127.524
50	60	1091.5	3888.42	1091.5	3888.42	76.4052
60	70	324.457	...	324.457	...	22.712
70	80	90.9231	177.852	90.9231	177.852	6.36462
Total		8974	67341.43	8974	67341.43	628

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	4927.5	-250	...	...	Liquid 10
10	Nozl	27.2103	-422.019	...	-1240	N1 6"
20	Sadl	553.473	365	1597.5	...	LeftSaddle
20	Plat	4820.89	4250	1586.2	...	PL
20	Liqd	26877.3	1500	...	...	Liquid 20
20	Ring	179.854	2000	...	...	Ring 1
20	Ring	486.572	365	...	...	SR1
20	Nozl	11.8383	200	1525.4	...	N3 2"
20	Nozl	88.167	500	1548.59	...	N4 4"
20	Nozl	6.77732	200	847.243	1260.83	N5 1.5"
20	Nozl	13.9324	1350	1538.96	...	N6 3"
20	Nozl	204.464	1050	1550.8	...	N7 4"
20	Wght	70	950	750	1299.04	Lifting lug 1
20	Wght	70	950	750	-1299.04	Lifting lug 2
30	Liqd	26877.3	1500	...	...	Liquid 30
30	Ring	179.627	1000	...	...	Ring 2
30	Nozl	384.019	1500	1804.8	...	M1 24"
30	Wght	500	1200	...	...	Cleat weight
30	Wght	50	1500	...	...	Manway davit
40	Sadl	553.473	2085	1597.5	...	New Saddle
40	Liqd	21501.8	1200	...	...	Liquid 40
40	Ring	179.627	500	...	...	Ring 3
40	Ring	486.573	2085	...	...	SR, 2085.0 Fr40
40	Nozl	579.553	1235	1830.2	...	N8 26"
40	Nozl	361.063	0.00058217	-2008	...	Nozzle boot
40	Wght	70	1450	750	1299.04	Lifting lug 3
40	Wght	70	1450	750	-1299.04	Lifting lug 4
50	Liqd	4927.5	300	...	...	Liquid 50
60	Nozl	16.8121	...	169.189	...	N2 3"
70	Nozl	12.8079	...	-451	...	N9 3"

Total Weight of Each Detail Type:

Saddles	1106.9
Platforms	4820.9
Liquid	85111.4
Stiffeners	1512.3
Nozzles	1706.6
Weights	830.0

Sum of the Detail Weights 95088.1 kgm

Note:

Because some nozzles were attached to a head or flange, their weights include the attached head or flange.



Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	9158.4	9158.4	9158.4	9158.4	9158.4	9158.4
Saddles	1106.9	1106.9	1106.9	1106.9	1106.9	1106.9
Stif. Rings	1512.3	1512.3	1512.3	1512.3	1512.3	1512.3
Nozzles	1706.6	1706.6	1706.6	1706.6	1706.6	1706.6
Wld Weights	830.0	830.0	830.0	830.0	830.0	830.0
Platforms	...	...	4820.9	4820.9	4820.9	4820.9
Ope. Liquid	...	...	...	...	...	85111.4
Test Liquid	...	67300.3	...	...	...	...
Totals	14314.2	81614.5	19135.1	19135.1	19135.1	104246.5

Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	14314.2 kgm
Shop Test Wt.	- Fabricated weight + water (full)	81614.5 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	19135.1 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	19135.1 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	19135.1 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	104246.5 kgm
Oper. Wt. + CA	- Corr wt. + operating liquid	104246.5 kgm
Field Test Wt.	- Empty weight + test medium (full)	86879.9 kgm

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	10387596	10227107
20	30	28462830	28274334
30	40	28462830	28274334
40	50	22770266	22619470
50	60	10387596	10227107
60	70	780245	...
70	80	1316576	1278543

Total	102567952.0	100900904.0 mm ²
Total	102.6	100.9 m ²

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Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Op.	Ambient					UG-44	50%	DNV
N1 6"	3.56	4.11	95	300	GR 2.3	3.171	2.162	3.555	2.162
N3 2"	4.28	4.97	95	300	GR 2.2	5.668	0.754	4.281	0.754
N4 4"	1.36	1.58	95	150	GR 2.3	...	...	...	...
N5 1.5"	1.36	1.58	95	150	GR 2.3	...	...	...	...
N6 3"	3.56	4.11	95	300	GR 2.3	4.219	1.114	3.555	1.114
N7 4"	1.36	1.58	95	150	GR 2.3	...	...	...	...
M1 24"	1.36	1.58	95	150	GR 2.3	...	...	...	...
N8 26"	1.36	1.58	95	150	GR 2.3	...	...	...	...
N2 3"	1.64	1.91	95	150	GR 2.2	...	...	...	...
N9 3"	3.56	4.11	95	300	GR 2.3	...	...	...	...
Min Rating	0.754	1.580, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N1 6"	196.088	0.50	4500001	4000	0.328	2.162	Pass
N3 2"	77.851	0.50	500000	1300	0.328	0.754	Pass
N6 3"	111.125	0.50	1081665	2000	0.328	1.114	Pass

Evaluating Flange on Nozzle [N1 6"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(4500001) + 4 \cdot \text{abs}(4000)196.088 \leq \pi \cdot 196.088^3 \cdot [(3.555 - 0.328) + .500 \cdot 3.555]$$

$$= 0.75139E+08 \leq 0.11858E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1 6" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(3.555, 0.500 \cdot 3.555 + 3.555 - 75138680.0 / (\pi \cdot 196.088^3))$$

$$= \min(3.555, 2.162)$$

$$= 2.162 \text{ MPa}$$

Evaluating Flange on Nozzle [N3 2"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(500000) + 4 \cdot \text{abs}(1300)77.851 \leq \pi \cdot 77.851^3 \cdot [(4.281 - 0.328) + .500 \cdot 4.281]$$

$$= 0.84050E+07 \leq 0.90358E+07 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3 2" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.281, 0.500 \cdot 4.281 + 4.281 - 8404991.0 / (\pi \cdot 77.851^3))$$

$$= \min(4.281, 0.754)$$

$$= 0.754 \text{ MPa}$$

Evaluating Flange on Nozzle [N6 3"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(1081665) + 4 \cdot \text{abs}(2000)111.125 \leq \pi \cdot 111.125^3 \cdot [(3.555 - 0.328) + .500 \cdot 3.555]$$

$$= 0.18196E+08 \leq 0.21582E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N6 3" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$



$$= \min(3.555, 0.500 \cdot 3.555 + 3.555 - 18196008.0/(\pi \cdot 111.125^3))$$
$$= \min(3.555, 1.114)$$
$$= 1.114 \text{ MPa}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (Part 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	5	
Base Elevation	0	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Length to Diameter ratio: 1.481

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50.0 • 1.0 • 1.0 • 1.0 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1983.503, 1.0 • 0.88 • 1.0 • 0.6 • 57.5²)

= 1748.1 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 2.0 • 0.7 • 1748.06

= 2427.5 N

From	Height mm	k1	k2	k3	Vz m/sec	Cf
10	1715.00	1.0000	1.0000	1.0000	57.50	0.7000
20	1715.00	1.0000	1.0000	1.0000	57.50	0.7000
30	1715.00	1.0000	1.0000	1.0000	57.50	0.7000
40	1715.00	1.0000	1.0000	1.0000	57.50	0.7000
50	1715.00	1.0000	1.0000	1.0000	57.50	0.7000
60	1766.60	1.0000	1.0000	1.0000	57.50	0.7000
70	2376.27	1.0000	1.0000	1.0000	57.50	0.7000

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
PL	11640000.00	1715.00	1.75	20354.75	1.20

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Lifting lug 1	0.00	2465.00	1.75	0.00
Lifting lug 2	0.00	2465.00	1.75	0.00
Cleat weight	0.00	1715.00	1.75	0.00
Manway davit	0.00	1715.00	1.75	0.00



Lifting lug 3	0.00	2465.00	1.75	0.00
Lifting lug 4	0.00	2465.00	1.75	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	1715	3624	1983845	1.74873	2427.52
20	30	1715	3624	10872000	1.74873	33658.2
30	40	1715	3624	10872000	1.74873	13303.4
40	50	1715	3624	8697600	1.74873	10642.7
50	60	1715	3624	1983845	1.74873	2427.52
60	70	1766.6	1043.94	69737.3	1.74873	85.3334
70	80	2376.27	1233.6	313585	1.74873	383.716



Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2015 + 2018 Amendment 2

Input Data:

Importance Factor as Entered by User I 1.5000
 Zone Number 2
 Soil Type Rocky or Hard Soil
 Response reduction factor (Force factor) R 1.5000
 Time Period of Vibration (program or user) 0.0303 Seconds
 Value of Sa/g from user Spectrum Data 0.1023
 Percent Damping 2.0 %
 Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa / g \cdot DampVal) / (R / I)$
 $= (0.1023 \cdot 1.0) / 1.0$
 $= 0.102$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.081
0.050	0.123
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.104
0.800	0.088
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.0303 secs
 Computed Sa/g = 0.1023

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
 The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:
 $= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 144483.7 \cdot 0.1023 \cdot 1.3$
 $= 19218.3 \text{ N}$

Vertical load per 7.4:
 $= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.1023 \cdot 1011386 \cdot 1.3$
 $= 89685.4 \text{ N}$

Earthquake Load Calculation:

	Earthquake	Earthquake	Element
--	------------	------------	---------



From	To	Height mm	Weight N	Ope Load N
10	20	1500	144484	19218.3
20	30	1500	144484	19218.3
20	30	1500	144484	19218.3
30	40	1500	144484	19218.3
40	0	1500	144484	19218.3
40	50	1500	144484	19218.3
50	60	1500	144484	19218.3

The Earthquake Weight does not include the weight of the saddles.

The Earthquake Loads calculated and printed in the Earthquake Load calculation report have been factored by the input scalar/load reduction factor of: 1.300.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	4275.000 mm
Center of Gravity of the Platforms	4300.000 mm
Center of Gravity of the Liquid	4250.000 mm
Center of Gravity of the Stiffening Rings	4253.880 mm
Center of Gravity of the Nozzles	5062.451 mm
Center of Gravity of the Added Weights (Operating)	4268.072 mm
Center of Gravity of the Added Weights (Empty)	4268.072 mm
Center of Gravity of Bare Shell New and Cold	4375.662 mm
Center of Gravity of Bare Shell Corroded	4375.662 mm
Vessel CG in the Operating Condition	4277.537 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	4397.238 mm
Vessel CG in the Test Condition	4286.645 mm

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (350.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

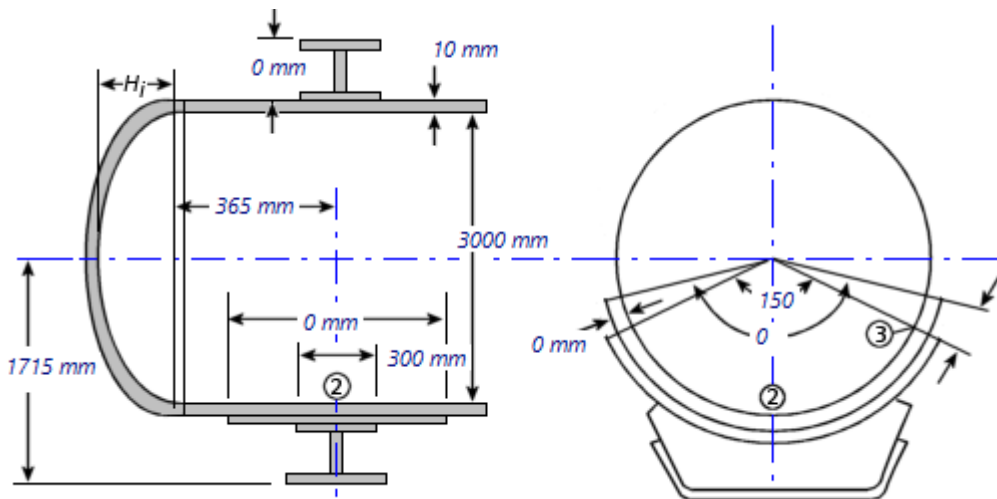
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(300.0 + 1.56 \cdot \sqrt{1505.0 \cdot 10.0}, 2 \cdot 365.0)$$

$$= 491.3783 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1505.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	8500.00	mm
Distance from Saddle to Vessel tangent	a1 or a	365.00	mm
Saddle Width	b1 or b	300.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Inside or Mean Depth of Head	Hi or hm	755.00	mm
Shell Allowable Stress used in Calculation		115.15	MPa
Head Allowable Stress used in Calculation		115.15	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1715.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	669277.50	N
Pressure used in Analysis	P	0.120 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	-6.25	-94.44
Long. Stress at Top of Midspan	6.25	115.15
Long. Stress at Bottom of Midspan	24.30	115.15
Long. Stress at Top of Saddles	8.68	115.15
Long. Stress at Bottom of Saddles	9.37	115.15
Tangential Shear in Shell	11.57	78.02



Circ. Compressive Stress in Shell	5.78	115.15
Stiffener Circ. Stress at Shell	45.23	143.93
Stiffener Circ. Stress at Tip	140.57	143.93

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(62928.4/2 + 0) \cdot 1715.0/2936.4146$$

$$= 55129.6 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(24358.08, 0) \cdot 1715.0/7720.0$$

$$= 5411.2 \text{ N}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(134528.12, 0) \cdot 1715.0/7720.0$$

$$= 29885.5 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(134528/2 + 0) \cdot 1715.0/2936.4146$$

$$= 117855.8 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 551422 + \max(5411, 55130, 29885, 117856)$$

$$= 669277.5 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.0 \cdot 1748.768 (3.614 + \pi/4 (3.02 \cdot 1.2)^2)$$

$$= 24359.340 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 674704.81 N

Transverse Shear Load 67264.06 N

Longitudinal Shear Load 134528.12 N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(3000.0/2 + 10.0 + 10.0) \sin(150.0/2), 2 \cdot 1505.0)$$

$$= 2936.415 \text{ mm}$$

The Computed K values from Table 4.15.1:

$$\text{K1} = 0.1607 \quad \text{K2} = 0.7988 \quad \text{K3} = 0.4851 \quad \text{K4} = 0.2952$$

$$\text{K5} = 0.6733 \quad \text{K6} = 0.0317 \quad \text{K7} = 0.0079 \quad \text{K8} = 0.3021$$

$$\text{K9} = 0.2177 \quad \text{K10} = 0.0355 \quad \text{K1*} = 0.2792$$

Note: Dimension a is less than Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2) / (2a \cdot L)) / (1 + (4\text{hm}) / (3L))]$$

$$= -669278 \cdot 365.0 [1 - (1 - 365.0/8500.0 + (1505.0^2 - 755.0^2) / (2 \cdot 365.0 \cdot 8500.0)) / (1 + (4 \cdot 755.0) / (3 \cdot 8500.0))]$$

$$= 24427876.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2 (\text{Rm}^2 - \text{hm}^2) / (L^2)) / (1 + (4\text{hm}^2) / (3L)) - 4a/L)$$

$$= 669278 \cdot 8500 / 4 ((1 + 2 (1505^2 - 755^2) / (8500^2)) / (1 + (4 \cdot 755) / (3 \cdot 8500)) - 4 \cdot 365 / 8500)$$

$$= 1087434752.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot \text{Rm} / (2t) - \text{M2} / (\pi \cdot \text{Rm}^2 t)$$



$$= 0.12 \cdot 1505.0 / (2 \cdot 10.0) - 0.10874E+10 / (\pi \cdot 1505.0^2 \cdot 10.0)$$

$$= -6.25 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.12 \cdot 1505.0 / (2 \cdot 10.0) + 0.10874E+10 / (\pi \cdot 1505.0^2 \cdot 10.0)$$

$$= 24.30 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.12 \cdot 1505.0 / (2 \cdot 10.0) - 24427876 / (\pi \cdot 1505.0^2 \cdot 10.0)$$

$$= 8.68 \text{ MPa}$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.12 \cdot 1505.0 / (2 \cdot 10.0) + 24427876 / (\pi \cdot 1505.0^2 \cdot 10.0)$$

$$= 9.37 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 669278(8500.0 - 2 \cdot 365.0) / (8500.0 + (4 \cdot 755.0 / 3))$$

$$= 547014.6 \text{ N}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$= T / (\pi \cdot R_m \cdot t)$$

$$= 547015 / (\pi \cdot 1505.0 \cdot 10.0)$$

$$= 11.57 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1505.0 \cdot 10.0)}$$

$$= 95.689 \text{ mm}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$= -K_5 \cdot Q \cdot k/A$$

$$= -0.673 \cdot 669278 \cdot 0.1 / 7793.78$$

$$= -5.78 \text{ MPa}$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$= -K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c_1/l$$

$$= -0.3021 \cdot 669278 / 7793.78 + 0.0317 \cdot 669278 \cdot 1505.0 \cdot 80.44 / 36063356$$

$$= 45.23 \text{ MPa}$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/l$$

$$= -0.3021 \cdot 669278 / 7793.78 - 0.0317 \cdot 669278 \cdot 1505.0 \cdot 129.56 / 36063356$$

$$= -140.57 \text{ MPa}$$

Distance between Saddle Supports [Ls]:

$$= 7720.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000016 \cdot 7720.0 (95.0 - 21.1)$$

$$= 9.149 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2945.0002	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	172.4	MPa
Height of Web at Center	Hw,c	179.0	mm
Friction Coefficient	mu	0.000	



Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1500.0 \cdot 10.0)} + 350.0) 10.0^3) / 12$$

$$= 45088.301 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 45088.301 + 5410.602 (93.396 - 5.0)^2$$

$$= 42322404.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	541.1	10.0	5.0	5410.6	27053.0	0.423E+08
Wearplate	350.0	10.0	15.0	3500.0	52500.0	0.215E+08
Web	10.0	173.0	106.5	1730.0	184245.0	0.461E+07
BasePlate	300.0	22.0	204.0	6600.0	1346400.0	0.810E+08
Totals	...	...	...	17240.6	1610197.9	0.149E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1610198 / 17240.602$$

$$= 93.396 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 150.0 / 2$$

$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(105.0) - 0.5 \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$$

$$= 0.2594$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.259 \cdot 669278$$

$$= 173592.0781 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 14.6751 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 103.4250 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1715.0 - 1500.0 \cdot \sin(150.0 / 2) / 1.309$$

$$= 608.130 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 105609448.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 65.9640 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 114.9167 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(669278 + 5427)300.0) / (4 \cdot 2945.0 \cdot 115.146))}$$

$$= 21.159 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 150.0 / 2) (1500.0 + 10.0 + 10.0)$$

$$= 2936.414 \text{ mm}$$



Distance between Ribs [e]:
 = Web Length / (Nribs - 1)
 = 2936.4143/(5 - 1)
 = 734.104 mm

Baseplate Pressure Area [Ap]:
 = e • Bpwid / 2
 = 734.1036 • 300.0/2
 = 110115.539 mm²

Bearing Pressure [Bp]:
 = (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 674705/(2945.0 • 300.0)
 = 0.764 N/mm²

Axial Load [P]:
 = Ap • Bp
 = 110115.5 • 0.76
 = 84092.219 N

Area of the Rib and Web [Ar]:
 = Rib Area + Web Area
 = 2900.0 + 3670.518
 = 6570.517 mm²

Compressive Stress [Sc]:
 = P/Ar
 = 84092.2/6570.5171
 = 12.800 MPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	300.0	...	3000.0	...	0.225E+08

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 300.0 - 10.0
 = 290.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0.0/6570.517
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 300.0/2
 = 150.000 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{22499978/6570.517}$
 = 58.518 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1715.0 - cos(150.0/2)(1500.0 + 10.0 + 10.0) - 22.0
 = 1299.595 mm

Intermediate Term [Cc]:
 = $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 199955/172.4}$
 = 151.319

Slenderness ratio [KL/r]:



$$= KL/r$$

$$= 1 \cdot 1299.595/58.518$$

$$= 22.208$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 134528.1 / (2 \cdot 2945.0) \cdot 734.104 \cdot 1299.59/2$$

$$= 10899566.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < Cc$ (22.2084 < 151.3192) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (22.21)^2 / (2 \cdot 151.32^2)) 172 / (5/3 + 3 \cdot (22.21) / (8 \cdot 151.32) - (22.21^3) / (8 \cdot 151.32^3))$$

$$= 99.06 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 12.8/99.06 + (10899566 \cdot 150.0/22500000) / 114.92$$

$$= 0.761$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	290.0	0.0	2900.0	0.0	0.225E+08
Web	734.1	10.0	0.0	7341.0	0.0	0.612E+05
Totals	...	...	...	10241.0	...	0.226E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/10241.035$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300.0/2$$

$$= 150.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1715.0 - \cos(1520.0/2) (22.0 + 0.0 + 0.0) - 0.0$$

$$= 173.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (22560320/10241.035) }$$

$$= 46.935 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 173.0/46.935$$

$$= 3.686$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 134528.125 / (2 \cdot 2945.0)$$

$$= 22.840 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 22.84 \cdot 734.104 \cdot 173.0$$

$$= 2901865.750 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 2901866/150402.266$$

$$= 19.288 \text{ MPa}$$

Compressive Allowable, $KL/r < Cc$ (3.6859 < 151.3192) per AISC E2-1 [Sca]:



$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (3.69)^2 / (2 \cdot 151.32^2)) 172 /$$

$$(5/3 + 3 \cdot (3.69) / (8 \cdot 151.32) - (3.69^3) / (8 \cdot 151.32^3))$$

$$= 102.8 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 16.84/102.83 + (2901866 \cdot 150.0/22560320) / 114.92$$

$$= 0.332$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B8
Bolt Allowable Stress	Stba	137.30 MPa
Bolt Corrosion Allowance	Bca	0.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	42.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	115.15 MPa
Area Available in a Single Bolt	BltArea	1018.2180 mm ²
Saddle Load Q0 (Weight)	Q0	556849.1 N
Saddle Load QL (Wind/Seismic contribution)	QL	29885.5 N
Maximum Transverse Force	Ft	67264.1 N
Maximum Longitudinal Force	Fl	134528.1 N
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 55943.6 \text{ N}$$

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= \max(Ff, Fl) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 134528.1 / (1018.218 \cdot 8)$$

$$= 16.5 \text{ MPa. Must be less than } 107.9 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 67264.1 / (1018.218 \cdot 8)$$

$$= 8.3 \text{ MPa. Must be less than } 107.9 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 134528.12 / (107.87 \cdot 8.0)$$

$$= 155.9004 \text{ mm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1715.0 \cdot 67264.06 + 0.0$$

$$= 115404640.00 \text{ N-mm}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 0.11540E+09/556849$$

$$= 207.16 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 155.9004, 0.0]$$



= 155.9004 mm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

*Warning - Distance to Saddle (a) > 0.25 * Tangent Distance (L) - 4.15.3.2*

Note:

Wear Pad Width (350.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(300.0 + 1.56 \cdot \sqrt{ 1505.0 \cdot 10.0 }, 2 \cdot 8135.0)$$

$$= 491.3783 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1505.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	8500.00	mm
Distance from Saddle to Vessel tangent	a1 or a	8135.00	mm
Saddle Width	b1 or b	300.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Inside or Mean Depth of Head	Hi or hm	755.00	mm
Shell Allowable Stress used in Calculation		115.15	MPa
Head Allowable Stress used in Calculation		115.15	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1715.00	mm
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		671864.00	N
Pressure used in Analysis	P	0.120	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	67.06	115.15
Long. Stress at Bottom of Midspan	-49.01	-94.44
Long. Stress at Bottom of Midspan	49.01	115.15
Long. Stress at Top of Saddles	82.05	115.15
Long. Stress at Bottom of Saddles	-64.00	-94.44
Long. Stress at Bottom of Saddles	64.00	115.15
Tangential Shear in Shell	11.62	78.02
Circ. Compressive Stress in Shell	5.80	115.15
Stiffener Circ. Stress at Shell	45.40	143.93
Stiffener Circ. Stress at Tip	141.11	143.93

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(62928.4 / 2 + 0) \cdot 1715.0 / 2936.4146$$

$$= 55129.6 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(24358.08, 0) \cdot 1715.0 / 7720.0$$

$$= 5411.2 \text{ N}$$

Saddle Reaction Force due to Earthquake FI [Fsl]:



$$\begin{aligned} &= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s \\ &= \max(134528.12, 0) \cdot 1715.0/7720.0 \\ &= 29885.5 \text{ N} \end{aligned}$$

Saddle Reaction Force due to Earthquake F_t [Fst]:

$$\begin{aligned} &= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E \\ &= 3.0(134528/2 + 0) \cdot 1715.0/2936.4146 \\ &= 117855.8 \text{ N} \end{aligned}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$\begin{aligned} &= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st}) \\ &= 554008 + \max(5411, 55130, 29885, 117856) \\ &= 671864.0 \text{ N} \end{aligned}$$

Longitudinal Wind Force [FI]:

$$\begin{aligned} &= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2)) \\ &= 1.0 \cdot 1748.768(3.614 + \pi/4(3.02 \cdot 1.2)^2) \\ &= 24359.340 \text{ N} \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	677291.31	N
Transverse Shear Load	67264.06	N
Longitudinal Shear Load	134528.12	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$\begin{aligned} &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m) \\ &= \min(2(3000.0/2 + 10.0 + 10.0) \sin(150.0/2), 2 \cdot 1505.0) \\ &= 2936.415 \text{ mm} \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0317	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned} &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h m^2)/(2a \cdot L)) / (1 + (4hm)/3L)] \\ &= -671864 \cdot 8135.0 [1 - (1 - 8135.0/8500.0 + (1505.0^2 - 755.0^2) / (2 \cdot 8135.0 \cdot 8500.0)) / (1 + (4 \cdot 755.0)/(3 \cdot 8500.0))] \\ &= -5197978624.0 \text{ N-mm} \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned} &= Q \cdot L/4 [(1 + 2(R_m^2 - h m^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L] \\ &= 671864 \cdot 8500/4 [(1 + 2(1505^2 - 755^2)/(8500^2)) / (1 + (4 \cdot 755)/(3 \cdot 8500)) - 4 \cdot 8135/8500] \\ &= -4130862592.0 \text{ N-mm} \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned} &= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t) \\ &= 0.12 \cdot 1505.0/(2 \cdot 10.0) - -0.4E+10/(\pi \cdot 1505.0^2 \cdot 10.0) \\ &= 67.06 \text{ MPa} \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned} &= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t) \\ &= 0.12 \cdot 1505.0/(2 \cdot 10.0) + -0.4E+10/(\pi \cdot 1505.0^2 \cdot 10.0) \\ &= -49.01 \text{ MPa} \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$\begin{aligned} &= P \cdot R_m/(2t) - M_1/(\pi \cdot R_m^2 \cdot t) \\ &= 0.12 \cdot 1505.0/(2 \cdot 10.0) - -0.5E+10/(\pi \cdot 1505.0^2 \cdot 10.0) \\ &= 82.05 \text{ MPa} \end{aligned}$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$\begin{aligned} &= P \cdot R_m/(2t) + M_1/(\pi \cdot R_m^2 \cdot t) \\ &= 0.12 \cdot 1505.0/(2 \cdot 10.0) + -0.5E+10/(\pi \cdot 1505.0^2 \cdot 10.0) \\ &= -64.00 \text{ MPa} \end{aligned}$$



Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L-2a)/(L+(4 \cdot hm/3))$
 $= 671864(8500.0 - 2 \cdot 8135.0)/(8500.0 + (4 \cdot 755.0/3))$
 $= -549128.8 \text{ N}$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:
 $= T / (\pi \cdot R_m \cdot t)$
 $= -549128.81/(\pi \cdot 1505.0 \cdot 10.0)$
 $= -11.62 \text{ MPa}$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{(R_m \cdot t)}$
 $= 0.78 \cdot \sqrt{(1505.0 \cdot 10.0)}$
 $= 95.689 \text{ mm}$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:
 $= -K_5 \cdot Q \cdot k/A$
 $= -0.673 \cdot 671864 \cdot 0.1/7793.79$
 $= -5.80 \text{ MPa}$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:
 $= -K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c_1/l$
 $= -0.3021 \cdot 671864/7793.79 + 0.0317 \cdot 671864 \cdot 1505.0 \cdot 80.44/36063448$
 $= 45.40 \text{ MPa}$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:
 $= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/l$
 $= -0.3021 \cdot 671864/7793.79 - 0.0317 \cdot 671864 \cdot 1505.0 \cdot 129.56/36063448$
 $= -141.11 \text{ MPa}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2945.0002	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	172.4	MPa
Height of Web at Center	Hw,c	175.0	mm
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3)/12$
 $= ((1.56 \cdot \sqrt{(1500.0 \cdot 10.0)} + 350.0)^3)/12$
 $= 45088.301 \text{ mm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$
 $= 45088.301 + 5410.602(93.396 - 5.0)^2$
 $= 42322404.000 \text{ mm}^4$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	541.1	10.0	5.0	5410.6	27053.0	0.423E+08
Wearplate	350.0	10.0	15.0	3500.0	52500.0	0.215E+08
Web	10.0	173.0	106.5	1730.0	184245.0	0.461E+07
BasePlate	300.0	22.0	204.0	6600.0	1346400.0	0.810E+08
Totals	...	...	...	17240.6	1610197.9	0.149E+09

Distance to Centroid [C1]:
 $= AY / A$
 $= 1610198/17240.602$



$$= 93.396 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 150.0/2$$

$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$$

$$= 0.2594$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.259 \cdot 671864$$

$$= 174262.9375 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 14.7318 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 103.4250 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1715.0 - 1500.0 \cdot \sin(150.0/2) / 1.309$$

$$= 608.130 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 106017592.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 66.2189 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 114.9167 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 \cdot \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(671864 + 5427) \cdot 300.0) / (4 \cdot 2945.0 \cdot 115.146)}$$

$$= 21.200 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 150.0/2) (1500.0 + 10.0 + 10.0)$$

$$= 2936.414 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2936.4143 / (5 - 1)$$

$$= 734.104 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 734.1036 \cdot 300.0 / 2$$

$$= 110115.539 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 677291 / (2945.0 \cdot 300.0)$$

$$= 0.767 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 110115.5 \cdot 0.77$$

$$= 84414.594 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 2900.0 + 3670.518$$

$$= 6570.517 \text{ mm}^2$$



Compressive Stress [Sc]:

$$= P/A_r$$

$$= 84414.6/6570.5171$$

$$= 12.849 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

B	D	Y	A	AY	Io
Rib+Web 10.0	300.0	...	3000.0	...	0.225E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 300.0 - 10.0$$

$$= 290.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/6570.517$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300.0/2$$

$$= 150.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(22499978/6570.517)}$$

$$= 58.518 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1715.0 - \cos(150.0/2) (1500.0 + 10.0 + 10.0) - 22.0$$

$$= 1299.595 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955/172.4)}$$

$$= 151.319$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1299.595/58.518$$

$$= 22.208$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 134528.1 / (2 \cdot 2945.0) \cdot 734.104 \cdot 1299.59/2$$

$$= 10899566.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ ($22.2084 < 151.3192$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (22.21)^2 / (2 \cdot 151.32^2)) 172 / (5/3 + 3 \cdot (22.21) / (8 \cdot 151.32) - (22.21^3) / (8 \cdot 151.32^3))$$

$$= 99.06 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C_1 / I) / Sba$$

$$= 12.85/99.06 + (10899566 \cdot 150.0/22500000) / 114.92$$

$$= 0.762$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

B	D	Y	A	AY	Io
---	---	---	---	----	----



Rib	10.0	290.0	0.0	2900.0	0.0	0.225E+08
Web	734.1	10.0	0.0	7341.0	0.0	0.612E+05
Totals	...	...	...	10241.0	...	0.226E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/10241.035$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300.0/2$$

$$= 150.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1715.0 - \cos(1520.0/2)(22.0 + 0.0 + 0.0) - 0.0$$

$$= 173.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(22560320/10241.035)}$$

$$= 46.935 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 173.0/46.935$$

$$= 3.686$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 134528.125/(2 \cdot 2945.0)$$

$$= 22.840 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 22.84 \cdot 734.104 \cdot 173.0$$

$$= 2901865.750 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 2901866/150402.266$$

$$= 19.288 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($3.6859 < 151.3192$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.69)^2 / (2 \cdot 151.32^2)) 172 / (5/3 + 3 \cdot (3.69) / (8 \cdot 151.32) - (3.69^3) / (8 \cdot 151.32^3))$$

$$= 102.8 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 16.91/102.83 + (2901866 \cdot 150.0/22560320) / 114.92$$

$$= 0.332$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B8
Bolt Allowable Stress	Stba	137.30 MPa
Bolt Corrosion Allowance	Bca	0.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	42.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	115.15 MPa
Area Available in a Single Bolt	BltArea	1018.2180 mm ²
Saddle Load Q0 (Weight)	Q0	559435.6 N
Saddle Load QL (Wind/Seismic contribution)	QL	29885.5 N



Maximum Transverse Force	Ft	67264.1 N
Maximum Longitudinal Force	F1	134528.1 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
= $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $134528.1 / (1018.218 \cdot 8)$
= 16.5 MPa. Must be less than 107.9 MPa.

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
= $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $67264.1 / (1018.218 \cdot 8)$
= 8.3 MPa. Must be less than 107.9 MPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F1 / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $134528.12 / (107.87 \cdot 8.0)$
= 155.9004 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot Ft + \text{Sum of X Moments}$
= $1715.0 \cdot 67264.06 + 0.0$
= 115404640.00 N-mm

Eccentricity (e):
= $Rmom / QO$
= $0.11540E+09 / 559436$
= 206.20 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0.0, 155.9004, 0.0]$
= 155.9004 mm²

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (350.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

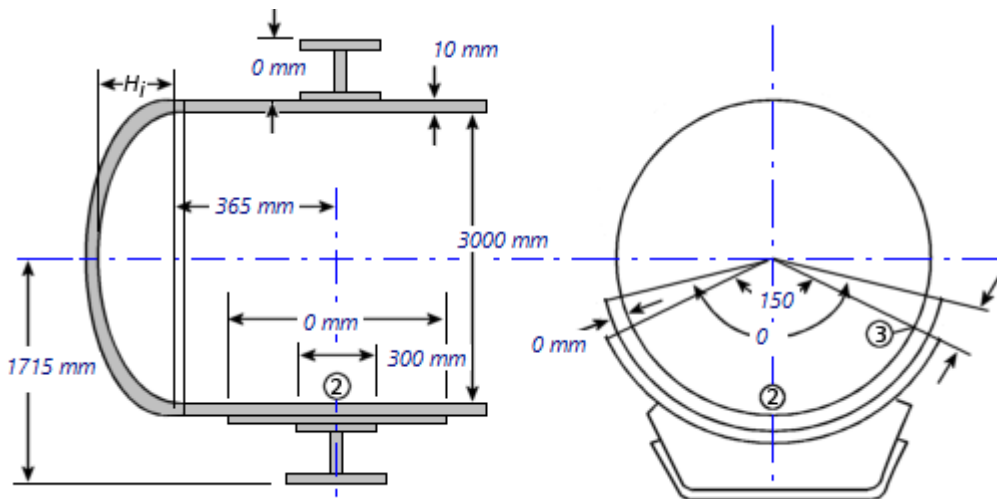
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(300.0 + 1.56 \cdot \sqrt{1505.0 \cdot 10.0}, 2 \cdot 365.0)$$

$$= 491.3783 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1505.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	8500.00	mm
Distance from Saddle to Vessel tangent	a1 or a	365.00	mm
Saddle Width	b1 or b	300.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Inside or Mean Depth of Head	Hi or hm	755.00	mm
Shell Allowable Stress used in Calculation		155.14	MPa
Head Allowable Stress used in Calculation		155.14	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1715.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	435876.00	N
Pressure used in Analysis	P	0.442 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	23.28	155.14
Long. Stress at Bottom of Midspan	43.17	155.14
Long. Stress at Top of Saddles	33.00	155.14
Long. Stress at Bottom of Saddles	33.45	155.14
Tangential Shear in Shell	7.54	91.88
Circ. Compressive Stress in Shell	3.77	155.14



Stiffener Circ. Stress at Shell	29.46	155.14
Stiffener Circ. Stress at Tip	91.55	155.14

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(20766.4/2 + 0) \cdot 1715.0/2936.4146$$

$$= 18192.8 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind; Sum of X Forces}) \cdot B / Ls$$

$$= \max(8038.17, 0) \cdot 1715.0/7720.0$$

$$= 1785.7 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl, Fwt, Fsl, Fst})$$

$$= 417683 + \max(1786, 18193, 0, 0)$$

$$= 435876.0 \text{ N}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.0 \cdot 1748.768(3.614 + \pi/4(3.02 \cdot 1.2)^2)$$

$$= 24359.340 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	441303.34	N
Transverse Shear Load	10383.19	N
Longitudinal Shear Load	8038.17	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(3000.0/2 + 10.0 + 10.0)\sin(150.0/2), 2 \cdot 1505.0)$$

$$= 2936.415 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0079	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	

Note: Dimension a is less than $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -435876 \cdot 365.0 [1 - (1 - 365.0/8500.0 + (1505.0^2 - 755.0^2)/(2 \cdot 365.0 \cdot 8500.0))/(1 + (4 \cdot 755.0)/(3 \cdot 8500.0))]$$

$$= 15908984.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 435876 \cdot 8500/4((1 + 2(1505^2 - 755^2)/(8500^2))/(1 + (4 \cdot 755)/(3 \cdot 8500)) - 4 \cdot 365/8500)$$

$$= 708206592.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 0.442 \cdot 1505.0/(2 \cdot 10.0) - 0.70821E+09/(\pi \cdot 1505.0^2 \cdot 10.0)$$

$$= 23.28 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 0.442 \cdot 1505.0/(2 \cdot 10.0) + 0.70821E+09/(\pi \cdot 1505.0^2 \cdot 10.0)$$

$$= 43.17 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot Rm/(2t) - M1/(\pi \cdot Rm^2 \cdot t)$$

$$= 0.442 \cdot 1505.0/(2 \cdot 10.0) - 15908984/(\pi \cdot 1505.0^2 \cdot 10.0)$$



= 33.00 MPa

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (\pi \cdot R_m^2 t)$$

$$= 0.442 \cdot 1505.0 / (2 \cdot 10.0) + 15908984 / (\pi \cdot 1505.0^2 \cdot 10.0)$$

$$= 33.45 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 435876(8500.0 - 2 \cdot 365.0) / (8500.0 + (4 \cdot 755.0 / 3))$$

$$= 356250.7 \text{ N}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$= T / (\pi \cdot R_m \cdot t)$$

$$= 356250.66 / (\pi \cdot 1505.0 \cdot 10.0)$$

$$= 7.54 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1505.0 \cdot 10.0)}$$

$$= 95.689 \text{ mm}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$= -K_5 \cdot Q \cdot k/A$$

$$= -0.673 \cdot 435876 \cdot 0.1 / 7793.78$$

$$= -3.77 \text{ MPa}$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$= -K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c_1/l$$

$$= -0.3021 \cdot 435876 / 7793.78 + 0.0317 \cdot 435876 \cdot 1505.0 \cdot 80.44 / 36063356$$

$$= 29.46 \text{ MPa}$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/l$$

$$= -0.3021 \cdot 435876 / 7793.78 - 0.0317 \cdot 435876 \cdot 1505.0 \cdot 129.56 / 36063356$$

$$= -91.55 \text{ MPa}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2945.0002	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	172.4	MPa
Height of Web at Center	Hw,c	179.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1500.0 \cdot 10.0)} + 350.0) 10.0^3) / 12$$

$$= 45088.301 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 45088.301 + 5410.602(93.396 - 5.0)^2$$

$$= 42322404.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	541.1	10.0	5.0	5410.6	27053.0	0.423E+08
Wearplate	350.0	10.0	15.0	3500.0	52500.0	0.215E+08
Web	10.0	173.0	106.5	1730.0	184245.0	0.461E+07
BasePlate	300.0	22.0	204.0	6600.0	1346400.0	0.810E+08



Totals	...	...	...	17240.6	1610197.9	0.149E+09
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Distance to Centroid [C1]:

$$= AY / A$$

$$= 1610198/17240.602$$

$$= 93.396 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 150.0/2$$

$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$$

$$= 0.2594$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.259 \cdot 435876.0$$

$$= 113054.1797 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 9.5574 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 103.4250 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1715.0 - 1500.0 \cdot \sin(150.0/2) / 1.309$$

$$= 608.130 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 68779576.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 42.9599 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 114.9167 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 \cdot \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(435876 + 5427)300.0 / (4 \cdot 2945.0 \cdot 115.146)}$$

$$= 17.112 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 150.0/2) (1500.0 + 10.0 + 10.0)$$

$$= 2936.414 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2936.4143 / (5 - 1)$$

$$= 734.104 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 734.1036 \cdot 300.0/2$$

$$= 110115.539 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 441303.375 / (2945.0 \cdot 300.0)$$

$$= 0.499 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 110115.5 \cdot 0.5$$

$$= 55002.094 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 2900.0 + 3670.518$$

$$= 6570.517 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 55002.1/6570.5171$$

$$= 8.372 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	300.0	...	3000.0	...	0.225E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 300.0 - 10.0$$

$$= 290.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/6570.517$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300.0/2$$

$$= 150.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{22499978/6570.517}$$

$$= 58.518 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1715.0 - \cos(150.0/2) (1500.0 + 10.0 + 10.0) - 22.0$$

$$= 1299.595 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955/172.4)}$$

$$= 151.319$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1299.595/58.518$$

$$= 22.208$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 8038.2 / (2 \cdot 2945.0) \cdot 734.104 \cdot 1299.59/2$$

$$= 651258.062 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ ($22.2084 < 151.3192$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (22.21)^2 / (2 \cdot 151.32^2)) 172 / (5/3 + 3 \cdot (22.21) / (8 \cdot 151.32) - (22.21^3) / (8 \cdot 151.32^3))$$

$$= 99.06 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 8.37/99.06 + (651258 \cdot 150.0/22500000) / 114.92$$

$$= 0.122$$



Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	290.0	0.0	2900.0	0.0	0.225E+08
Web	734.1	10.0	0.0	7341.0	0.0	0.612E+05
Totals	...	...	...	10241.0	...	0.226E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/10241.035$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300.0/2$$

$$= 150.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1715.0 - \cos(1520.0/2)(22.0 + 0.0 + 0.0) - 0.0$$

$$= 173.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{22560320/10241.035}$$

$$= 46.935 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 173.0/46.935$$

$$= 3.686$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 8038.167/(2 \cdot 2945.0)$$

$$= 1.365 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.365 \cdot 734.104 \cdot 173.0$$

$$= 173388.859 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 173388.859/150402.266$$

$$= 1.152 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (3.6859 < 151.3192) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.69)^2 / (2 \cdot 151.32^2)) 172 / (5/3 + 3 \cdot (3.69) / (8 \cdot 151.32) - (3.69^3) / (8 \cdot 151.32^3))$$

$$= 102.8 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 11.02/102.83 + (173388.88 \cdot 150.0/22560320) / 114.92$$

$$= 0.117$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B8
Bolt Allowable Stress	Stba	129.63 MPa
Bolt Corrosion Allowance	Bca	0.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	42.0000 mm
Thread Series	Series	TEMA Metric



BasePlate Allowable Stress	S	115.15 MPa
Area Available in a Single Bolt	BltArea	1018.2180 mm ²
Saddle Load QO (Weight)	QO	423110.6 N
Saddle Load QL (Wind/Seismic contribution)	QL	1785.7 N
Maximum Transverse Force	Ft	10383.2 N
Maximum Longitudinal Force	Fl	8038.2 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
= $Fl / (Bolt\ Area \cdot Number\ of\ Bolts)$
= $8038.2 / (1018.218 \cdot 8)$
= 1.0 MPa. Must be less than 77.8 MPa.

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
= $Ft / (Bolt\ Area \cdot Number\ of\ Bolts)$
= $10383.2 / (1018.218 \cdot 8)$
= 1.3 MPa. Must be less than 77.8 MPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $Fl / (BoltShearAllowable \cdot Nbolts)$
= $8038.17 / (77.78 \cdot 8.0)$
= 12.9199 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot Ft + Sum\ of\ X\ Moments$
= $1715.0 \cdot 10383.19 + 0.0$
= 17814396.00 N-mm

Eccentricity (e):
= $Rmom / QO$
= $17814396 / 423110.56$
= 42.09 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0.0, 12.9199, 0.0]$
= 12.9199 mm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

*Warning - Distance to Saddle (a) > 0.25 * Tangent Distance (L) - 4.15.3.2*

Note:
Wear Pad Width (350.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
= $\min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
= $\min(300.0 + 1.56 \cdot \sqrt{1505.0 \cdot 10.0}, 2 \cdot 8135.0)$
= 491.3783 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	1505.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	8500.00	mm
Distance from Saddle to Vessel tangent	a1 or a	8135.00	mm
Saddle Width	b1 or b	300.00	mm



Saddle Bearing Angle	delta or theta	150.00	degrees
Inside or Mean Depth of Head	Hi or hm	755.00	mm
Shell Allowable Stress used in Calculation		155.14	MPa
Head Allowable Stress used in Calculation		155.14	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1715.00	mm
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	441598.75	N
Pressure used in Analysis	P	0.442 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	71.37	155.14
Long. Stress at Bottom of Midspan	-4.92	-155.14
Long. Stress at Bottom of Midspan	4.92	155.14
Long. Stress at Top of Saddles	81.22	155.14
Long. Stress at Bottom of Saddles	-14.77	-155.14
Long. Stress at Bottom of Saddles	14.77	155.14
Tangential Shear in Shell	7.63	91.88
Circ. Compressive Stress in Shell	3.82	155.14
Stiffener Circ. Stress at Shell	29.84	155.14
Stiffener Circ. Stress at Tip	92.75	155.14

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$$

$$= 3.0(20766.4/2 + 0) \cdot 1715.0/2936.4146$$

$$= 18192.8\ N$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(8038.17, 0) \cdot 1715.0/7720.0$$

$$= 1785.7\ N$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 423406 + \max(1786, 18193, 0, 0)$$

$$= 441598.8\ N$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$$

$$= 1.0 \cdot 1748.768(3.614 + \pi/4(3.02 \cdot 1.2)^2)$$

$$= 24359.340\ N$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	447026.12	N
Transverse Shear Load	10383.19	N
Longitudinal Shear Load	8038.17	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(ShellID/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(3000.0/2 + 10.0 + 10.0) \sin(150.0/2), 2 \cdot 1505.0)$$

$$= 2936.415\ mm$$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0317	K8 = 0.3021



K9 = 0.2177 K10 = 0.0355 K1* = 0.2792

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)] \\
 &= -441599 \cdot 8135.0 [1 - (1 - 8135.0/8500.0 + (1505.0^2 - 755.0^2)/(2 \cdot 8135.0 \cdot 8500.0))/(1 + (4 \cdot 755.0)/(3 \cdot 8500.0))] \\
 &= -3416496128.0 \text{ N-mm}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4 \cdot ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L) \\
 &= 441599 \cdot 8500/4 \cdot ((1 + 2(1505^2 - 755^2)/(8500^2))/(1 + (4 \cdot 755)/(3 \cdot 8500)) - 4 \cdot 8135/8500) \\
 &= -2715109120.0 \text{ N-mm}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t) \\
 &= 0.442 \cdot 1505.0/(2 \cdot 10.0) - 0.3E+10/(\pi \cdot 1505.0^2 \cdot 10.0) \\
 &= 71.37 \text{ MPa}
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t) \\
 &= 0.442 \cdot 1505.0/(2 \cdot 10.0) + 0.3E+10/(\pi \cdot 1505.0^2 \cdot 10.0) \\
 &= -4.92 \text{ MPa}
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) - M1/(\pi \cdot Rm^2 \cdot t) \\
 &= 0.442 \cdot 1505.0/(2 \cdot 10.0) - 0.3E+10/(\pi \cdot 1505.0^2 \cdot 10.0) \\
 &= 81.22 \text{ MPa}
 \end{aligned}$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$\begin{aligned}
 &= P \cdot Rm/(2t) + M1/(\pi \cdot Rm^2 t) \\
 &= 0.442 \cdot 1505.0/(2 \cdot 10.0) + 0.3E+10/(\pi \cdot 1505.0^2 \cdot 10.0) \\
 &= -14.77 \text{ MPa}
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a)/(L + (4 \cdot hm/3)) \\
 &= 441599(8500.0 - 2 \cdot 8135.0)/(8500.0 + (4 \cdot 755.0/3)) \\
 &= -360928.1 \text{ N}
 \end{aligned}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$\begin{aligned}
 &= T / (\pi \cdot Rm \cdot t) \\
 &= -360928.09/(\pi \cdot 1505.0 \cdot 10.0) \\
 &= -7.63 \text{ MPa}
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(1505.0 \cdot 10.0)} \\
 &= 95.689 \text{ mm}
 \end{aligned}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k/A \\
 &= -0.673 \cdot 441599 \cdot 0.1/7793.79 \\
 &= -3.82 \text{ MPa}
 \end{aligned}$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$\begin{aligned}
 &= -K8 \cdot Q/A + K6 \cdot Q \cdot Rm \cdot c1/l \\
 &= -0.3021 \cdot 441599/7793.79 + 0.0317 \cdot 441599 \cdot 1505.0 \cdot 80.44/36063448 \\
 &= 29.84 \text{ MPa}
 \end{aligned}$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$\begin{aligned}
 &= -K8 \cdot Q/A - K6 \cdot Q \cdot Rm \cdot c2/l \\
 &= -0.3021 \cdot 441599/7793.79 - 0.0317 \cdot 441599 \cdot 1505.0 \cdot 129.56/36063448 \\
 &= -92.75 \text{ MPa}
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:



Baseplate Length	Bplen	2945.0002	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	172.4	MPa
Height of Web at Center	Hw,c	175.0	mm
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$
 $= ((1.56 \cdot \sqrt{(1500.0 \cdot 10.0)} + 350.0)10.0^3)/12$
 $= 45088.301 \text{ mm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 45088.301 + 5410.602(93.396 - 5.0)^2$
 $= 42322404.000 \text{ mm}^4$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	541.1	10.0	5.0	5410.6	27053.0	0.423E+08
Wearplate	350.0	10.0	15.0	3500.0	52500.0	0.215E+08
Web	10.0	173.0	106.5	1730.0	184245.0	0.461E+07
BasePlate	300.0	22.0	204.0	6600.0	1346400.0	0.810E+08
Totals	...	...	...	17240.6	1610197.9	0.149E+09

Distance to Centroid [C1]:
 $= AY / A$
 $= 1610198/17240.602$
 $= 93.396 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 150.0/2$
 $= 105.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$
 $= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0)\cos(105.0))$
 $= 0.2594$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.259 \cdot 441598.75$
 $= 114538.5078 \text{ N}$

Tension Stress, St = (Fh/As) = 9.6829 MPa
 Allowed Stress, Sa = 0.6 * Yield Str = 103.4250 MPa

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1715.0 - 1500.0 \cdot \sin(150.0/2) / 1.309$
 $= 608.130 \text{ mm}$

Bending Moment, M = Fh * d = 69682608.0000 N-mm

Bending Stress, Sb = (M * C1 / I) = 43.5240 MPa
 Allowed Stress, Sa = 2/3 * Yield Str = 114.9167 MPa

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.



$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(441599 + 5427)300.0 / (4 \cdot 2945.0 \cdot 115.146)}$$

$$= 17.223 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 150.0/2) (1500.0 + 10.0 + 10.0)$$

$$= 2936.414 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2936.4143 / (5 - 1)$$

$$= 734.104 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 734.1036 \cdot 300.0 / 2$$

$$= 110115.539 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 447026.156 / (2945.0 \cdot 300.0)$$

$$= 0.506 \text{ N/mm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 110115.5 \cdot 0.51$$

$$= 55715.359 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 2900.0 + 3670.518$$

$$= 6570.517 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 55715.4 / 6570.5171$$

$$= 8.480 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	300.0	...	3000.0	...	0.225E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 300.0 - 10.0$$

$$= 290.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0.0 / 6570.517$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300.0 / 2$$

$$= 150.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(22499978 / 6570.517)}$$

$$= 58.518 \text{ mm}$$

Length of Outer Rib [L]:



$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1715.0 - \cos(150.0/2) (1500.0 + 10.0 + 10.0) - 22.0$$

$$= 1299.595 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955 / 172.4)}$$

$$= 151.319$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1299.595 / 58.518$$

$$= 22.208$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 8038.2 / (2 \cdot 2945.0) \cdot 734.104 \cdot 1299.59 / 2$$

$$= 651258.062 \text{ N-mm}$$

Compressive Allowable, $KL/r < Cc$ ($22.2084 < 151.3192$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (22.21)^2 / (2 \cdot 151.32^2)) 172 /$$

$$(5/3 + 3 \cdot (22.21) / (8 \cdot 151.32) - (22.21^3) / (8 \cdot 151.32^3))$$

$$= 99.06 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 8.48 / 99.06 + (651258 \cdot 150.0 / 22500000) / 114.92$$

$$= 0.123$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	290.0	0.0	2900.0	0.0	0.225E+08
Web	734.1	10.0	0.0	7341.0	0.0	0.612E+05
Totals	...	...	...	10241.0	...	0.226E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 10241.035$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300.0 / 2$$

$$= 150.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1715.0 - \cos(1520.0/2) (22.0 + 0.0 + 0.0) - 0.0$$

$$= 173.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(22560320 / 10241.035)}$$

$$= 46.935 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 173.0 / 46.935$$

$$= 3.686$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 8038.167 / (2 \cdot 2945.0)$$

$$= 1.365 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:



$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 1.365 \cdot 734.104 \cdot 173.0 \\
 &= 173388.859 \text{ N-mm}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 173388.859 / 150402.266 \\
 &= 1.152 \text{ MPa}
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($3.6859 < 151.3192$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (3.69)^2 / (2 \cdot 151.32^2)) 172 / \\
 &\quad (5/3 + 3 \cdot (3.69) / (8 \cdot 151.32) - (3.69^3) / (8 \cdot 151.32^3)) \\
 &= 102.8 \text{ MPa}
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba} \\
 &= 11.16 / 102.83 + (173388.88 \cdot 150.0 / 22560320) / 114.92 \\
 &= 0.119
 \end{aligned}$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B8
Bolt Allowable Stress	Stba	129.63 MPa
Bolt Corrosion Allowance	Bca	0.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	42.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	115.15 MPa
Area Available in a Single Bolt	BltArea	1018.2180 mm ²
Saddle Load QO (Weight)	QO	428833.3 N
Saddle Load QL (Wind/Seismic contribution)	QL	1785.7 N
Maximum Transverse Force	Ft	10383.2 N
Maximum Longitudinal Force	Fl	8038.2 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [$\tau_{b,l}$]:

$$\begin{aligned}
 &= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 8038.2 / (1018.218 \cdot 8) \\
 &= 1.0 \text{ MPa. Must be less than } 77.8 \text{ MPa.}
 \end{aligned}$$

Shear Stress in a Single Bolt, Transverse Direction [$\tau_{b,t}$]:

$$\begin{aligned}
 &= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 10383.2 / (1018.218 \cdot 8) \\
 &= 1.3 \text{ MPa. Must be less than } 77.8 \text{ MPa.}
 \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$\begin{aligned}
 &= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts}) \\
 &= 8038.17 / (77.78 \cdot 8.0) \\
 &= 12.9199 \text{ mm}^2
 \end{aligned}$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$\begin{aligned}
 &= B \cdot F_t + \text{Sum of X Moments} \\
 &= 1715.0 \cdot 10383.19 + 0.0 \\
 &= 17814396.00 \text{ N-mm}
 \end{aligned}$$

Eccentricity (e):

$$\begin{aligned}
 &= R_{mom} / QO \\
 &= 17814396 / 428833.34 \\
 &= 41.52 \text{ mm} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}
 \end{aligned}$$



Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 12.9199, 0.0]
= 12.9199 mm²

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Note:

Wear Pad Width (350.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

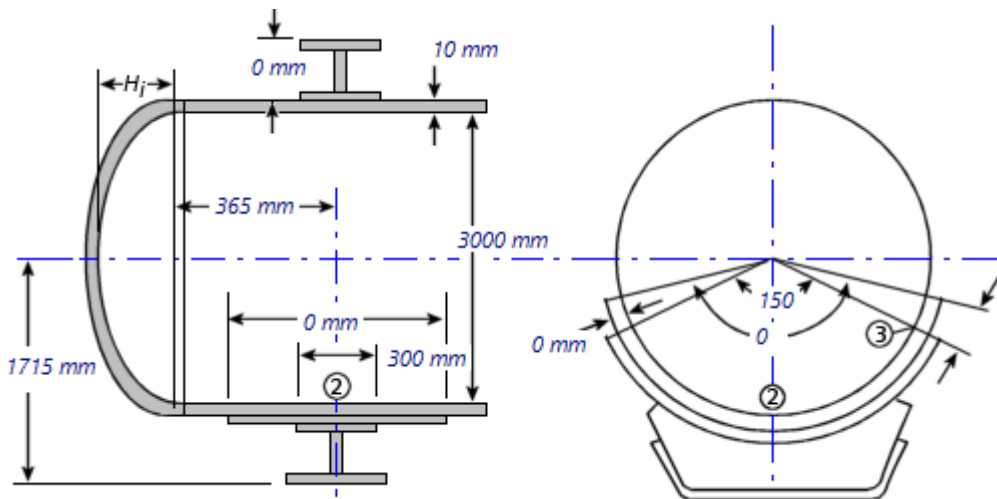
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(300.0 + 1.56 \cdot \sqrt{1505.0 \cdot 10.0}, 2 \cdot 365.0)$$

$$= 491.3783 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1505.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	8500.00	mm
Distance from Saddle to Vessel tangent	a1 or a	365.00	mm
Saddle Width	b1 or b	300.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Inside or Mean Depth of Head	Hi or hm	755.00	mm
Shell Allowable Stress used in Calculation		115.15	MPa
Head Allowable Stress used in Calculation		115.15	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1715.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	669277.50	N
Pressure used in Analysis	P	-0.074 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	-20.88	-94.44
Long. Stress at Top of Midspan	20.88	115.15
Long. Stress at Bottom of Midspan	9.67	115.15
Long. Stress at Top of Saddles	-5.95	-94.44
Long. Stress at Top of Saddles	5.95	115.15
Long. Stress at Bottom of Saddles	-5.26	-94.44
Long. Stress at Bottom of Saddles	5.26	115.15



Tangential Shear in Shell	11.57	78.02
Circ. Compressive Stress in Shell	5.78	115.15
Stiffener Circ. Stress at Shell	45.23	143.93
Stiffener Circ. Stress at Tip	140.57	193.92

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (62928.4 / 2 + 0) \cdot 1715.0 / 2936.4146$
 $= 55129.6 \text{ N}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(24358.08, 0) \cdot 1715.0 / 7720.0$
 $= 5411.2 \text{ N}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(134528.12, 0) \cdot 1715.0 / 7720.0$
 $= 29885.5 \text{ N}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (134528 / 2 + 0) \cdot 1715.0 / 2936.4146$
 $= 117855.8 \text{ N}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 551422 + \max(5411, 55130, 29885, 117856)$
 $= 669277.5 \text{ N}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi / 4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.0 \cdot 1748.768 (3.614 + \pi / 4 (3.02 \cdot 1.2)^2)$
 $= 24359.340 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	674704.81	N
Transverse Shear Load	67264.06	N
Longitudinal Shear Load	134528.12	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2 (ShellID / 2 + t + \text{WearPadThickness}) \sin(\theta / 2), 2 \cdot R_m)$
 $= \min(2 (3000.0 / 2 + 10.0 + 10.0) \sin(150.0 / 2), 2 \cdot 1505.0)$
 $= 2936.415 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0079	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	

Note: Dimension a is less than $R_m / 2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a / L + (R_m^2 - h_m^2) / (2a \cdot L)) / (1 + (4h_m) / (3L))]$
 $= -669278 \cdot 365.0 [1 - (1 - 365.0 / 8500.0 + (1505.0^2 - 755.0^2) / (2 \cdot 365.0 \cdot 8500.0)) / (1 + (4 \cdot 755.0) / (3 \cdot 8500.0))]$
 $= 24427876.0 \text{ N-mm}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2) / (L^2)) / (1 + (4h_m^2) / (3L)) - 4a / L)$
 $= 669278 \cdot 8500 / 4 ((1 + 2(1505^2 - 755^2) / (8500^2)) / (1 + (4 \cdot 755) / (3 \cdot 8500)) - 4 \cdot 365 / 8500)$
 $= 1087434752.0 \text{ N-mm}$



Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -0.074 \cdot 1505.0 / (2 \cdot 10.0) - 0.10874E+10 / (\pi \cdot 1505.0^2 \cdot 10.0)$$

$$= -20.88 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -0.074 \cdot 1505.0 / (2 \cdot 10.0) + 0.10874E+10 / (\pi \cdot 1505.0^2 \cdot 10.0)$$

$$= 9.67 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= -0.074 \cdot 1505.0 / (2 \cdot 10.0) - 24427876 / (\pi \cdot 1505.0^2 \cdot 10.0)$$

$$= -5.95 \text{ MPa}$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= -0.074 \cdot 1505.0 / (2 \cdot 10.0) + 24427876 / (\pi \cdot 1505.0^2 \cdot 10.0)$$

$$= -5.26 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 669278(8500.0 - 2 \cdot 365.0) / (8500.0 + (4 \cdot 755.0 / 3))$$

$$= 547014.6 \text{ N}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$= T / (\pi \cdot R_m \cdot t)$$

$$= 547015 / (\pi \cdot 1505.0 \cdot 10.0)$$

$$= 11.57 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1505.0 \cdot 10.0)}$$

$$= 95.689 \text{ mm}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$= -K_5 \cdot Q \cdot k/A$$

$$= -0.673 \cdot 669278 \cdot 0.1/7793.78$$

$$= -5.78 \text{ MPa}$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$= -K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c_1/l$$

$$= -0.3021 \cdot 669278/7793.78 + 0.0317 \cdot 669278 \cdot 1505.0 \cdot 80.44/36063356$$

$$= 45.23 \text{ MPa}$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/l$$

$$= -0.3021 \cdot 669278/7793.78 - 0.0317 \cdot 669278 \cdot 1505.0 \cdot 129.56/36063356$$

$$= -140.57 \text{ MPa}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2945.0002	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	172.4	MPa
Height of Web at Center	Hw,c	179.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1500.0 \cdot 10.0)} + 350.0)^3) / 12$$



$$= 45088.301 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 45088.301 + 5410.602 (93.396 - 5.0)^2$
 $= 42322404.000 \text{ mm}^4$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	541.1	10.0	5.0	5410.6	27053.0	0.423E+08
Wearplate	350.0	10.0	15.0	3500.0	52500.0	0.215E+08
Web	10.0	173.0	106.5	1730.0	184245.0	0.461E+07
BasePlate	300.0	22.0	204.0	6600.0	1346400.0	0.810E+08
Totals	...	...	...	17240.6	1610197.9	0.149E+09

Distance to Centroid [C1]:
 $= AY / A$
 $= 1610198 / 17240.602$
 $= 93.396 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 150.0 / 2$
 $= 105.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(105.0) - 0.5 \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$
 $= 0.2594$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.259 \cdot 669278$
 $= 173592.0781 \text{ N}$

Tension Stress, $St = (Fh / As) = 14.6751 \text{ MPa}$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 103.4250 \text{ MPa}$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1715.0 - 1500.0 \cdot \sin(150.0/2) / 1.309$
 $= 608.130 \text{ mm}$

Bending Moment, $M = Fh \cdot d = 105609448.0000 \text{ N-mm}$

Bending Stress, $Sb = (M \cdot C1 / I) = 65.9640 \text{ MPa}$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 114.9167 \text{ MPa}$

Minimum Thickness of Baseplate per Moss:

The maximum of $2/3 \cdot \text{Yield}$ and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(669278 + 5427)300.0 / (4 \cdot 2945.0 \cdot 115.146))}$$

$$= 21.159 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 150.0/2) (1500.0 + 10.0 + 10.0)$
 $= 2936.414 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (N_{\text{ribs}} - 1)$
 $= 2936.4143 / (5 - 1)$
 $= 734.104 \text{ mm}$



Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot Bpwid / 2 \\ &= 734.1036 \cdot 300.0 / 2 \\ &= 110115.539 \text{ mm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth) \\ &= 674705 / (2945.0 \cdot 300.0) \\ &= 0.764 \text{ N/mm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 110115.5 \cdot 0.76 \\ &= 84092.219 \text{ N} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= Rib \text{ Area} + Web \text{ Area} \\ &= 2900.0 + 3670.518 \\ &= 6570.517 \text{ mm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P / Ar \\ &= 84092.2 / 6570.5171 \\ &= 12.800 \text{ MPa} \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	300.0	...	3000.0	...	0.225E+08

Rib dimension [D]:

$$\begin{aligned} &= Saddle \text{ Width} - Web \text{ Thickness} \\ &= 300.0 - 10.0 \\ &= 290.000 \text{ mm} \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} &= AY / A \\ &= 0.0 / 6570.517 \\ &= 0.000 \text{ mm} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} &= Saddle \text{ Width} / 2 \\ &= 300.0 / 2 \\ &= 150.000 \text{ mm} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) } \\ &= \sqrt{ (22499978 / 6570.517) } \\ &= 58.518 \text{ mm} \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned} &= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk \\ &= 1715.0 - \cos(150.0 / 2) (1500.0 + 10.0 + 10.0) - 22.0 \\ &= 1299.595 \text{ mm} \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned} &= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) } \\ &= \sqrt{ (2 \cdot \pi^2 \cdot 199955 / 172.4) } \\ &= 151.319 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} &= KL / r \\ &= 1 \cdot 1299.595 / 58.518 \\ &= 22.208 \end{aligned}$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$



$$= 134528.1 / (2 \cdot 2945.0) \cdot 734.104 \cdot 1299.59 / 2$$

$$= 10899566.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ (22.2084 < 151.3192) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (22.21)^2 / (2 \cdot 151.32^2)) 172 /$$

$$(5/3 + 3 \cdot (22.21) / (8 \cdot 151.32) - (22.21^3) / (8 \cdot 151.32^3))$$

$$= 99.06 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 12.8 / 99.06 + (10899566 \cdot 150.0 / 22500000) / 114.92$$

$$= 0.761$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	290.0	0.0	2900.0	0.0	0.225E+08
Web	734.1	10.0	0.0	7341.0	0.0	0.612E+05
Totals	...	...	...	10241.0	...	0.226E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 10241.035$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300.0 / 2$$

$$= 150.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1715.0 - \cos(1520.0/2) (22.0 + 0.0 + 0.0) - 0.0$$

$$= 173.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{22560320 / 10241.035}$$

$$= 46.935 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 173.0 / 46.935$$

$$= 3.686$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 134528.125 / (2 \cdot 2945.0)$$

$$= 22.840 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 22.84 \cdot 734.104 \cdot 173.0$$

$$= 2901865.750 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 2901866 / 150402.266$$

$$= 19.288 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (3.6859 < 151.3192) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.69)^2 / (2 \cdot 151.32^2)) 172 /$$

$$(5/3 + 3 \cdot (3.69) / (8 \cdot 151.32) - (3.69^3) / (8 \cdot 151.32^3))$$

$$= 102.8 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)



$$= Sc/Sca + (Mbase,c \cdot C1/I)/Sba$$

$$= 16.84/102.83 + (2901866 \cdot 150.0/22560320)/114.92$$

$$= 0.332$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

*Warning - Distance to Saddle (a) > 0.25 * Tangent Distance (L) - 4.15.3.2*

Note:

Wear Pad Width (350.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ Rm \cdot t }, 2a)$$

$$= \min(300.0 + 1.56 \cdot \sqrt{ 1505.0 \cdot 10.0 }, 2 \cdot 8135.0)$$

$$= 491.3783 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1505.00	mm
Shell Thickness used in this Case	t	10.000	mm
Stiffened Vessel Length per 4.15.6	L	8500.00	mm
Distance from Saddle to Vessel tangent	a1 or a	8135.00	mm
Saddle Width	b1 or b	300.00	mm
Saddle Bearing Angle	delta or theta	150.00	degrees
Inside or Mean Depth of Head	Hi or hm	755.00	mm
Shell Allowable Stress used in Calculation		115.15	MPa
Head Allowable Stress used in Calculation		115.15	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1715.00	mm
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		671864.00	N
Pressure used in Analysis	P	-0.074	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	52.43	115.15
Long. Stress at Bottom of Midspan	-63.64	-94.44
Long. Stress at Bottom of Midspan	63.64	115.15
Long. Stress at Top of Saddles	67.42	115.15
Long. Stress at Bottom of Saddles	-78.63	-94.44
Long. Stress at Bottom of Saddles	78.63	115.15
Tangential Shear in Shell	11.62	78.02
Circ. Compressive Stress in Shell	5.80	115.15
Stiffener Circ. Stress at Shell	45.40	143.93
Stiffener Circ. Stress at Tip	141.11	193.92

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3.0(62928.4/2 + 0) \cdot 1715.0/2936.4146$$

$$= 55129.6 \text{ N}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(24358.08, 0) \cdot 1715.0/7720.0$$

$$= 5411.2 \text{ N}$$



Saddle Reaction Force due to Earthquake FI [Fsl]:

$$= \max(\text{FI,eq; Sum of X Forces}) \cdot B / L_s$$

$$= \max(134528.12, 0) \cdot 1715.0/7720.0$$

$$= 29885.5 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(134528/2 + 0) \cdot 1715.0/2936.4146$$

$$= 117855.8 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl, Fwt, Fsl, Fst})$$

$$= 554008 + \max(5411, 55130, 29885, 117856)$$

$$= 671864.0 \text{ N}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.0 \cdot 1748.768(3.614 + \pi/4(3.02 \cdot 1.2)^2)$$

$$= 24359.340 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	677291.31	N
Transverse Shear Load	67264.06	N
Longitudinal Shear Load	134528.12	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(3000.0/2 + 10.0 + 10.0)\sin(150.0/2), 2 \cdot 1505.0)$$

$$= 2936.415 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1607	K2 = 0.7988	K3 = 0.4851	K4 = 0.2952
K5 = 0.6733	K6 = 0.0317	K7 = 0.0317	K8 = 0.3021
K9 = 0.2177	K10 = 0.0355	K1* = 0.2792	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -671864 \cdot 8135.0 [1 - (1 - 8135.0/8500.0 + (1505.0^2 - 755.0^2)/(2 \cdot 8135.0 \cdot 8500.0))/(1 + (4 \cdot 755.0)/(3 \cdot 8500.0))]$$

$$= -5197978624.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 671864 \cdot 8500/4((1 + 2(1505^2 - 755^2)/(8500^2))/(1 + (4 \cdot 755)/(3 \cdot 8500)) - 4 \cdot 8135/8500)$$

$$= -4130862592.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 \cdot t)$$

$$= -0.074 \cdot 1505.0/(2 \cdot 10.0) - -0.4E+10/(\pi \cdot 1505.0^2 \cdot 10.0)$$

$$= 52.43 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= -0.074 \cdot 1505.0/(2 \cdot 10.0) + -0.4E+10/(\pi \cdot 1505.0^2 \cdot 10.0)$$

$$= -63.64 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m/(2t) - M1/(\pi \cdot R_m^2 \cdot t)$$

$$= -0.074 \cdot 1505.0/(2 \cdot 10.0) - -0.5E+10/(\pi \cdot 1505.0^2 \cdot 10.0)$$

$$= 67.42 \text{ MPa}$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m/(2t) + M1/(\pi \cdot R_m^2 \cdot t)$$

$$= -0.074 \cdot 1505.0/(2 \cdot 10.0) + -0.5E+10/(\pi \cdot 1505.0^2 \cdot 10.0)$$

$$= -78.63 \text{ MPa}$$



Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L-2a)/(L+(4 \cdot hm/3))$
 $= 671864(8500.0 - 2 \cdot 8135.0)/(8500.0 + (4 \cdot 755.0/3))$
 $= -549128.8 \text{ N}$

Shear Stress in the shell with a single ring (4.15.13) [τ1]:
 $= T / (\pi \cdot R_m \cdot t)$
 $= -549128.81/(\pi \cdot 1505.0 \cdot 10.0)$
 $= -11.62 \text{ MPa}$

Decay Length (4.15.20) [x1,x2]:
 $= 0.78 \cdot \sqrt{(R_m \cdot t)}$
 $= 0.78 \cdot \sqrt{(1505.0 \cdot 10.0)}$
 $= 95.689 \text{ mm}$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ6*]:
 $= -K5 \cdot Q \cdot k/A$
 $= -0.673 \cdot 671864 \cdot 0.1/7793.79$
 $= -5.80 \text{ MPa}$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ8*]:
 $= -K8 \cdot Q/A + K6 \cdot Q \cdot R_m \cdot c1/l$
 $= -0.3021 \cdot 671864/7793.79 + 0.0317 \cdot 671864 \cdot 1505.0 \cdot 80.44/36063448$
 $= 45.40 \text{ MPa}$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ9*]:
 $= -K8 \cdot Q/A - K6 \cdot Q \cdot R_m \cdot c2/l$
 $= -0.3021 \cdot 671864/7793.79 - 0.0317 \cdot 671864 \cdot 1505.0 \cdot 129.56/36063448$
 $= -141.11 \text{ MPa}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2945.0002	mm
Baseplate Thickness	Bpthk	22.0000	mm
Baseplate Width	Bpwid	300.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	172.4	MPa
Height of Web at Center	Hw,c	175.0	mm
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3)/12$
 $= ((1.56 \cdot \sqrt{(1500.0 \cdot 10.0)} + 350.0)^3)/12$
 $= 45088.301 \text{ mm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 45088.301 + 5410.602(93.396 - 5.0)^2$
 $= 42322404.000 \text{ mm}^4$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	541.1	10.0	5.0	5410.6	27053.0	0.423E+08
Wearplate	350.0	10.0	15.0	3500.0	52500.0	0.215E+08
Web	10.0	173.0	106.5	1730.0	184245.0	0.461E+07
BasePlate	300.0	22.0	204.0	6600.0	1346400.0	0.810E+08
Totals	...	...	...	17240.6	1610197.9	0.149E+09

Distance to Centroid [C1]:
 $= AY / A$



$$= 1610198/17240.602$$
$$= 93.396 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$
$$= 180 - 150.0/2$$
$$= 105.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$
$$= (1 + \cos(105.0) - 0.5 \cdot \sin(105.0)^2) / (\pi - 1.833 + \sin(105.0) \cos(105.0))$$
$$= 0.2594$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$
$$= 0.259 \cdot 671864$$
$$= 174262.9375 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 14.7318 \text{ MPa}$$
$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 103.4250 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$
$$= 1715.0 - 1500.0 \cdot \sin(150.0/2) / 1.309$$
$$= 608.130 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 106017592.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 66.2189 \text{ MPa}$$
$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 114.9167 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

The maximum of 2/3*Yield and the Saddle Allowable Stress value was used to determine the baseplate thickness.

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$
$$= \sqrt{3(671864 + 5427)300.0 / (4 \cdot 2945.0 \cdot 115.146)}$$
$$= 21.200 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$
$$= 2 \cdot \cos(90 - 150.0/2) (1500.0 + 10.0 + 10.0)$$
$$= 2936.414 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$
$$= 2936.4143 / (5 - 1)$$
$$= 734.104 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$
$$= 734.1036 \cdot 300.0/2$$
$$= 110115.539 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$
$$= 677291 / (2945.0 \cdot 300.0)$$
$$= 0.767 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$
$$= 110115.5 \cdot 0.77$$
$$= 84414.594 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$
$$= 2900.0 + 3670.518$$



$$= 6570.517 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 84414.6/6570.5171$$

$$= 12.849 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	300.0	...	3000.0	...	0.225E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 300.0 - 10.0$$

$$= 290.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/6570.517$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300.0/2$$

$$= 150.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(22499978/6570.517)}$$

$$= 58.518 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1715.0 - \cos(150.0/2) (1500.0 + 10.0 + 10.0) - 22.0$$

$$= 1299.595 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199955/172.4)}$$

$$= 151.319$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1299.595/58.518$$

$$= 22.208$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 134528.1 / (2 \cdot 2945.0) \cdot 734.104 \cdot 1299.59/2$$

$$= 10899566.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ ($22.2084 < 151.3192$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (22.21)^2 / (2 \cdot 151.32^2)) 172 /$$

$$(5/3 + 3 \cdot (22.21) / (8 \cdot 151.32) - (22.21^3) / (8 \cdot 151.32^3))$$

$$= 99.06 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 12.85/99.06 + (10899566 \cdot 150.0/22500000) / 114.92$$

$$= 0.762$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
--	---	---	---	---	----	----



Rib	10.0	290.0	0.0	2900.0	0.0	0.225E+08
Web	734.1	10.0	0.0	7341.0	0.0	0.612E+05
Totals	...	...	...	10241.0	...	0.226E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/10241.035$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 300.0/2$$

$$= 150.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1715.0 - \cos(1520.0/2)(22.0 + 0.0 + 0.0) - 0.0$$

$$= 173.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(22560320/10241.035)}$$

$$= 46.935 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 173.0/46.935$$

$$= 3.686$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 134528.125/(2 \cdot 2945.0)$$

$$= 22.840 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 22.84 \cdot 734.104 \cdot 173.0$$

$$= 2901865.750 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 2901866/150402.266$$

$$= 19.288 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (3.6859 < 151.3192) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.69)^2 / (2 \cdot 151.32^2)) 172 / (5/3 + 3 \cdot (3.69) / (8 \cdot 151.32) - (3.69^3) / (8 \cdot 151.32^3))$$

$$= 102.8 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 16.91/102.83 + (2901866 \cdot 150.0/22560320) / 114.92$$

$$= 0.332$$

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N1 6"

From: 10

Pressure for Reinforcement Calculations	P	0.1200	MPa
Temperature for Internal Pressure	Temp	95	°C
Design External Pressure	Pext	0.07	MPa
Temperature for External Pressure	Tempex	95	°C

Parent Material		SA-240 316L	
Parent Allowable Stress at Temperature	Sv	115.15	MPa
Parent Allowable Stress At Ambient	Sva	115.15	MPa

Inside Diameter of Elliptical Head	D	3000.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	10.0000	mm
Head Internal Corrosion Allowance	cas	0.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1240.0000	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP316L +IT	
Material UNS Number		S31603	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	115.10	MPa
Allowable Stress At Ambient	Sna	115.10	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		6.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		40	

Flange Material		SA-182 F316L	
Flange Type		Weld Neck Flange	

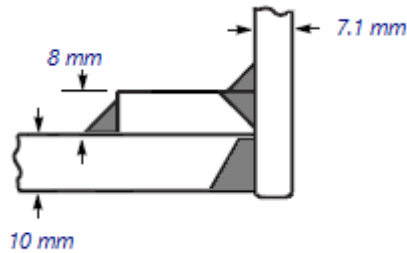
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.3500	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-240 316L	
Pad Allowable Stress at Temperature	Sp	115.10	MPa
Pad Allowable Stress At Ambient	Spa	115.10	MPa
Diameter of Pad along vessel surface	Dp	288.2750	mm
Thickness of Pad	te	8.0000	mm
Weld leg size between Pad and Shell	Wp	6.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	8.0000	mm
Reinforcing Pad Width		60.0000	mm

Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N1 6"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	6.065 in.
Actual Thickness Used in Calculation	0.280 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.12 \cdot 3000.0 \cdot 1.0 / (2 \cdot 115.15 \cdot 1.0 - 0.2 \cdot 0.12)$
 $= 1.5630 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.12 \cdot 77.03 / (115 \cdot 1.0 - 0.6 \cdot 0.12)$
 $= 0.0803 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6594 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	384.7444 mm
Parallel to Vessel Wall, opening length	d	192.3722 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		25.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115.1 / 115.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115.1 / 115.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 115.1 / 115.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.)$
 $= 1.000$

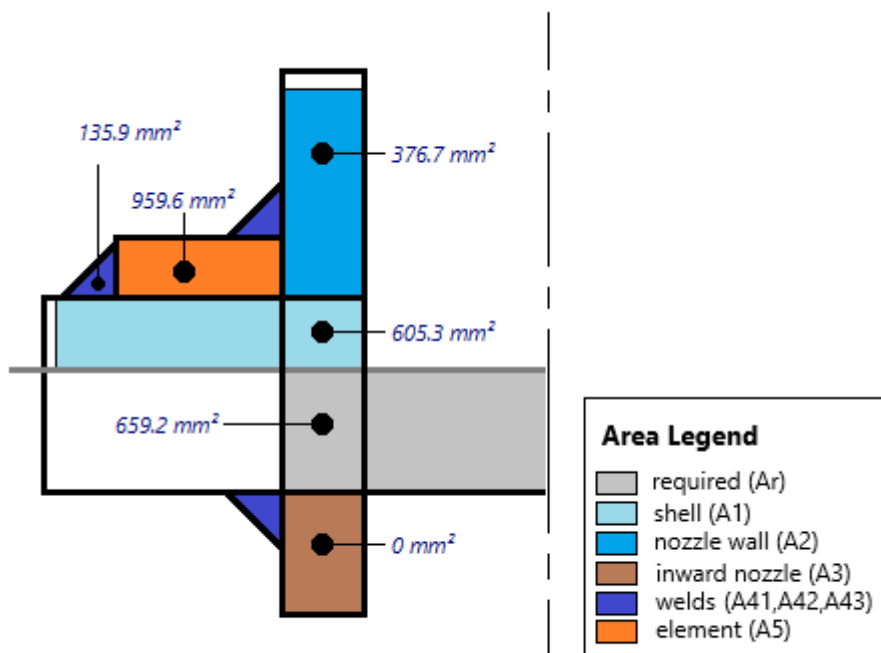


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	300.685	659.245	NA
Area in Shell	A1	1622.997	605.253	NA
Area in Nozzle Wall	A2	410.528	376.723	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	135.945	135.945	NA
Area in Element	A5	959.612	959.612	NA
TOTAL AREA AVAILABLE	Atot	3129.083	2077.533	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 53.21 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(192.3722 \cdot 6.8536 \cdot 1 + 2 \cdot 7.112 \cdot 6.8536 \cdot 1(1 - 1.))$
 $= 659.245 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 192.372(1.0 \cdot 10.0 - 1.0 \cdot 6.854) - 2 \cdot 7.112$
 $(1.0 \cdot 10.0 - 1.0 \cdot 6.8536) \cdot (1 - 1.)$
 $= 605.253 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha 3)$
 $= (2 \cdot 25.0) \cdot (7.11 - 0.66) \cdot 0.9996 / \sin(58.9)$
 $= 376.723 \text{ mm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:



$$= W_o^2 \cdot fr_3 + (W_i\text{-can}/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1. + 6.0^2 \cdot 1.$$

$$= 135.945 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$$

$$= (330.1345 - 210.1345) \cdot 8.0 \cdot 0.9996$$

$$= 959.612 \text{ mm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N1 6"

Intermediate Calc. for nozzle/shell Welds	Tmin	7.1120	mm
Intermediate Calc. for pad/shell Welds	TminPad	8.0000	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.9784 = 0.7 * tmin.	7.0700 = 0.7 * W _o mm
Pad Weld	4.0000 = 0.5 * TminPad	4.2420 = 0.7 * W _p mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (659.2452 - 605.2527 + 2 \cdot 7.112 \cdot 0.9996 \cdot (1.0 \cdot 10.0 - 6.8536)) \cdot 115)$$

$$= 11367.24 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_5 + A_4 - (W_i\text{-Can}/0.707)^2 \cdot fr_2) \cdot Sv$$

$$= (376.7225 + 959.6124 + 135.9451 - 0.0 \cdot 1.) \cdot 115$$

$$= 169513.48 \text{ N}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot Sv$$

$$= (376.7225 + 0.0 + 99.9596 + (142.1826)) \cdot 115$$

$$= 71254.04 \text{ N}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (376.7225 + 0.0 + 135.9451 + 959.6124 + (142.1826)) \cdot 115$$

$$= 185883.92 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 210.1345 \cdot 10.0 \cdot 0.49 \cdot 115$$



= 186145. N

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.1416/2.0) \cdot 288.275 \cdot 6.0 \cdot 0.49 \cdot 115 \\ &= 153219. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 100.6267) \cdot (7.112 - 0.0) \cdot 0.7 \cdot 115 \\ &= 181130. \text{ N} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.1416/2) \cdot 210.1345 \cdot 8.0 \cdot 0.74 \cdot 115 \\ &= 224894. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 210.1345 \cdot (6.35 - 0.0) \cdot 0.74 \cdot 115 \\ &= 178582. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (153219 + 181130) = 334349 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (186145 + 224894 + 178582 + 0) = 589620 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ &= (153219 + 178582 + 0) = 331801 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 334349 N	, must exceed W = 11367 N	or W1 = 169513 N
Path 2-2 = 589620 N	, must exceed W = 11367 N	or W2 = 71254 N
Path 3-3 = 331800 N	, must exceed W = 11367 N	or W3 = 185883 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Nozzle is O.K. for the External Pressure: 0.1 MPa

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N1 6"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	6.065 in.
Actual Thickness Used in Calculation	0.280 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$\begin{aligned} &= P \cdot D \cdot K / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P) \text{ per Appendix 1-4(c)} \\ &= 0.12 \cdot 3000.0 \cdot 1.0 / (2 \cdot 115.15 \cdot 1.0 \cdot 0.2 \cdot 0.12) \\ &= 1.5630 \text{ mm} \end{aligned}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$\begin{aligned} &= P \cdot R / (Sn \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= 0.12 \cdot 77.03 / (115 \cdot 1.0 \cdot 0.6 \cdot 0.12) \\ &= 0.0803 \text{ mm} \end{aligned}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.6594 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 308.1020 mm
 Parallel to Vessel Wall, opening length d 154.0510 mm
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 25.0000 mm

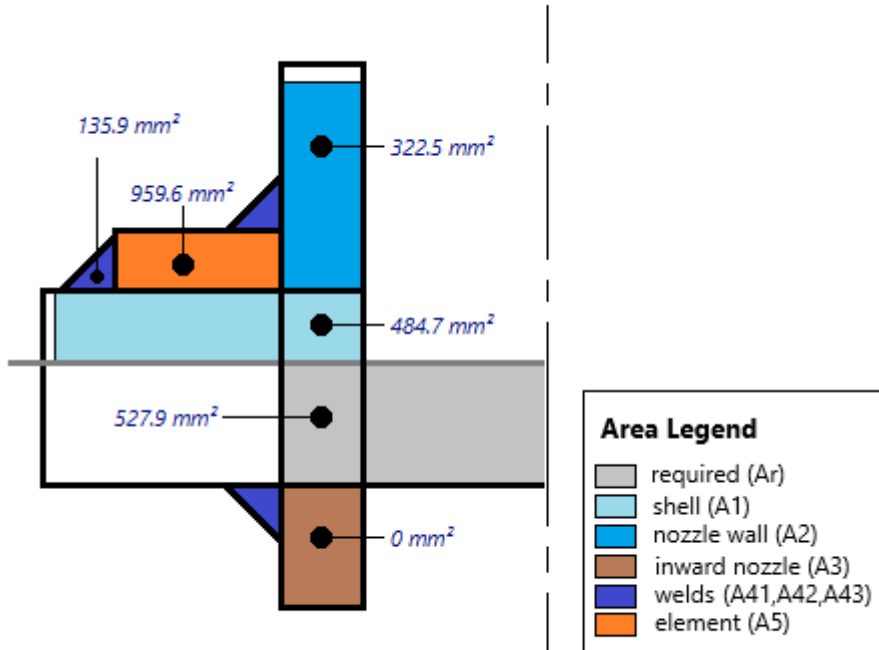


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	240.789	527.925	NA
Area in Shell	A1	1299.681	484.681	NA
Area in Nozzle Wall	A2	351.441	322.501	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	135.945	135.945	NA
Area in Element	A5	959.612	959.612	NA
TOTAL AREA AVAILABLE	Atot	2746.680	1902.739	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= 0.5 (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(154.051 \cdot 6.8536 \cdot 1 + 2 \cdot 7.112 \cdot 6.8536 \cdot (1 - 1.))$
 $= 527.925 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 154.051(1.0 \cdot 10.0 - 1.0 \cdot 6.854) - 2 \cdot 7.112$
 $(1.0 \cdot 10.0 - 1.0 \cdot 6.8536) \cdot (1 - 1.)$
 $= 484.681 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 25.0) \cdot (7.11 - 0.66) \cdot 0.9996$



$$= 322.501 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr3 + (W_i\text{-can}/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 $= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1. + 6.0^2 \cdot 1.$
 $= 135.945 \text{ mm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 $= (288.275 - 168.275) \cdot 8.0 \cdot 0.9996$
 $= 959.612 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 0.6594 \text{ mm}$
Wall Thickness per UG16(b),	$tr16b = 1.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 1.5630 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 1.5630 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb2 = 0.9700 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 1.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 6.2200 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.22, \max(1.563, 1.5)]$
 $= 1.5630 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(0.6594, 1.563)$
 $= 1.5630 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 7.112 = 6.223 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 31.8,	Allowable	: 115.1 MPa	Passed
Expansion	: 0.0,	Allowable	: 255.9 MPa	Passed
Occasional	: 0.6,	Allowable	: 153.1 MPa	Passed
Shear	: 21.9,	Allowable	: 80.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: N1 6"

Intermediate Calc. for nozzle/shell Welds	T_{min}	7.1120	mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	8.0000	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9784 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$4.0000 = 0.5 \cdot T_{minPad}$	$4.2420 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (527.9252 - 484.6807 + 2 \cdot 7.112 \cdot 0.9996 \cdot (1.0 \cdot 10.0 - 6.8536)) \cdot 115)$
 $= 10129.75 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i\text{-Can}/0.707)^2 \cdot fr2) \cdot S_v$
 $= (322.5013 + 959.6124 + 135.9451 - 0.0 \cdot 1.) \cdot 115$
 $= 163270.61 \text{ N}$

Weld Load [W2]:



$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv \\ &= (322.5013 + 0.0 + 99.9596 + (142.1826)) \cdot 115 \\ &= 65011.19 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S \\ &= (322.5013 + 0.0 + 135.9451 + 959.6124 + (142.1826)) \cdot 115 \\ &= 179641.05 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 115 \\ &= 149064. \text{ N} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.1416/2.0) \cdot 288.275 \cdot 6.0 \cdot 0.49 \cdot 115 \\ &= 153219. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 80.5815) \cdot (7.112 - 0.0) \cdot 0.7 \cdot 115 \\ &= 145049. \text{ N} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.1416/2) \cdot 168.275 \cdot 8.0 \cdot 0.74 \cdot 115 \\ &= 180094. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 168.275 \cdot (6.35 - 0.0) \cdot 0.74 \cdot 115 \\ &= 143007. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (153219 + 145049) = 298268 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (149064 + 180094 + 143007 + 0) = 472166 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (153219 + 143007 + 0) = 296226 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 298267 N	, must exceed W = 10129 N	or W1 = 163270 N
Path 2-2 = 472166 N	, must exceed W = 10129 N	or W2 = 65011 N
Path 3-3 = 296226 N	, must exceed W = 10129 N	or W3 = 179641 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 45.2323 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 406.4754 mm

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N1 6" :



Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	0.0000	mm
Vessel Diameter	Dv	5400.000	mm
Vessel Thickness	Tv	10.000	mm

Design Temperature	T1	95.0	C
Vessel Material		SA-240 316L	
Vessel UNS Number		S31603	
Vessel Cold S.I. Allowable	Smc	115.15	MPa
Vessel Hot S.I. Allowable	Smh	115.15	MPa

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow

Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	0.0000	mm
Nozzle Diameter	Dn	154.051	mm
Nozzle Thickness	Tn	7.112	mm
Nozzle Material		SA-312 TP316L	
Nozzle UNS Number		S31603	
Nozzle Cold S.I. Allowable	SNmc	115.10	MPa
Nozzle Hot S.I. Allowable	SNmh	115.10	MPa

Thickness of Reinforcing Pad	Tpad	8.000	mm
Diameter of Reinforcing Pad	Dpad	288.275	mm

Design Internal Pressure	Dp	0.120	MPa
Include Pressure Thrust		No	

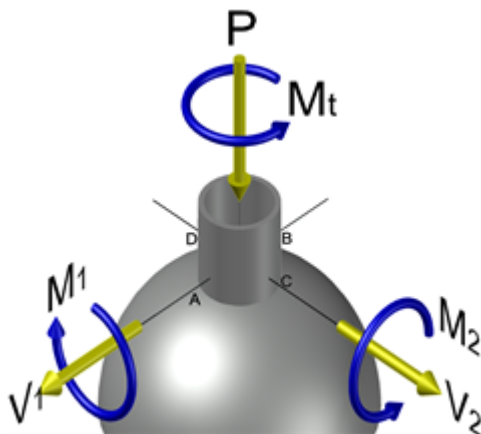
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-4000.0	N
Longitudinal Shear (SUS)	(V1) V1	6000.0	N
Circumferential Shear (SUS)	(Vc) V2	6000.0	N
Circumferential Moment (SUS)	(Mc) M1	3600000.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	2700000.0	N-mm
Torsional Moment (SUS)	Mt	4500000.5	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".





and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(2705.0 (10.0 - 0))}$$

$$= 711.022 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load		P	-4000.0	N
Circumferential Shear	(VC)	V2	6000.0	N
Longitudinal Shear	(VL)	V1	6000.0	N
Circumferential Moment	(MC)	M1	3600000.0	N-mm
Longitudinal Moment	(ML)	M2	2700000.0	N-mm
Torsional Moment		MT	4500000.5	N-mm

Dimensionless Parameters: U = 0.38 TAU = 11.33 RHO = 2.53

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 3	0.06972	(A,B,C,D)
M(x) / P	SP 3	0.06045	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 3	0.17816	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 3	0.17343	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 3	0.17816	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 3	0.17343	(A,B,C,D)
N(y) * T / P	SP 3	0.25807	(A,B,C,D)
M(y) / P	SP 3	0.07776	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 3	0.18911	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 3	0.28043	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 3	0.18911	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 3	0.28043	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Rad.	Bend. P	4.5	-4.5	4.5	-4.5	4.5	-4.5	4.5	-4.5
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-9.0	-9.0	9.0	9.0
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-52.3	52.3	52.3	-52.3
Rad.	Memb. ML	-6.7	-6.7	6.7	6.7	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-39.3	39.3	39.3	-39.3	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-40.6	28.9	51.3	-36.2	-56.0	39.8	66.6	-47.0
Tang.	Memb. P	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2
Tang.	Bend. P	5.8	-5.8	5.8	-5.8	5.8	-5.8	5.8	-5.8
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-9.5	-9.5	9.5	9.5
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-84.6	84.6	84.6	-84.6
Tang.	Memb. ML	-7.1	-7.1	7.1	7.1	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-63.5	63.5	63.5	-63.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-61.7	53.8	79.6	-58.9	-85.2	72.6	103.1	-77.7
Shear VC		1.3	1.3	-1.3	-1.3	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.3	-1.3	1.3	1.3
Shear MT		5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6
Tot. Shear		6.9	6.9	4.4	4.4	4.4	4.4	6.9	6.9
Str. Int.		63.7	55.5	80.2	59.7	85.8	73.1	104.4	79.2

Unitless Prm: U = 0.88 TAU = 0.00 (19.77) RHO = 0.00 (1.41)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.07704	(A,B,C,D)
M(x) / P	SR 2	0.04126	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.09065	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.09386	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.09065	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.09386	(A,B,C,D)
N(y) * T / P	SR 2	0.02339	(A,B,C,D)
M(y) / P	SR 2	0.01285	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.02698	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.02815	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.02698	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.02815	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1
Rad. Bend. P		9.9	-9.9	9.9	-9.9	9.9	-9.9	9.9	-9.9
Rad. Memb. MC		0.0	0.0	0.0	0.0	-19.8	-19.8	19.8	19.8
Rad. Bend. MC		0.0	0.0	0.0	0.0	-123.2	123.2	123.2	-123.2
Rad. Memb. ML		-14.9	-14.9	14.9	14.9	0.0	0.0	0.0	0.0
Rad. Bend. ML		-92.4	92.4	92.4	-92.4	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-94.3	70.7	120.3	-84.4	-130.1	96.6	156.1	-110.2
Tang. Memb. P		0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Tang. Bend. P		3.1	-3.1	3.1	-3.1	3.1	-3.1	3.1	-3.1
Tang. Memb. MC		0.0	0.0	0.0	0.0	-5.9	-5.9	5.9	5.9
Tang. Bend. MC		0.0	0.0	0.0	0.0	-37.0	37.0	37.0	-37.0
Tang. Memb. ML		-4.4	-4.4	4.4	4.4	0.0	0.0	0.0	0.0
Tang. Bend. ML		-27.7	27.7	27.7	-27.7	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-28.1	21.1	36.2	-25.4	-38.8	28.9	46.9	-33.2
Shear VC		1.3	1.3	-1.3	-1.3	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.3	-1.3	1.3	1.3
Shear MT		3.4	3.4	3.4	3.4	3.4	3.4	3.4	3.4
Tot. Shear		4.8	4.8	2.1	2.1	2.1	2.1	4.8	4.8
Str. Int.		94.7	71.2	120.3	84.4	130.1	96.6	156.3	110.5

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		9.0	9.0	9.0	9.0	9.0	9.0	9.0	9.0
Rad. Pl (SUS)		-5.9	-5.9	7.6	7.6	-8.1	-8.1	9.8	9.8
Rad. Q (SUS)		-34.8	34.8	43.7	-43.7	-47.9	47.9	56.8	-56.8
Long. Pm (SUS)		9.0	9.0	9.0	9.0	9.0	9.0	9.0	9.0
Long. Pl (SUS)		-3.9	-3.9	10.3	10.3	-6.3	-6.3	12.7	12.7
Long. Q (SUS)		-57.7	57.7	69.2	-69.2	-78.9	78.9	90.4	-90.4

Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	1.3	1.3	-1.3	-1.3	-1.3	-1.3	1.3	1.3
Shear Q (SUS)	5.6	5.6	5.6	5.6	5.6	5.6	5.6	5.6
Pm (SUS)	9.0	9.0	9.0	9.0	9.0	9.0	9.0	9.0
Pm+P1 (SUS)	5.7	5.7	19.8	19.8	3.3	3.3	22.2	22.2
Pm+P1+Q (Total)	54.7	64.5	89.2	50.7	76.8	82.1	113.3	70.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	9.00	115.15	Passed
Pm+P1 (SUS)	22.17	172.72	Passed
Pm+P1+Q (TOTAL)	113.35	345.44	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	16.2	16.2	16.2	16.2	16.2	16.2	16.2	16.2
Rad.	P1 (SUS)	-11.8	-11.8	18.0	18.0	-16.8	-16.8	22.9	22.9
Rad.	Q (SUS)	-82.5	82.5	102.3	-102.3	-113.3	113.3	133.1	-133.1
Long.	Pm (SUS)	16.2	16.2	16.2	16.2	16.2	16.2	16.2	16.2
Long.	P1 (SUS)	-3.5	-3.5	5.4	5.4	-5.0	-5.0	6.8	6.8
Long.	Q (SUS)	-24.6	24.6	30.8	-30.8	-33.9	33.9	40.0	-40.0
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	P1 (SUS)	1.3	1.3	-1.3	-1.3	-1.3	-1.3	1.3	1.3
Shear	Q (SUS)	3.4	3.4	3.4	3.4	3.4	3.4	3.4	3.4
	Pm (SUS)	16.2	16.2	16.2	16.2	16.2	16.2	16.2	16.2
	Pm+P1 (SUS)	12.9	12.9	34.3	34.3	12.1	12.1	39.2	39.2
	Pm+P1+Q (Total)	78.5	87.4	136.5	68.2	113.9	112.8	172.5	94.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	16.20	115.15	Passed
Pm+P1 (SUS)	39.22	172.72	Passed
Pm+P1+Q (TOTAL)	172.45	345.44	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: N3 2"

From: 20

Pressure for Reinforcement Calculations	P	0.1013	MPa
Temperature for Internal Pressure	Temp	95	°C
Design External Pressure	Pext	0.07	MPa
Temperature for External Pressure	Tempex	95	°C

Parent Material		SA-240 316L	
Parent Allowable Stress at Temperature	Sv	115.15	MPa
Parent Allowable Stress At Ambient	Sva	115.15	MPa

Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	664.9997	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent 250.00 mm

User Entered Minimum Design Metal Temperature 11.40 °C

Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP316	
Material UNS Number		S31600	
Material Specification/Type		Wld. pipe	
Allowable Stress at Temperature	Sn	137.88	MPa
Allowable Stress At Ambient	Sna	137.88	MPa

Note: The allowables above have been divided by 0.85, see UG-37 [S].

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm

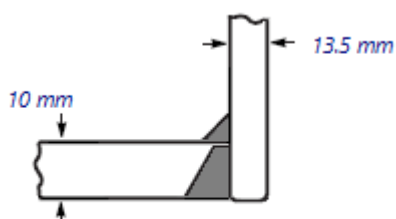
Flange Material		SA-182 F316	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	4.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 2.2	

The Pressure Design option was Design Pressure + static head.





ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.530 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.1 \cdot 1500.0 / (115 \cdot 1.0 - 0.6 \cdot 0.1)$
= 1.3206 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $25.4 \cdot (\exp([0.1 / (137.88 \cdot 1.0)] - 1))$
= 0.0187 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3942 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 101.6000 mm
Parallel to Vessel Wall, opening length d 50.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad Tlnp 25.0000 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N3 2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 0.3942 mm
Wall Thickness per UG16(b), tr16b = 1.5000 mm
Wall Thickness, shell/head, internal pressure trb1 = 1.3206 mm
Wall Thickness tb1 = max(trb1, tr16b) = 1.5000 mm
Wall Thickness, shell/head, external pressure trb2 = 0.9703 mm
Wall Thickness tb2 = max(trb2, tr16b) = 1.5000 mm
Wall Thickness per table UG-45 tb3 = 4.8000 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[4.8, max(1.5, 1.5)]
= 1.5000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.3942, 1.5)
= 1.5000 mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,
(see 319.4.4 and 302.3.5):**

Sustained	:	12.9,	Allowable	:	137.9 MPa	Passed
Expansion	:	0.0,	Allowable	:	331.8 MPa	Passed
Occasional	:	0.1,	Allowable	:	183.4 MPa	Passed
Shear	:	11.8,	Allowable	:	96.5 MPa	Passed

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of
the expansion stress allowable.*

SA-240 316L, Min Metal Temp without impact per UHA-51: -196 °C

SA-312 TP316, Min Metal Temp without impact per UHA-51: -196 °C

Weld Size Calculations, Description: N3 2"



Intermediate Calc. for nozzle/shell Welds Tmin 10.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5035 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 310.5034 mm

Input Echo, WRC297 Item 1, Description: N3 2" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3000.000	mm
Shell Thickness	10.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240	316L
Vessel Cold S.I. Allowable	Smc	115.15 MPa
Vessel Hot S.I. Allowable	Smh	115.15 MPa
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	50.800	mm
Nozzle Thickness	13.4620	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312	TP316
Nozzle Cold S.I. Allowable	SNmc	137.88 MPa
Nozzle Hot S.I. Allowable	SNmh	137.88 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	0.10	MPa
Radial Load	P	-1300.00	N
Circumferential Shear	Vc	2000.00	N
Longitudinal Shear	Vl	2000.00	N
Circumferential Moment	Mc	300000.00	N-mm
Longitudinal Moment	Ml	400000.03	N-mm
Torsional Moment	Mt	500000.00	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (5.774) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))}$$

$$= 77.724 + 2 \cdot 1.65 \cdot \sqrt{(1505.0(10.0 - 0.0))}$$

$$= 482.563 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3010.000 mm
Nozzle Outside Diameter	(d) =	77.724 mm
Vessel Thickness used	(T) =	10.000 mm
Nozzle Thickness used	(t) =	13.462 mm
T / t	=	0.743
d / t	=	10.000



$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.448$$

Nr/P	=	0.163
Mr/P	=	0.320
M0/P	=	0.103
N0/P	=	0.181
MrD/Mc	=	0.303
NrDL/Mc	=	0.109
M0d/Mc	=	0.109
N0DL/Mc	=	-0.070
MrD/M1	=	0.272
NrDL/M1	=	0.112
M0D/M1	=	0.097
N0DL/M1	=	0.007

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-30	30	30	-30
Normal Pressure Stress	15	15	15	15
Outplane Stress Summary	-4	39	55	-20

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-84	84	84	-84
Inplane Pressure Stress	7	7	7	7
Inplane Stress Summary	-55	62	124	-93

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	6	6	3	3

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	56	64	124	93

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2



Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	2	2	-2	-2
Outplane Bending (Mc)	-25	25	25	-25
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	7	7	7	7

Outplane Stress Summary	-4	29	40	-26
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-70	70	70	-70
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	15	15	15	15

Inplane Stress Summary	-32	58	116	-73
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	3	3	6	6
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	32	58	117	74
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	0	0	0

Hoop Stress Summary	2	2	2	3
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	10	-10	10	-10
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-10	-10	10	10
Axial Bending (ML)	-36	36	36	-36
Axial Pressure Stress	0	0	0	0



Axial Stress Summary -36 16 58 -35

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	3	3	3	3
Shear Stress Summary	5	5	2	2

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	40	18	58	39

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	2	2	-2	-2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	0	0	0
Hoop Stress Summary	5	5	0	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	10	-10	10	-10
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-31	31	31	-31
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-28	14	50	-33

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	3	3	3	3
Shear Stress Summary	2	2	5	5

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	34	14	51	34

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int.	Stress Values at
------------------------	------------------

Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15	15	15	15	15	15	15	15
Circ. Pl (SUS)	2	2	2	2	-2	-2	6	6
Circ. Q (SUS)	-22	22	38	-38	-46	46	94	-94
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-3	-3	7	7	4	4	0	0
Long. Q (SUS)	-60	60	108	-108	-17	17	33	-33
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0
Pm+Pl (SUS)	17.1	17.1	17.3	17.3	13.4	13.4	21.1	21.1
Pm+Pl+Q (Total)	56.7	65.4	122.2	94.2	33.6	59.5	115.5	73.8

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	15.00	115.15	Passed
Pm+Pl (SUS)	21.07	172.72	Passed
Pm+Pl+Q (TOTAL)	122.24	345.44	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	0	0	0	0	0	0	0	0
Circ. Pl (SUS)	2	2	2	2	4	4	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. Pl (SUS)	-10	-10	10	10	-7	-7	7	7
Long. Q (SUS)	-26	26	46	-46	-21	21	41	-41
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)	12.2	12.2	10.1	10.1	11.2	11.2	7.3	7.3
Pm+Pl+Q (Total)	38.8	17.1	56.1	38.2	32.2	14.4	48.7	34.9

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	137.88	Passed
Pm+Pl (SUS)	12.17	206.82	Passed
Pm+Pl+Q (TOTAL)	56.07	413.65	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N4 4"

From: 20

Pressure for Reinforcement Calculations	P	0.1013	MPa
Temperature for Internal Pressure	Temp	95	°C
Design External Pressure	Pext	0.07	MPa
Temperature for External Pressure	Tempex	95	°C

Parent Material		SA-240 316L	
Parent Allowable Stress at Temperature	Sv	115.15	MPa
Parent Allowable Stress At Ambient	Sva	115.15	MPa

Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	1634.9998	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		550.00	mm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP316L +IT	
Material UNS Number		S31603	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	115.10	MPa
Allowable Stress At Ambient	Sna	115.10	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		4.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

Flange Material		SA-182 F316L	
Flange Type		Weld Neck Flange	

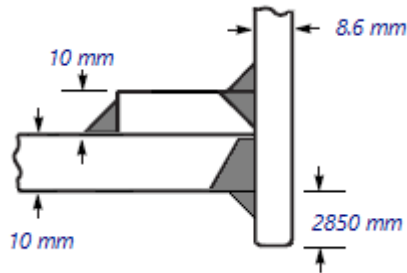
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	2850.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-240 316L +IT	
Pad Allowable Stress at Temperature	Sp	115.10	MPa
Pad Allowable Stress At Ambient	Spa	115.10	MPa
Diameter of Pad along vessel surface	Dp	214.3000	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		50.0000	mm

Flange Class		150	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N4 4"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.826 in.
Actual Thickness Used in Calculation	0.337 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.1 \cdot 1500.0 / (115 \cdot 1.0 - 0.6 \cdot 0.1)$
 $= 1.3206 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.1 \cdot 48.59 / (115 \cdot 1.0 - 0.6 \cdot 0.1)$
 $= 0.0428 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4923 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	194.3608 mm
Parallel to Vessel Wall, opening length	d	97.1804 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		25.0000 mm
Normal to Vessel Wall, Inward		21.3995 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115.1 / 115.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115.1 / 115.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 115.1 / 115.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.)$
 $= 1.000$

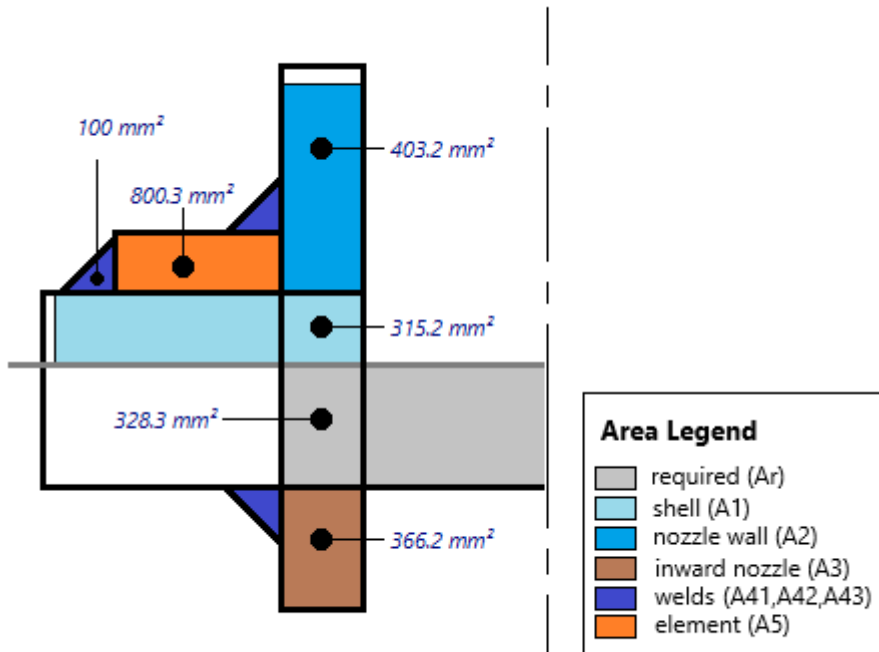


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	128.350	328.330	NA
Area in Shell	A1	843.403	315.169	NA
Area in Nozzle Wall	A2	425.678	403.210	NA
Area in Inward Nozzle	A3	366.203	366.203	NA
Area in Welds	A41+A42+A43	99.960	99.960	NA
Area in Element	A5	800.285	800.285	NA
TOTAL AREA AVAILABLE	Atot	2535.528	1984.825	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(97.1804 \cdot 6.7566 \cdot 1 + 2 \cdot 8.5598 \cdot 6.7566 \cdot 1(1-1.))$
 $= 328.330 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 97.18(1.0 \cdot 10.0 - 1.0 \cdot 6.757) - 2 \cdot 8.56$
 $(1.0 \cdot 10.0 - 1.0 \cdot 6.7566) \cdot (1 - 1.)$
 $= 315.169 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 25.0) \cdot (8.56 - 0.49) \cdot 0.9996$
 $= 403.210 \text{ mm}^2$

Area Available in Inward Nozzle [A3]:
 $= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$
 $= 2 \cdot 8.5598 \cdot (21.3995) \cdot 0.9996$



= 366.203 mm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr_3 + (W_i\text{-can}/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
= $10.0^2 \cdot 1. + (0.0)^2 \cdot 1. + 0.0^2 \cdot 1.$
= 99.960 mm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
= $(194.3608 - 114.3) \cdot 10.0 \cdot 0.9996$
= 800.285 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 0.4923 mm
Wall Thickness per UG16(b),	tr16b = 1.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 1.3206 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 1.5000 mm
Wall Thickness, shell/head, external pressure	trb2 = 0.9703 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 1.5000 mm
Wall Thickness per table UG-45	tb3 = 5.2578 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[5.258, max(1.5, 1.5)]
= 1.5000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.4923, 1.5)
= 1.5000 mm

Available Nozzle Neck Thickness = $0.875 \cdot 8.560 = 7.490$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle	:	-104 °C
Gov. MDMT of the Reinforcement Pad	:	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-104 °C

Weld Size Calculations, Description: N4 4"

Intermediate Calc. for nozzle/shell Welds	Tmin	8.5598 mm
Intermediate Calc. for pad/shell Welds	TminPad	10.0000 mm

Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	5.9919 = $0.7 \cdot t_{min}$	7.0700 = $0.7 \cdot W_o$ mm
Pad Weld	5.0000 = $0.5 \cdot T_{minPad}$	5.6560 = $0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (328.3298 - 315.1686 + 2 \cdot 8.5598 \cdot 0.9996 \cdot (1.0 \cdot 10.0 - 6.7566)) \cdot 115)$
 $= 7905.73 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (403.2096 + 800.2846 + 99.9596 - 0.0 \cdot 1.) \cdot 115$
 $= 150075.39 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (403.2096 + 366.203 + 99.9596 + (171.1269)) \cdot 115$
 $= 119799.63 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (403.2096 + 366.203 + 99.9596 + 800.2846 + (171.1269)) \cdot 115$
 $= 211941.78 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 115$
 $= 101251. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 214.3 \cdot 8.0 \cdot 0.49 \cdot 115$
 $= 151868. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 52.8701) \cdot (8.5598 - 0.0) \cdot 0.7 \cdot 115$
 $= 114541. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 114.3 \cdot 10.0 \cdot 0.74 \cdot 115$
 $= 152910. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 114.3 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 115$
 $= 152972. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (S_{PEW} + S_{NW}) = (151868 + 114541) = 266409 \text{ N}$

Failure Path [Path 2-2]:
 $= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$
 $= (101251 + 152910 + 152972 + 0) = 407133 \text{ N}$

Failure Path [Path 3-3]:
 $= (S_{pew} + T_{ngw} + S_{inw})$
 $= (151868 + 152972 + 0) = 304840 \text{ N}$



Summary of Failure Path Calculations:

Path 1-1 = 266408 N	, must exceed W = 7905 N	or W1 = 150075 N
Path 2-2 = 407133 N	, must exceed W = 7905 N	or W2 = 119799 N
Path 3-3 = 304839 N	, must exceed W = 7905 N	or W3 = 211941 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 1.0891 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 3160.0000 mm

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Input, Nozzle Desc: N5 1.5"

From: 20

Pressure for Reinforcement Calculations	P	0.1096	MPa
Temperature for Internal Pressure	Temp	95	°C
Design External Pressure	Pext	0.07	MPa
Temperature for External Pressure	Tempex	95	°C

Parent Material		SA-240 316L	
Parent Allowable Stress at Temperature	Sv	115.15	MPa
Parent Allowable Stress At Ambient	Sva	115.15	MPa

Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	664.9997	mm
Shell Finished (Minimum) Thickness	t	6.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	1250.0000	mm
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Distance from Bottom/Left Tangent		250.00	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L +IT	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.10	MPa
Allowable Stress At Ambient	Sna	115.10	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		326.10	deg
Diameter		1.5000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2240	mm

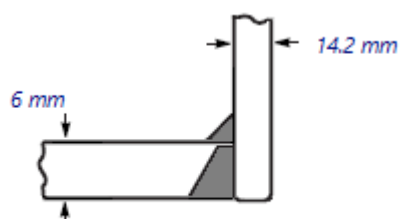
Flange Material		SA-182 F316L +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	211.1370	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	4.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.



Reinforcement CALCULATION, Description: N5 1.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.500 in.
Actual Thickness Used in Calculation	0.560 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.11 \cdot 1500.0 / (115 \cdot 1.0 - 0.6 \cdot 0.11)$
= 1.4282 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $19.05 \cdot (\exp([0.11 / (115 \cdot 1.0)] - 1))$
= 0.0181 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3106 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	137.4656	mm
Parallel to Vessel Wall, opening length	d	68.7328	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N5 1.5".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N5 1.5"

Intermediate Calc. for nozzle/shell Welds Tmin 10.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: N5 1.5"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.500 in.
Actual Thickness Used in Calculation	0.560 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.11 \cdot 1500.0 / (115 \cdot 1.0 - 0.6 \cdot 0.11)$
= 1.4282 mm



Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R(\exp([P/(Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $19.05(\exp([0.11/(115.1 \cdot 1.0)] - 1))$
= 0.0181 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3106 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	78.5480	mm
Parallel to Vessel Wall	Rn+tn+t	39.2740	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N5 1.5".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.3106	mm
Wall Thickness per UG16(b),	tr16b	= 1.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 1.4282	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 1.5000	mm
Wall Thickness, shell/head, external pressure	trb2	= 0.9703	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000	mm
Wall Thickness per table UG-45	tb3	= 4.5200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[4.52, max(1.5, 1.5)]
= 1.5000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.3106, 1.5)
= 1.5000 mm

Available Nozzle Neck Thickness = 14.2240 mm -> OK

Weld Size Calculations, Description: N5 1.5"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 52.4917 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 274.4343 mm

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Input, Nozzle Desc: N6 3"

From: 20

Pressure for Reinforcement Calculations	P	0.1013	MPa
Temperature for Internal Pressure	Temp	95	°C
Design External Pressure	Pext	0.07	MPa
Temperature for External Pressure	Tempex	95	°C

Parent Material		SA-240 316L	
Parent Allowable Stress at Temperature	Sv	115.15	MPa
Parent Allowable Stress At Ambient	Sva	115.15	MPa

Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	1634.9998	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1400.00	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP316	
Material UNS Number		S31600	
Material Specification/Type		Wld. pipe	
Allowable Stress at Temperature	Sn	137.88	MPa
Allowable Stress At Ambient	Sna	137.88	MPa

Note: The allowables above have been divided by 0.85, see UG-37 [S].

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		40	

Flange Material		SA-182 F316L	
Flange Type		Weld Neck Flange	

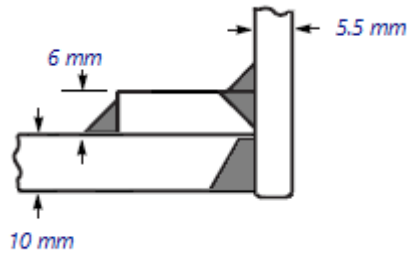
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-240 316L +IT	
Pad Allowable Stress at Temperature	Sp	115.10	MPa
Pad Allowable Stress At Ambient	Spa	115.10	MPa
Diameter of Pad along vessel surface	Dp	188.9000	mm
Thickness of Pad	te	6.0000	mm
Weld leg size between Pad and Shell	Wp	6.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	6.0000	mm
Reinforcing Pad Width		50.0000	mm

Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N6 3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.068 in.
Actual Thickness Used in Calculation	0.216 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.1 \cdot 1500.0 / (115 \cdot 1.0 - 0.6 \cdot 0.1)$
 $= 1.3206 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.1 \cdot 38.96 / (138 \cdot 1.0 - 0.6 \cdot 0.1)$
 $= 0.0286 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4257 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	155.8544 mm
Parallel to Vessel Wall, opening length	d	77.9272 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		19.7160 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N6 3".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 0.4257 mm
Wall Thickness per UG16(b),	tr16b = 1.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 1.3206 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 1.5000 mm
Wall Thickness, shell/head, external pressure	trb2 = 0.9703 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 1.5000 mm
Wall Thickness per table UG-45	tb3 = 4.8000 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(1.5, 1.5)]$
 $= 1.5000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.4257, 1.5)$
 $= 1.5000 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 5.486 = 4.801 \text{ mm} \rightarrow \text{OK}$

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,
 (see 319.4.4 and 302.3.5):**



Sustained :	37.2,	Allowable :	137.9 MPa	Passed
Expansion :	0.0,	Allowable :	307.5 MPa	Passed
Occasional :	0.3,	Allowable :	183.4 MPa	Passed
Shear :	27.3,	Allowable :	96.5 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Pad Weld for Nozzle skipped, as Nozzle is not a Carbon Steel material.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Gov. MDMT of the Reinforcement Pad	:	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-104 °C

Weld Size Calculations, Description: N6 3"

Intermediate Calc. for nozzle/shell Welds	Tmin	5.4864	mm
Intermediate Calc. for pad/shell Welds	TminPad	6.0000	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	3.8405 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	3.0000 = 0.5*TminPad	4.2420 = 0.7 * Wp mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.6588 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 310.6588 mm

Input Echo, WRC297 Item 1, Description: N6 3" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3000.000	mm
Shell Thickness	10.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240	316L
Vessel Cold S.I. Allowable	Smc	115.15 MPa
Vessel Hot S.I. Allowable	Smh	115.15 MPa
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	77.927	mm
Nozzle Thickness	5.4864	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312	TP316
Nozzle Cold S.I. Allowable	SNmc	137.88 MPa
Nozzle Hot S.I. Allowable	SNmh	137.88 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	188.900	mm
Thickness of Reinforcing Pad	6.0000	mm



Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	0.10	MPa
Radial Load	P	-2000.00	N
Circumferential Shear	Vc	3000.00	N
Longitudinal Shear	Vl	3000.00	N
Circumferential Moment	Mc	600000.00	N-mm
Longitudinal Moment	Ml	900000.06	N-mm
Torsional Moment	Mt	0.110E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(1508.0(10.0 - 0.0))}$
 $= 494.142 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3016.000 mm
Nozzle Outside Diameter	(d) =	88.900 mm
Vessel Thickness used	(T) =	16.000 mm
Nozzle Thickness used	(t) =	5.486 mm
T / t	=	2.916
d / t	=	16.204

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.405$$

Nr/P	=	0.174
Mr/P	=	0.179
M0/P	=	0.168
N0/P	=	0.242
MrD/Mc	=	0.193
NrDL/Mc	=	0.163
M0d/Mc	=	0.214
N0DL/Mc	=	-0.128
MrD/Ml	=	0.186
NrDL/Ml	=	0.161
M0D/Ml	=	0.203
N0DL/Ml	=	-0.058

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	2	2	-2	-2
Outplane Bending (Ml)	-48	48	48	-48
Normal Pressure Stress	9	9	9	9
Outplane Stress Summary	-26	53	64	-46

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1



Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-44	44	44	-44
Inplane Pressure Stress	4	4	4	4

Inplane Stress Summary	-36	35	64	-40
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5

Shear Stress Summary	6	6	4	4
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	39	56	69	48
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	3	3	-3	-3
Outplane Bending (Mc)	-33	33	33	-33
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	4	4	4	4

Outplane Stress Summary	-15	35	44	-38
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-30	30	30	-30
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	9	9	9	9

Inplane Stress Summary	-15	28	54	-23
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	4	4	6	6
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stress Intensities	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	19	37	57	41

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	2	2	-2	-2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	0	0	0
Hoop Stress Summary	4	5	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	36	-36	36	-36
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-31	-31	31	31
Axial Bending (ML)	-213	213	213	-213
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-206	146	283	-216

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	16	16	16	16
Shear Stress Summary	20	20	12	12

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	215	149	284	218

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	3	3	-3	-3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	0	0	0	0
Hoop Stress Summary	6	6	0	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	1	1	1	1
Axial Bending (P)	36	-36	36	-36
Axial Membrane (Mc)	-21	-21	21	21
Axial Bending (Mc)	-150	150	150	-150
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	0	0	0	0
Axial Stress Summary	-132	93	209	-163

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	16	16	16	16
Shear Stress Summary	12	12	20	20

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	140	95	214	168

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(1505.0(10.0 - 0.0))}$
 $= 493.739 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	3010.000 mm
Nozzle Outside Diameter (d)	=	188.900 mm
Vessel Thickness used (T)	=	10.000 mm
Nozzle Thickness used (t)	=	55.486 mm
T / t	=	0.180
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.089$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.108
Mr/P	=	0.217
M0/P	=	0.063
N0/P	=	0.105
MrD/Mc	=	0.310
NrDL/Mc	=	0.076
M0d/Mc	=	0.095
N0DL/Mc	=	0.048
MrD/Ml	=	0.183
NrDL/Ml	=	0.102
M0D/Ml	=	0.057
N0DL/Ml	=	0.103

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0



Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-16	16	16	-16
Normal Pressure Stress	15	15	15	15
Outplane Stress Summary	3	21	45	-1

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	26	-26	26	-26
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-52	52	52	-52
Inplane Pressure Stress	7	7	7	7
Inplane Stress Summary	-21	31	92	-63

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	25	31	92	63

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-18	18	18	-18
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	7	7	7	7
Outplane Stress Summary	-2	18	36	-14

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	26	-26	26	-26
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-59	59	59	-59
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	15	15	15	15
Inplane Stress Summary	-18	48	104	-65

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			1	1	1	1
Shear Stress Summary			0	0	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress			18	48	105	65

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9	9	9	9	9	9	9	9
Circ. Pl (SUS)	3	3	-1	-1	-3	-3	5	5
Circ. Q (SUS)	-41	41	55	-55	-22	22	38	-38
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-5	-5	7	7	4	4	-2	-2
Long. Q (SUS)	-36	36	52	-52	-26	26	40	-40
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	9.0	9.0	9.0	9.0	9.0	9.0	9.0	9.0
Pm+Pl (SUS)	13.2	13.2	11.3	11.3	8.4	8.4	14.1	14.1
Pm+Pl+Q (Total)	40.2	54.8	67.0	49.0	21.1	36.0	54.8	40.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	9.00	115.15	Passed
Pm+Pl (SUS)	14.08	172.72	Passed
Pm+Pl+Q (TOTAL)	67.00	345.44	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15	15	15	15	15	15	15	15
Circ. Pl (SUS)	-2	-2	6	6	0	0	4	4
Circ. Q (SUS)	-9	9	23	-23	-33	33	85	-85
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-2	-2	6	6	1	1	3	3
Long. Q (SUS)	-26	26	78	-78	-11	11	25	-25

Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0
Pm+P1 (SUS)	13.1	13.1	21.1	21.1	15.1	15.1	19.1	19.1
Pm+P1+Q (Total)	25.3	31.4	91.0	65.0	18.0	48.0	104.1	66.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	15.00	115.15	Passed
Pm+P1 (SUS)	21.12	172.72	Passed
Pm+P1+Q (TOTAL)	104.06	345.44	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	0	0	0	0	0	0	0	0
Circ. P1 (SUS)	3	3	-1	-1	4	4	-2	-2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	0	0	0	0	0	0	0	0
Long. P1 (SUS)	-30	-30	32	32	-20	-20	22	22
Long. Q (SUS)	-177	177	249	-249	-114	114	186	-186
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	16	16	16	16	16	16	16	16
Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+P1 (SUS)	33.5	33.5	33.5	33.5	24.7	24.7	24.7	24.7
Pm+P1+Q (Total)	213.4	149.5	283.2	217.8	140.4	95.8	213.4	166.4

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	137.88	Passed
Pm+P1 (SUS)	33.54	206.82	Passed
Pm+P1+Q (TOTAL)	283.20	413.65	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: N7 4"

From: 20

Pressure for Reinforcement Calculations	P	0.1013	MPa
Temperature for Internal Pressure	Temp	95	°C
Design External Pressure	Pext	0.07	MPa
Temperature for External Pressure	Tempex	95	°C

Parent Material	SA-240 316L
Parent Allowable Stress at Temperature	Sv 115.15 MPa
Parent Allowable Stress At Ambient	Sva 115.15 MPa

Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	1634.9998	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent 1100.00 mm

User Entered Minimum Design Metal Temperature 11.40 °C

Type of Element Connected to the Parent : Nozzle

Material	SA-312 TP316L +IT
Material UNS Number	S31603
Material Specification/Type	Smls. & wld. pipe
Allowable Stress at Temperature	Sn 115.10 MPa
Allowable Stress At Ambient	Sna 115.10 MPa

Diameter Basis (for tr calc only)	Inside
Layout Angle	270.00 deg
Diameter	4.0000 in.

Size and Thickness Basis	Actual
Actual Thickness	tn 19.0500 mm

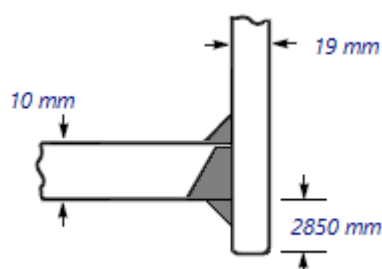
Flange Material	SA-182 F316L +IT
Flange Type	Long Weld Neck

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		6.0000	mm
Inside Projection	h	2850.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class	150
Flange Grade	GR 2.3

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N7 4"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 4.000 in.
 Actual Thickness Used in Calculation 0.750 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.1 \cdot 1500.0 / (115 \cdot 1.0 - 0.6 \cdot 0.1)$
 $= 1.3206 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.1 \cdot 50.8 / (115 \cdot 1.0 - 0.6 \cdot 0.1)$
 $= 0.0447 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5552 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	203.2000	mm
Parallel to Vessel Wall, opening length	d	101.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	25.0000	mm
Normal to Vessel Wall, Inward		25.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115.1 / 115.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115.1 / 115.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$

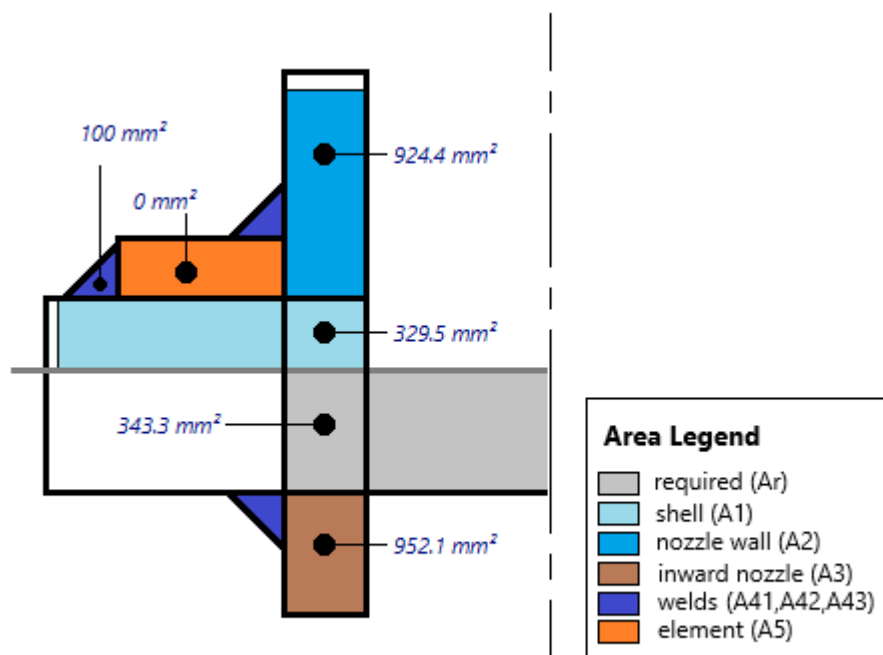


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)



AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	134.198	343.289	NA
Area in Shell	A1	881.689	329.475	NA
Area in Nozzle Wall	A2	949.879	924.367	NA
Area in Inward Nozzle	A3	952.115	952.115	NA
Area in Welds	A41+A42+A43	99.960	99.960	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	2883.643	2305.917	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(101.6 \cdot 6.7566 \cdot 1 + 2 \cdot 19.05 \cdot 6.7566 \cdot 1(1-1.))$
 $= 343.289 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 101.6(1.0 \cdot 10.0 - 1.0 \cdot 6.757) - 2 \cdot 19.05$
 $(1.0 \cdot 10.0 - 1.0 \cdot 6.7566) \cdot (1 - 1.)$
 $= 329.475 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn) fr2$
 $= (2 \cdot 25.0)(19.05 - 0.56) 0.9996$
 $= 924.367 \text{ mm}^2$

Area Available in Inward Nozzle [A3]:
 $= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$
 $= 2 \cdot 19.05 \cdot (25.0) \cdot 0.9996$
 $= 952.115 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi-can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 0.9996 + (0.0)^2 \cdot 0.9996$
 $= 99.960 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 0.5552 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 1.5000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 1.3206 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 1.5000 \text{ mm}$
 Wall Thickness, shell/head, external pressure $trb2 = 0.9703 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 1.5000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 5.7300 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.73, \max(1.5, 1.5)]$
 $= 1.5000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.5552, 1.5)$
 $= 1.5000 \text{ mm}$

Available Nozzle Neck Thickness = 19.0500 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :



Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N7 4"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.0000	mm
Intermediate Calc. for Inward Weld	TminIns	10.0000	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Inward Weld	6.0000 = Min per Code	0.0000 = 0.7 * Wi-Can mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))•Sv)
= max(0, (343.2893 - 329.4754 + 2 • 19.05 • 0.9996 •
(1.0 • 10.0 - 6.7566))115)
= 15812.41 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (924.3666 + 0.0 + 99.9596 - 0.0 • 1.) • 115
= 117937.55 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (924.3666 + 952.1154 + 99.9596 + (380.8461)) • 115
= 271410.34 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (924.3666 + 952.1154 + 99.9596 + 0.0 + (380.8461)) • 115
= 271410.34 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 139.7 • 10.0 • 0.49 • 115
= 123752. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 60.325) • (19.05 - 0.0) • 0.7 • 115
= 290856. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 139.7 • (6.0 - 0.0) • 0.74 • 115
= 112179. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (123752 + 290856) = 414608 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (123752 + 0 + 112179 + 0) = 235931 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)



= (123752 + 112179 + 0) = 235931 N

Summary of Failure Path Calculations:

Path 1-1 = 414607 N	, must exceed W = 15812 N	or W1 = 117937 N
Path 2-2 = 235930 N	, must exceed W = 15812 N	or W2 = 271410 N
Path 3-3 = 235930 N	, must exceed W = 15812 N	or W3 = 271410 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 1.6272 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 3160.0000 mm

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Input, Nozzle Desc: M1 24"

From: 30

Pressure for Reinforcement Calculations	P	0.1013	MPa
Temperature for Internal Pressure	Temp	95	°C
Design External Pressure	Pext	0.07	MPa
Temperature for External Pressure	Tempex	95	°C

Parent Material		SA-240 316L	
Parent Allowable Stress at Temperature	Sv	115.15	MPa
Parent Allowable Stress At Ambient	Sva	115.15	MPa

Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	2499.9998	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		4550.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-240 316L +IT	
Material UNS Number		S31603	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	115.10	MPa
Allowable Stress At Ambient	Sna	115.10	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-182 F316L	
Flange Type		Weld Neck Flange	

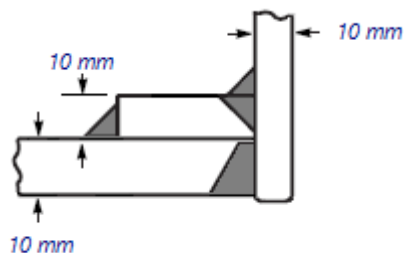
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-240 316L	
Pad Allowable Stress at Temperature	Sp	115.10	MPa
Pad Allowable Stress At Ambient	Spa	115.10	MPa
Diameter of Pad along vessel surface	Dp	729.6000	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	6.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		50.0000	mm
This is a Manway, Coupling or Access Opening.			

Flange Class		150	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: M1 24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.1 \cdot 1500.0 / (115 \cdot 1.0 - 0.6 \cdot 0.1)$
 $= 1.3206 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.1 \cdot 304.8 / (115 \cdot 1.0 - 0.6 \cdot 0.1)$
 $= 0.2685 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.2429 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1219.2000 mm
Parallel to Vessel Wall, opening length	d	609.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		25.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115.1 / 115.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115.1 / 115.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 115.1 / 115.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.)$
 $= 1.000$

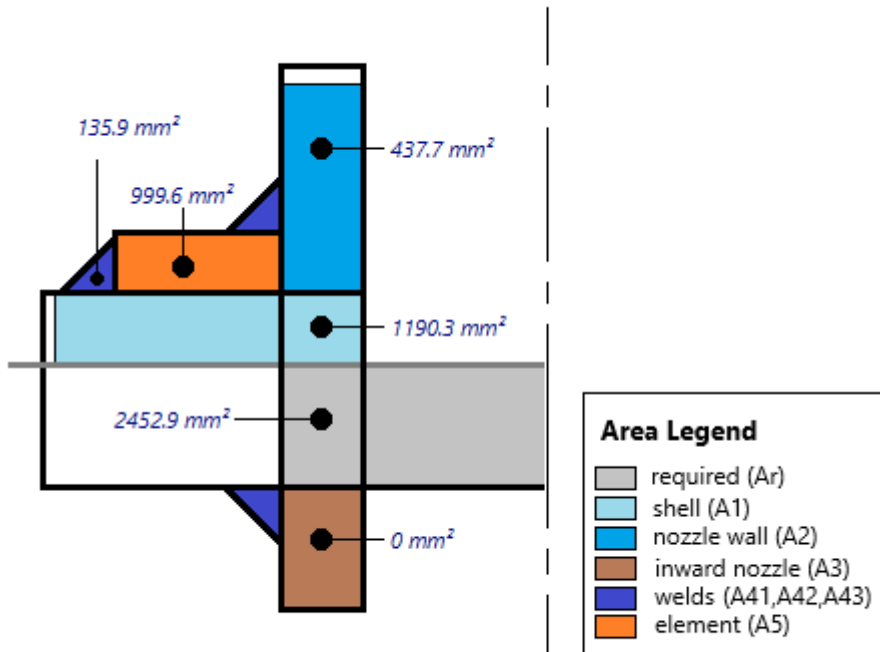


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	805.077	2452.878	NA
Area in Shell	A1	5290.864	1190.292	NA
Area in Nozzle Wall	A2	486.380	437.679	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	135.945	135.945	NA
Area in Element	A5	999.596	999.596	NA
TOTAL AREA AVAILABLE	Atot	6912.785	2763.513	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 698.5239 10.0000 mm
 Based on given Pad Diameter: 729.6000 6.8924 mm
 Based on Shell or Nozzle Thickness: 698.5239 10.0000 mm

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1))$ per UG-37(d)
 $= 0.5(609.6 \cdot 8.0474 \cdot 1 + 2 \cdot 10.0 \cdot 8.0474 \cdot 1(1-1.))$
 $= 2452.878 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 609.6(1.0 \cdot 10.0 - 1.0 \cdot 8.047) - 2 \cdot 10.0$
 $(1.0 \cdot 10.0 - 1.0 \cdot 8.0474) \cdot (1 - 1.)$
 $= 1190.292 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 25.0) \cdot (10.0 - 1.24) \cdot 0.9996$



= 437.679 mm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
= $10.0^2 \cdot 1. + (0.0)^2 \cdot 1. + 6.0^2 \cdot 1.$
= 135.945 mm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
= $(729.6 - 629.6) \cdot 10.0 \cdot 0.9996$
= 999.596 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 1.2429$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= t_a
= 1.2429 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: M1 24"

Intermediate Calc. for nozzle/shell Welds T_{min} 10.0000 mm
Intermediate Calc. for pad/shell Welds T_{minPad} 10.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$5.0000 = 0.5 * T_{minPad}$	$4.2420 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= $\max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
= $\max(0, (2452.8782 - 1190.2924 + 2 \cdot 10.0 \cdot 0.9996 \cdot (1.0 \cdot 10.0 - 8.0474)) \cdot 115)$
= 149864.48 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= $(A2 + A5 + A4 - (W_i - Can/0.707)^2 \cdot fr_2) \cdot Sv$
= $(437.6794 + 999.5961 + 135.9451 - 0.0 \cdot 1.) \cdot 115$
= 181135.45 N

Weld Load [W2]:
= $(A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot Sv$
= $(437.6794 + 0.0 + 99.9596 + (199.9192)) \cdot 115$



= 84920.02 N

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S \\ &= (437.6794 + 0.0 + 135.9451 + 999.5961 + (199.9192)) \cdot 115 \\ &= 204153.48 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 629.6 \cdot 10.0 \cdot 0.49 \cdot 115 \\ &= 557724. \text{ N} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW} \\ &= (3.1416/2.0) \cdot 729.6 \cdot 6.0 \cdot 0.49 \cdot 115 \\ &= 387784. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 309.8) \cdot (10.0 - 0.0) \cdot 0.7 \cdot 115 \\ &= 784093. \text{ N} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg} \\ &= (3.1416/2) \cdot 629.6 \cdot 10.0 \cdot 0.74 \cdot 115 \\ &= 842277. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 629.6 \cdot (6.0 - 0.0) \cdot 0.74 \cdot 115 \\ &= 505570. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (387785 + 784093) = 1171878 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw}) \\ &= (557724 + 842277 + 505570 + 0) = 1905570 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (S_{pew} + T_{ngw} + S_{inw}) \\ &= (387785 + 505570 + 0) = 893355 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1171877 N	, must exceed W = 149864 N	or W1 = 181135 N
Path 2-2 = 1905570 N	, must exceed W = 149864 N	or W2 = 84920 N
Path 3-3 = 893354 N	, must exceed W = 149864 N	or W3 = 204153 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0 MPa

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 33.4049 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 293.4049 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.614 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.



Input, Nozzle Desc: N8 26"

From: 40

Pressure for Reinforcement Calculations	P	0.1013	MPa
Temperature for Internal Pressure	Temp	95	°C
Design External Pressure	Pext	0.07	MPa
Temperature for External Pressure	Tempex	95	°C

Parent Material		SA-240 316L	
Parent Allowable Stress at Temperature	Sv	115.15	MPa
Parent Allowable Stress At Ambient	Sva	115.15	MPa

Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	1585.0000	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		7285.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		11.40	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-240 316L +IT	
Material UNS Number		S31603	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	115.10	MPa
Allowable Stress At Ambient	Sna	115.10	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		26.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-182 F316L	
Flange Type		Weld Neck Flange	

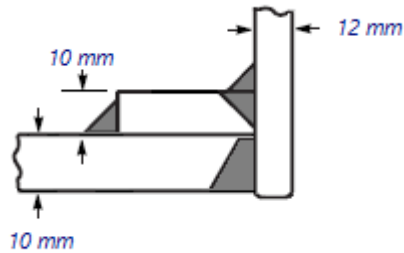
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-240 316L	
Pad Allowable Stress at Temperature	Sp	115.10	MPa
Pad Allowable Stress At Ambient	Spa	115.10	MPa
Diameter of Pad along vessel surface	Dp	762.0000	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		38.8000	mm

Flange Class		150	
Flange Grade		GR 2.3	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N8 26"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	26.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.1 \cdot 1500.0 / (115 \cdot 1.0 - 0.6 \cdot 0.1)$
 $= 1.3206 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.1 \cdot 330.2 / (115 \cdot 1.0 - 0.6 \cdot 0.1)$
 $= 0.2908 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.4059 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1320.7999	mm
Parallel to Vessel Wall, opening length	d	660.4000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		25.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115.1 / 115.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115.1 / 115.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 115.1 / 115.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.)$
 $= 1.000$

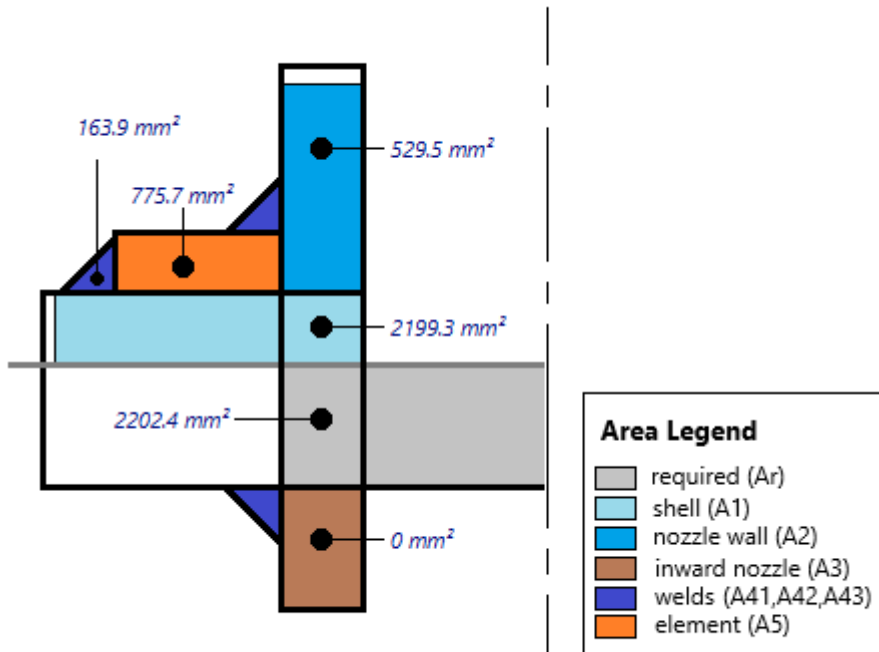


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	872.167	2202.371	NA
Area in Shell	A1	5731.760	2199.289	NA
Area in Nozzle Wall	A2	585.222	529.493	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	163.934	163.934	NA
Area in Element	A5	775.687	775.687	NA
TOTAL AREA AVAILABLE	Atot	7256.603	3668.403	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(660.4 \cdot 6.6697 \cdot 1 + 2 \cdot 12.0 \cdot 6.6697 \cdot 1(1 - 1.))$
 $= 2202.371 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 660.4(1.0 \cdot 10.0 - 1.0 \cdot 6.67) - 2 \cdot 12.0$
 $(1.0 \cdot 10.0 - 1.0 \cdot 6.6697) \cdot (1 - 1.)$
 $= 2199.289 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 25.0) \cdot (12.0 - 1.41) \cdot 0.9996$
 $= 529.493 \text{ mm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1. + 8.0^2 \cdot 1.$



= 163.934 mm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) • fr4
= (762.0 - 684.4) • 10.0 • 0.9996
= 775.687 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.4059 mm
Wall Thickness per UG16(b),	tr16b = 1.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 1.3206 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 1.5000 mm
Wall Thickness, shell/head, external pressure	trb2 = 0.9703 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 1.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[8.331, max(1.5, 1.5)]
= 1.5000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(1.4059, 1.5)
= 1.5000 mm

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N8 26"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	10.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	5.0000 = 0.5*TminPad	5.6560 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (2202.3713 - 2199.2893 + 2 • 12.0 • 0.9996 •
(1.0 • 10.0 - 6.6697))115)
= 9553.64 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:



$$\begin{aligned} &= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv \\ &= (529.4926 + 775.6867 + 163.9337 - 0.0 \cdot 1.) \cdot 115 \\ &= 169148.88 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv \\ &= (529.4926 + 0.0 + 99.9596 + (239.9031)) \cdot 115 \\ &= 100094.70 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S \\ &= (529.4926 + 0.0 + 163.9337 + 775.6867 + (239.9031)) \cdot 115 \\ &= 196770.52 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 684.4 \cdot 10.0 \cdot 0.49 \cdot 115 \\ &= 606268. \text{ N} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.1416/2.0) \cdot 762.0 \cdot 8.0 \cdot 0.49 \cdot 115 \\ &= 540007. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 336.2) \cdot (12.0 - 0.0) \cdot 0.7 \cdot 115 \\ &= 1021093. \text{ N} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.1416/2) \cdot 684.4 \cdot 10.0 \cdot 0.74 \cdot 115 \\ &= 915588. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 684.4 \cdot (6.0 - 0.0) \cdot 0.74 \cdot 115 \\ &= 549575. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (540007 + 1021093) = 1561100 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (606268 + 915588 + 549575 + 0) = 2071430 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (540007 + 549575 + 0) = 1089582 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1561099 N	, must exceed W = 9553 N	or W1 = 169148 N
Path 2-2 = 2071430 N	, must exceed W = 9553 N	or W2 = 100094 N
Path 3-3 = 1089581 N	, must exceed W = 9553 N	or W3 = 196770 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0 MPa

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 39.5551 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 349.5551 mm

Percent Elongation Calculations:

PV Elite 28 U2 Licensee: SL User
FileName : 411-V-001_7%
Nozzle Calcs.: N8 26"

Nozl: 30 5:39pm Aug 11,2026

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HEXAGON

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.785 %
Note: Please Check Requirements of Table UHA-44 for Elongation limits.

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Input, Nozzle Desc: Nozzle boot

From: 40

Pressure for Reinforcement Calculations	P	0.1386	MPa
Temperature for Internal Pressure	Temp	95	°C
Design External Pressure	Pext	0.07	MPa
Temperature for External Pressure	Tempex	95	°C

Parent Material		SA-240 316L	
Parent Allowable Stress at Temperature	Sv	115.15	MPa
Parent Allowable Stress At Ambient	Sva	115.15	MPa

Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	2499.9998	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		6050.00	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-240 316L +IT	
Material UNS Number		S31603	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	115.10	MPa
Allowable Stress At Ambient	Sna	115.10	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		40.0000	in.

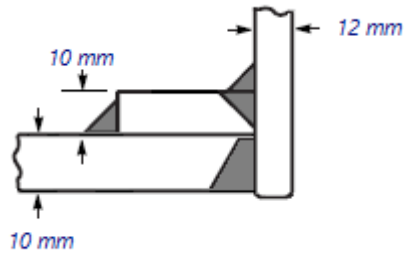
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	700.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.90	MPa
Pad Allowable Stress At Ambient	Spa	137.90	MPa
Diameter of Pad along vessel surface	Dp	1220.0000	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		90.0000	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: Nozzle boot

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	40.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.14 \cdot 1500.0 / (115 \cdot 1.0 - 0.6 \cdot 0.14)$
 $= 1.8070 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.14 \cdot 508.0 / (115 \cdot 1.0 - 0.6 \cdot 0.14)$
 $= 0.6122 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 2.5560 mm

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	2032.0000 mm
Parallel to Vessel Wall, opening length	d	1016.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		25.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115.1 / 115.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115.1 / 115.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 137.9 / 115.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.0)$
 $= 1.000$

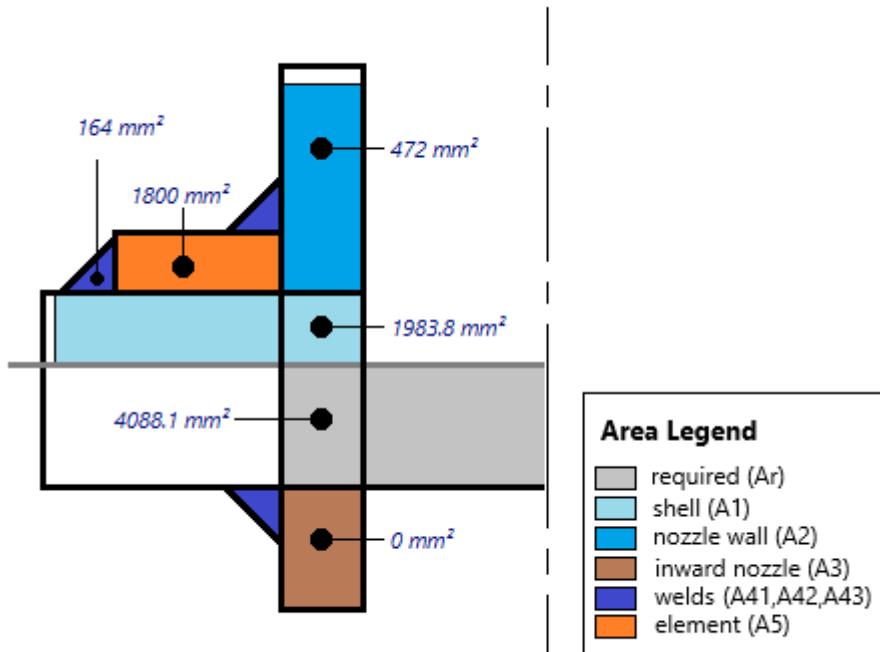


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1835.944	4088.115	NA
Area in Shell	A1	8323.993	1983.828	NA
Area in Nozzle Wall	A2	569.159	472.010	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	163.960	163.960	NA
Area in Element	A5	1800.000	1800.000	NA
TOTAL AREA AVAILABLE	Atot	10857.112	4419.798	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 1186.8317 10.0000 mm
 Based on given Pad Diameter: 1220.0000 8.1573 mm
 Based on Shell or Nozzle Thickness: 1186.8317 10.0000 mm

Area Required [A]:
 $= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1))$ per UG-37(d)
 $= 0.5(1016.0 \cdot 8.0474 \cdot 1 + 2 \cdot 12.0 \cdot 8.0474 \cdot 1(1 - 1.))$
 $= 4088.115 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 1016.0(1.0 \cdot 10.0 - 1.0 \cdot 8.047) - 2 \cdot 12.0$
 $(1.0 \cdot 10.0 - 1.0 \cdot 8.0474) \cdot (1 - 1.)$
 $= 1983.828 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 25.0) \cdot (12.0 - 2.56) \cdot 0.9996$



= 472.010 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - c_n / 0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10.0^2 \cdot 1. + (0.0)^2 \cdot 1. + 8.0^2 \cdot 1.0$
 = 163.960 mm²

Area Available in Element [A5]:
 = $(\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$
 = $(1220.0 - 1040.0) \cdot 10.0 \cdot 1.0$
 = 1800.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.5560 mm
Wall Thickness per UG16(b),	tr16b = 1.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 1.8070 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 1.8070 mm
Wall Thickness, shell/head, external pressure	trb2 = 0.9703 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 1.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[8.331, max(1.807, 1.5)]
 = 1.8070 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(2.556, 1.807)
 = 2.5560 mm

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 10.0, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = -1.0$, Temp. Reduction = 0 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 10.0, tr = 1.807, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.181$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1524.0000 mm
Parallel to Vessel Wall, opening length	d	762.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		25.0000 mm



Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (mm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [External Pressure]	AR	2725.546	2725.546
Area Available in Shell	A1	991.905	991.905
Area Available in Nozzle Wall	A2	472.010	472.010
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	99.960	163.960
Area Available in Pad	A5	0.000	1800.000
Area Available in Hub	A6	0.000	0.000
TOTAL AREA AVAILABLE	Atot	1563.874	3427.875

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

The following Calculation is performed using the internal design pressure (P) in the corroded condition.

Warning: The flange could not be considered in the moment of inertia calculations.

Area/Inertia Contributions from Fig.1-7-2 (for Bending stress only):

	Area	Distance	Inertia	d	I+Ad ²
Nozzle	2544.0	106.0	0.9528E+07	48.275	0.1546E+08
Shell	1600.0	5.0	0.1333E+05	52.725	0.4461E+07
Pad	900.0	15.0	0.7500E+04	42.725	0.1650E+07
Sum	5044.0				0.2157E+08

Membrane stress per Appendix 1-7 (1 or 2) Sm	42.48	MPa
Allowable Membrane Stress S	115.15	MPa
Bending stress at the inside of the Nozzle Sb	23.01	MPa
Combined Membrane plus Bending Stress Sm + Sb	65.49	MPa
Allowable Membrane + Bending Stress 1.5 * S	172.72	MPa
M.A.W.P for the Appendix 1-7 Stress calc	0.37	MPa

Area contributions from Fig. 1-7-1 for Membrane stress per 1-7(b)(4):

Area from Nozzle =	1182.439 mm ²
Area from Shell =	1226.784 mm ²
Area from Pad =	900.000 mm ²
Total Area, As =	3309.224 mm ²

Membrane Stress Calc for Large Nozzles per 1-7(b)(4) [Sm]:

$$= P [R(Rn + tn + \sqrt{(rm \cdot t)}) + Rn(t + te + \sqrt{(Rnm \cdot tn)})] / As$$

$$= 0.139 [1500.0(508.0 + 12.0 + 122.678) + 508.0(98.537)] / 3309.224$$

$$= 42.477 \text{ MPa}$$

The area As was computed per figure 1-7-1 for the Sm calculation.

Moment [M]:

$$= (Rn^3/6 + R \cdot Rn \cdot e) P$$

$$= (508.0^3/6 + 1500.0 \cdot 508.0 \cdot 52.725) 0.139$$

$$= 8600393 \text{ N-mm}$$

Bending Stress Calc for Large Nozzles per 1-7(b)(4)(5) [Sb]:

$$= (M(e + t/2)) / I$$

$$= (8600393(52.725 + 10.0/2)) / 21568432$$

$$= 23.010 \text{ MPa}$$

Weld Size Calculations, Description: Nozzle boot

Intermediate Calc. for nozzle/shell Welds Tmin	10.0000	mm
Intermediate Calc. for pad/shell Welds TminPad	10.0000	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm



Pad Weld $5.0000 = 0.5 \cdot T_{minPad}$ $5.6560 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (4088.1152 - 1983.828 + 2 \cdot 12.0 \cdot 0.9996 \cdot (1.0 \cdot 10.0 - 8.0474)) \cdot 115)$
 $= 247674.14 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (472.0101 + 1800.0005 + 163.9596 - 0.0 \cdot 1.) \cdot 115$
 $= 280469.62 \text{ N}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (472.0101 + 0.0 + 99.9596 + (239.9031)) \cdot 115$
 $= 93476.37 \text{ N}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (472.0101 + 0.0 + 163.9596 + 1800.0005 + (239.9031)) \cdot 115$
 $= 308091.28 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 1040.0 \cdot 10.0 \cdot 0.49 \cdot 115$
 $= 921272. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 1220.0 \cdot 8.0 \cdot 0.49 \cdot 115$
 $= 864927. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 514.0) \cdot (12.0 - 0.0) \cdot 0.7 \cdot 115$
 $= 1561100. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 1040.0 \cdot 10.0 \cdot 0.74 \cdot 138$
 $= 1666910. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 1040.0 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 115$
 $= 1391870. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (S_{PEW} + S_{NW}) = (864927 + 1561100) = 2426027 \text{ N}$

Failure Path [Path 2-2]:
 $= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$
 $= (921272 + 1666910 + 1391870 + 0) = 3980052 \text{ N}$

Failure Path [Path 3-3]:
 $= (S_{pew} + T_{ngw} + S_{inw})$
 $= (864927 + 1391870 + 0) = 2256798 \text{ N}$

Summary of Failure Path Calculations:



Path 1-1 = 2426027 N , must exceed W = 247674 N or W1 = 280469 N

Path 2-2 = 3980052 N , must exceed W = 247674 N or W2 = 93476 N

Path 3-3 = 2256797 N , must exceed W = 247674 N or W3 = 308091 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0 MPa

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 93.0174 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 803.0173 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.167 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

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Input, Nozzle Desc: N2 3"

From: 60

Pressure for Reinforcement Calculations	P	0.1013	MPa
Temperature for Internal Pressure	Temp	95	°C
Design External Pressure	Pext	0.07	MPa
Temperature for External Pressure	Tempex	95	°C

Parent Material	SA-240 316L
Parent Allowable Stress at Temperature	Sv 115.15 MPa
Parent Allowable Stress At Ambient	Sva 115.15 MPa

Outside Diameter of Bolted Blind Flange	D	869.95	mm
Head Finished (Minimum) Thickness	t	66.8020	mm
Head Internal Corrosion Allowance	cas	0.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
-------------------------------	----	--------	----

User Entered Minimum Design Metal Temperature	11.40	°C
---	-------	----

Type of Element Connected to the Parent : Nozzle

Material	SA-312 TP316
Material UNS Number	S31600
Material Specification/Type	Wld. pipe
Allowable Stress at Temperature	Sn 137.88 MPa
Allowable Stress At Ambient	Sna 137.88 MPa

Note: The allowables above have been divided by 0.85, see UG-37 [S].

Diameter Basis (for tr calc only)	Inside
Layout Angle	0.00 deg
Diameter	3.0000 in.

Size and Thickness Basis	Actual
Actual Thickness	tn 15.7480 mm

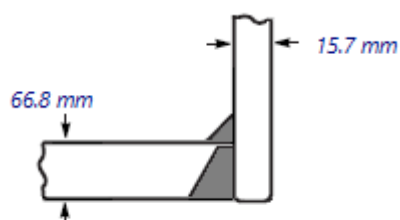
Flange Material	SA-182 F316
Flange Type	Long Weld Neck

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	204.7870	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	66.8020	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class	150
Flange Grade	GR 2.2

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N2 3"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.620 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.1 \cdot 38.1 / (138 \cdot 1.0 - 0.6 \cdot 0.1)$
 $= 0.0280$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4053 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 241.3000 mm
 Parallel to Vessel Wall Rn+tn+t 120.6500 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 39.3700 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n/S_v)$
 $= \min(1, 137.9/115.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n/S_v)$
 $= \min(1, 137.9/115.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

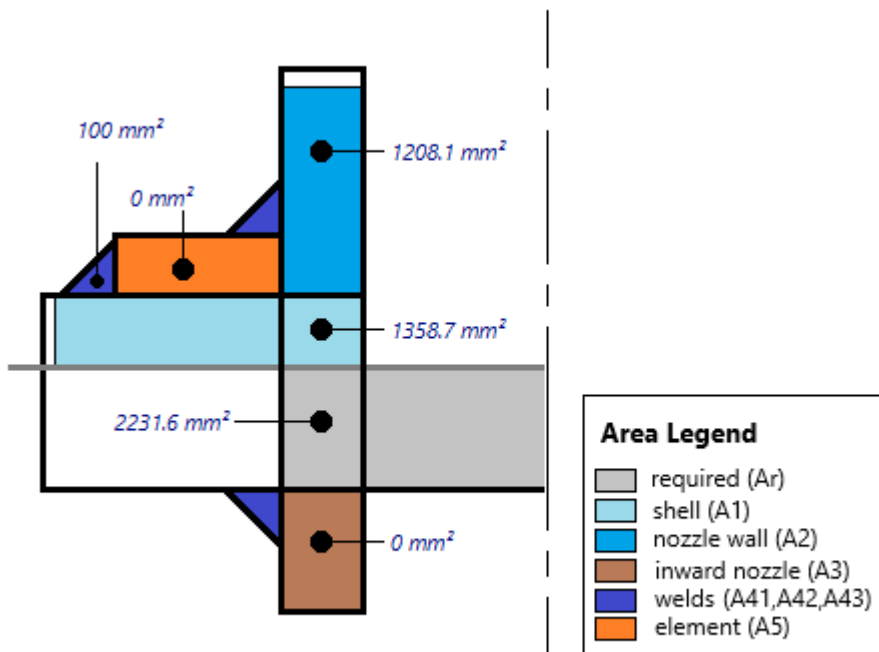


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	2231.357	2231.608	NA
Area in Shell	A1	1359.798	1358.708	NA
Area in Nozzle Wall	A2	1237.792	1208.085	NA



Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	2697.590	2666.792	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= 0.5 \cdot d \cdot t + t \cdot t_n(1 - fr_1) \text{ per UG-39(b)(1)}$$

$$= 0.5 \cdot 76.2 \cdot 58.5724 + 58.5724 \cdot 15.748(1 - 1.0)$$

$$= 2231.608 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E_1 \cdot t - F \cdot tr) - 2 \cdot t_n(E_1 \cdot t - F \cdot tr) \cdot (1 - fr_1)$$

$$= 165.1(1.0 \cdot 66.802 - 1.0 \cdot 58.572) - 2 \cdot 15.748$$

$$(1.0 \cdot 66.802 - 1.0 \cdot 58.5724) \cdot (1 - 1.0)$$

$$= 1358.708 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np})(t_n - tr_n)fr_2$$

$$= (2 \cdot 39.37)(15.75 - 0.41)1.0$$

$$= 1208.085 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can/0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 100.000 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 0.4053 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 1.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 58.5658 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 58.5658 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 1.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 5.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[5.258, \max(58.5658, 1.5)]$$

$$= 5.2578 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(0.4053, 5.2578)$$

$$= 5.2578 \text{ mm}$$

Available Nozzle Neck Thickness = 15.7480 mm -> OK

SA-240 316L, Min Metal Temp without impact per UHA-51: -196 °C

SA-312 TP316, Min Metal Temp without impact per UHA-51: -196 °C

Weld Size Calculations, Description: N2 3"

Intermediate Calc. for nozzle/shell Welds Tmin 15.7480 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:



$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (2231.6084 - 1358.7075 + 2 \cdot 15.748 \cdot 1.0 \cdot \\
 &\quad (1.0 \cdot 66.802 - 58.5724)) \cdot 115) \\
 &= 130346.33 \text{ N}
 \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (1208.0847 + 0.0 + 100.0 - 0.0 \cdot 1.0) \cdot 115 \\
 &= 150608.58 \text{ N}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (1208.0847 + 0.0 + 100.0 + (2103.9958)) \cdot 115 \\
 &= 392855.72 \text{ N}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (1208.0847 + 0.0 + 100.0 + 0.0 + (2103.9958)) \cdot 115 \\
 &= 392855.72 \text{ N}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 107.696 \cdot 10.0 \cdot 0.49 \cdot 115 \\
 &= 95440. \text{ N}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 45.974) \cdot (15.748 - 0.0) \cdot 0.7 \cdot 138 \\
 &= 219512. \text{ N}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 107.696 \cdot (66.802 - 0.0) \cdot 0.74 \cdot 138 \\
 &= 1152955. \text{ N}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (95440 + 219512) = 314951 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (95440 + 0 + 1152955 + 0) = 1248395 \text{ N}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\
 &= (95440 + 1152955 + 0) = 1248395 \text{ N}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 314951 N	, must exceed W = 130346 N	or W1 = 150608 N
Path 2-2 = 1248394 N	, must exceed W = 130346 N	or W2 = 392855 N
Path 3-3 = 1248394 N	, must exceed W = 130346 N	or W3 = 392855 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2 MPa

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Cut Length for this Nozzle is, Drop + Ho + H + T : 271.6398 mm

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Input, Nozzle Desc: N9 3"

From: 70

Pressure for Reinforcement Calculations	P	0.1013	MPa
Temperature for Internal Pressure	Temp	95	°C
Design External Pressure	Pext	0.07	MPa
Temperature for External Pressure	Tempex	95	°C

Parent Material		SA-240 316L	
Parent Allowable Stress at Temperature	Sv	115.15	MPa
Parent Allowable Stress At Ambient	Sva	115.15	MPa

Inside Diameter of Elliptical Head	D	1016.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	6.0000	mm
Head Internal Corrosion Allowance	cas	0.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
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User Entered Minimum Design Metal Temperature		11.40	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP316L +IT	
Material UNS Number		S31603	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	115.10	MPa
Allowable Stress At Ambient	Sna	115.10	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		40	

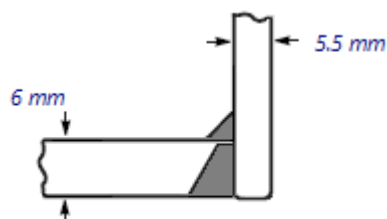
Flange Material		SA-182 F316	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N9 3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 3.068 in.
Actual Thickness Used in Calculation 0.216 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $0.1 \cdot 1016.0 \cdot 0.9 / (2 \cdot 115.15 \cdot 1.0 - 0.2 \cdot 0.1)$
= 0.4024 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.1 \cdot 38.96 / (115 \cdot 1.0 - 0.6 \cdot 0.1)$
= 0.0343 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4264 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	155.8544	mm
Parallel to Vessel Wall, opening length	d	77.9272	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	13.7160	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N9 3".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.4264	mm
Wall Thickness per UG16(b),	tr16b	= 1.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 0.4471	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 1.5000	mm
Wall Thickness, shell/head, external pressure	trb2	= 0.3285	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[4.8, max(1.5, 1.5)]
= 1.5000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.4264, 1.5)
= 1.5000 mm

Available Nozzle Neck Thickness = $0.875 \cdot 5.486 = 4.801$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N9 3"

Intermediate Calc. for nozzle/shell welds	Tmin	5.4864	mm
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Results Per UW-16.1:

PV Elite 28 U2 Licensee: SL User
FileName : 411-V-001_7%
Nozzle Calcs.: N9 3"

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HEXAGON

	Required Thickness	Actual Thickness
Nozzle Weld	$3.8405 = 0.7 * t_{min.}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0754 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 307.0754 mm

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: